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(54) **DEPTH PLANE SELECTION FOR
MULTI-DEPTH PLANE DISPLAY SYSTEMS
BY USER CATEGORIZATION**

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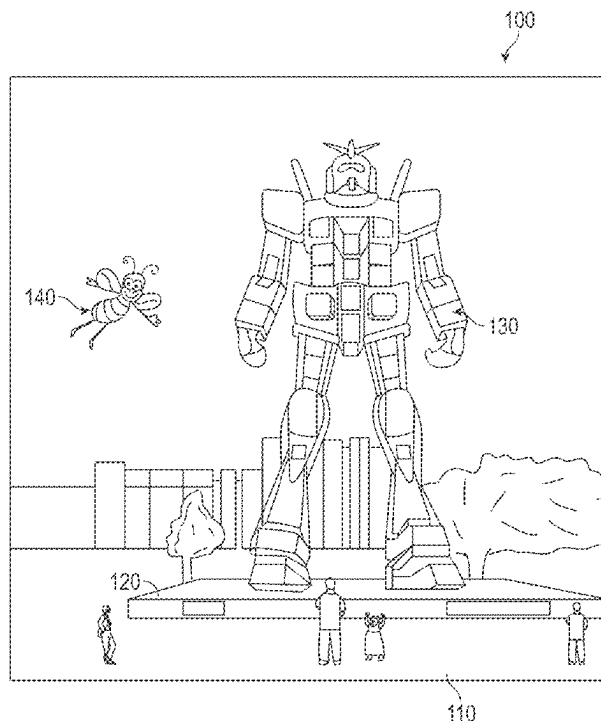
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(57) **ABSTRACT**

A display system includes a head-mounted display configured to project light, having different amounts of wavefront divergence, to an eye of a user to display virtual image content appearing to be disposed at different depth planes. The wavefront divergence may be changed in discrete steps, with the change in steps being triggered based upon whether the user is fixating on a particular depth plane. The display system may be calibrated for switching depth planes for a main user. Upon determining that a guest user is utilizing the system, rather than undergoing a full calibration, the display system may be configured to switch depth planes based on a rough determination of the virtual content that the user is looking at. The virtual content may have an associated depth plane and the display system may be configured to switch to the depth plane of that virtual content.



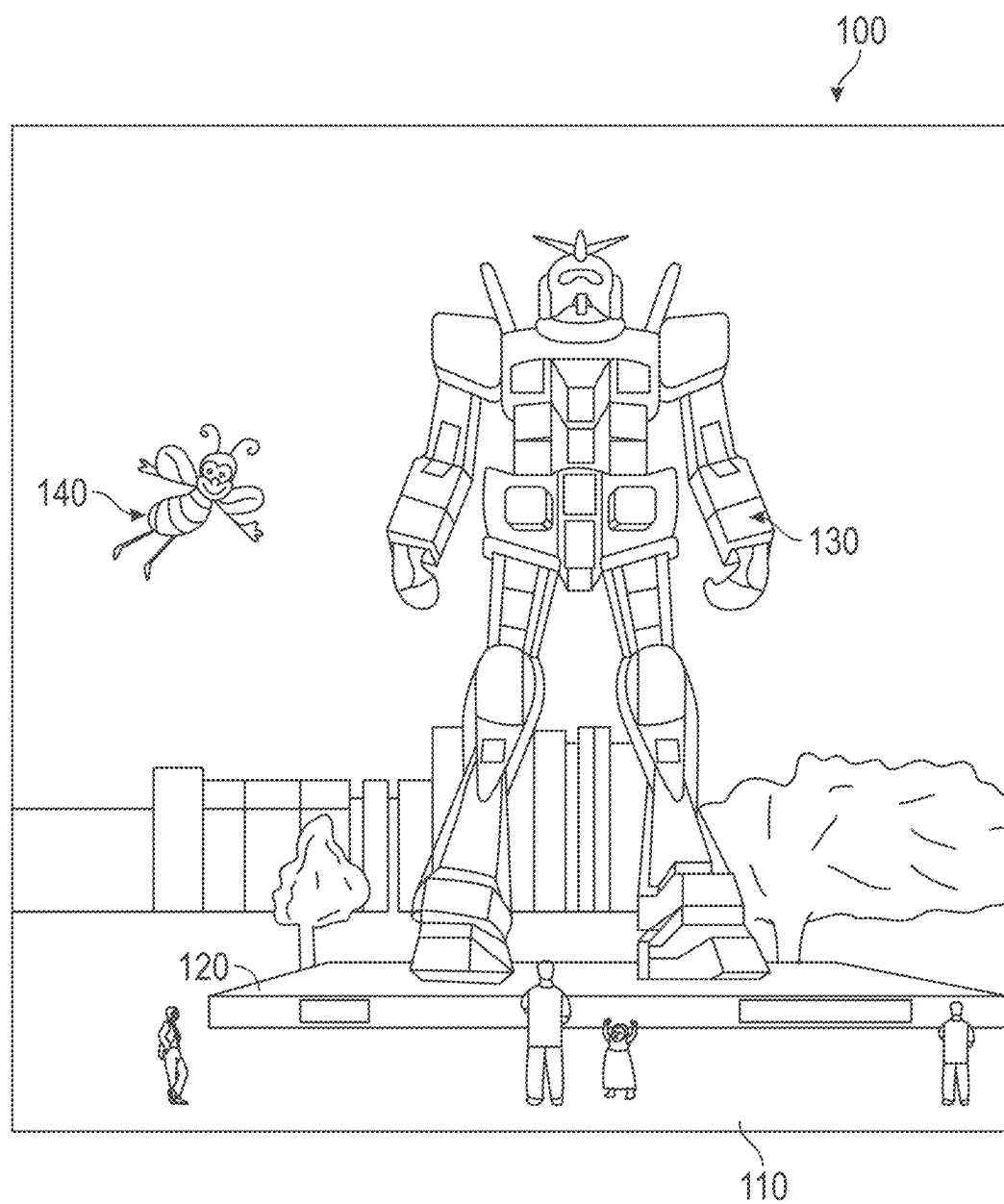


FIG. 1

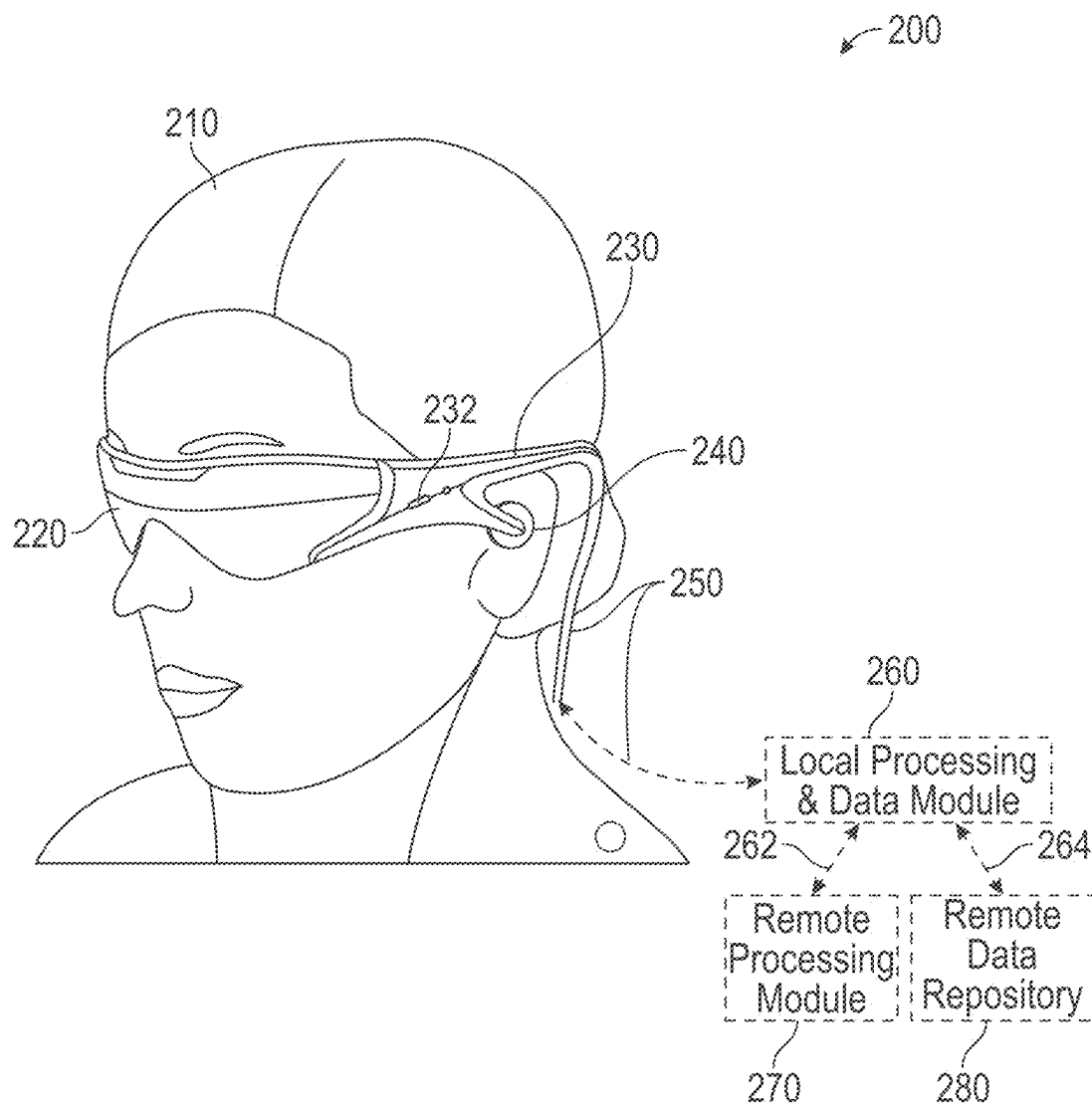
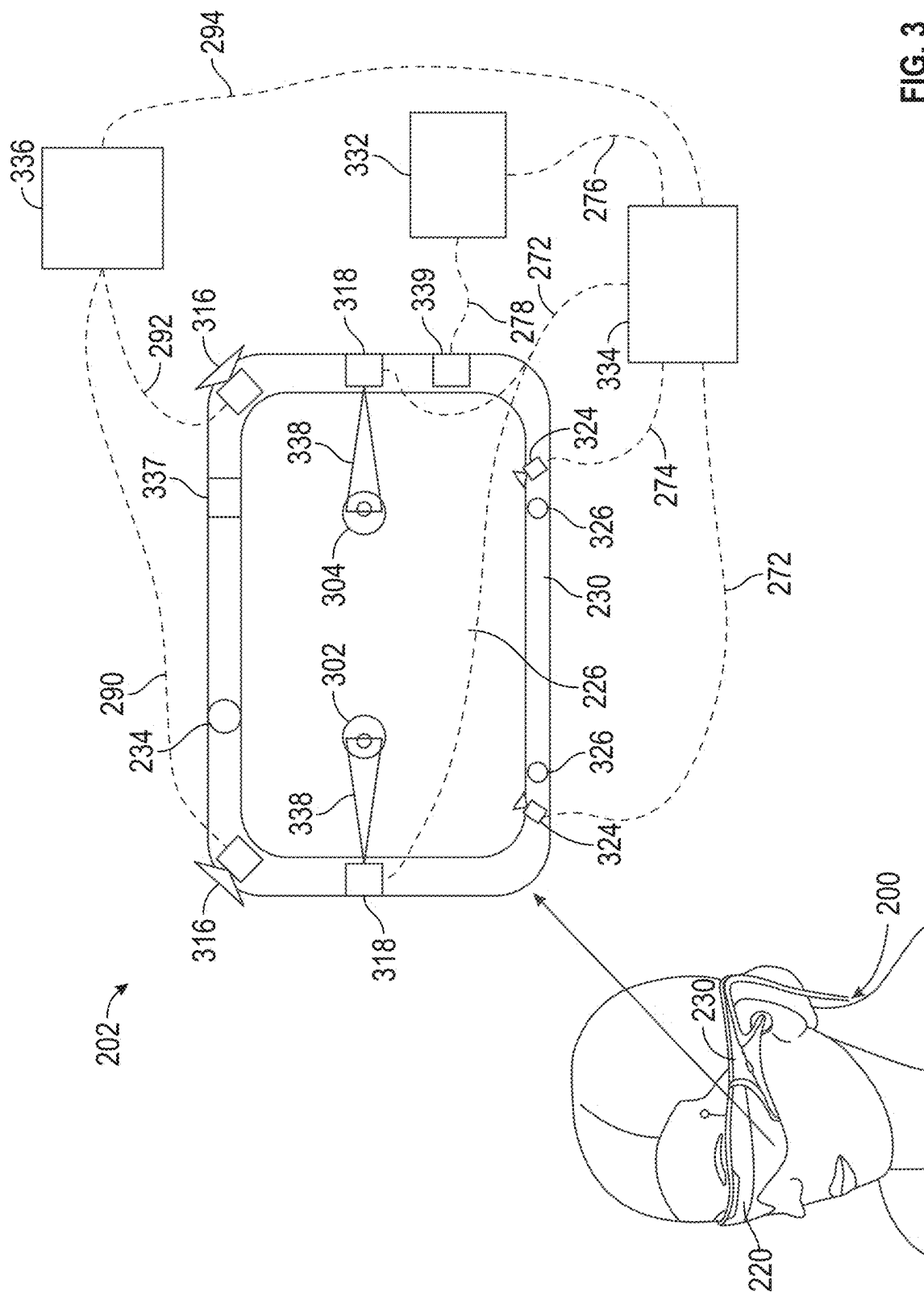


FIG. 2



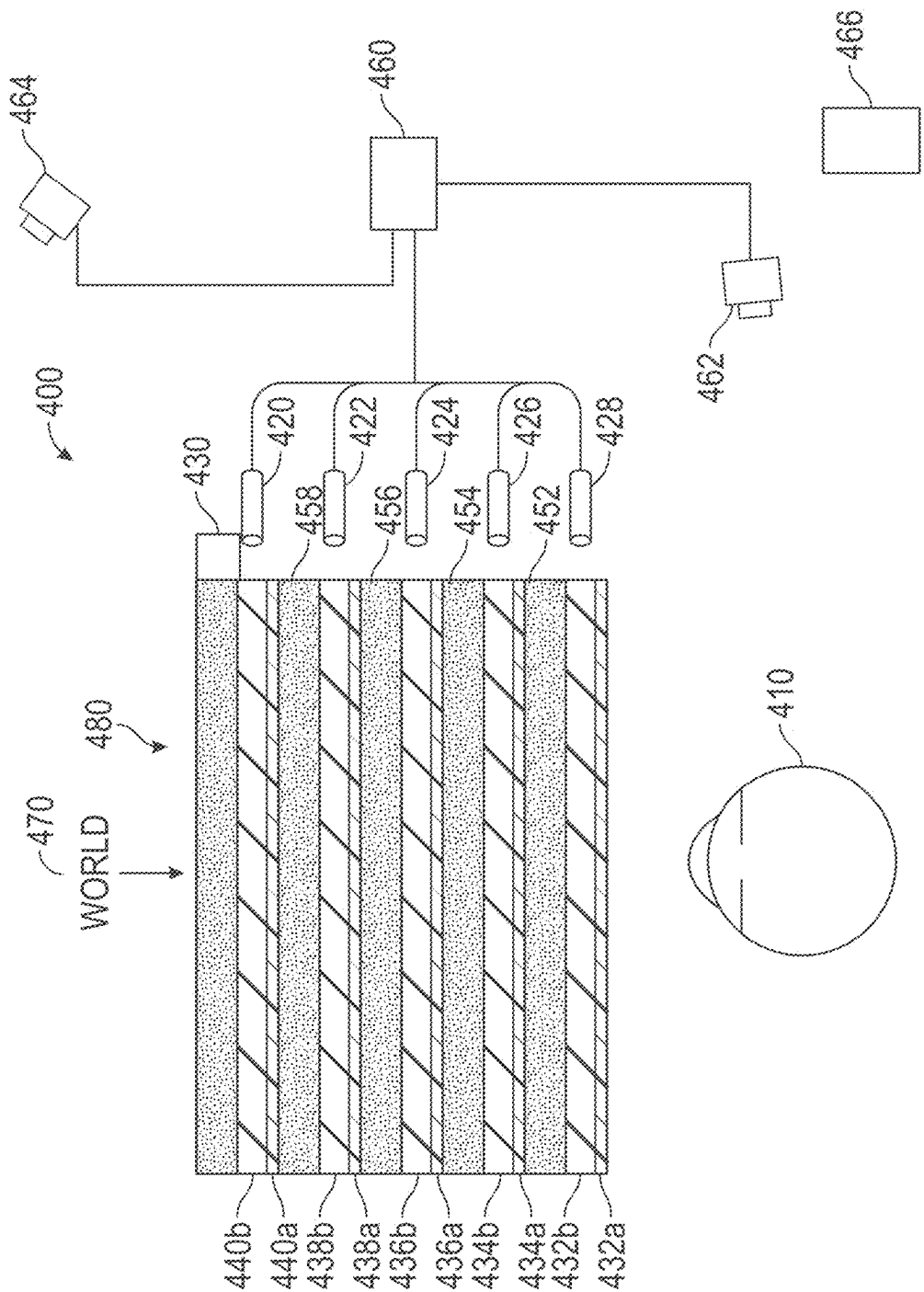


FIG. 4

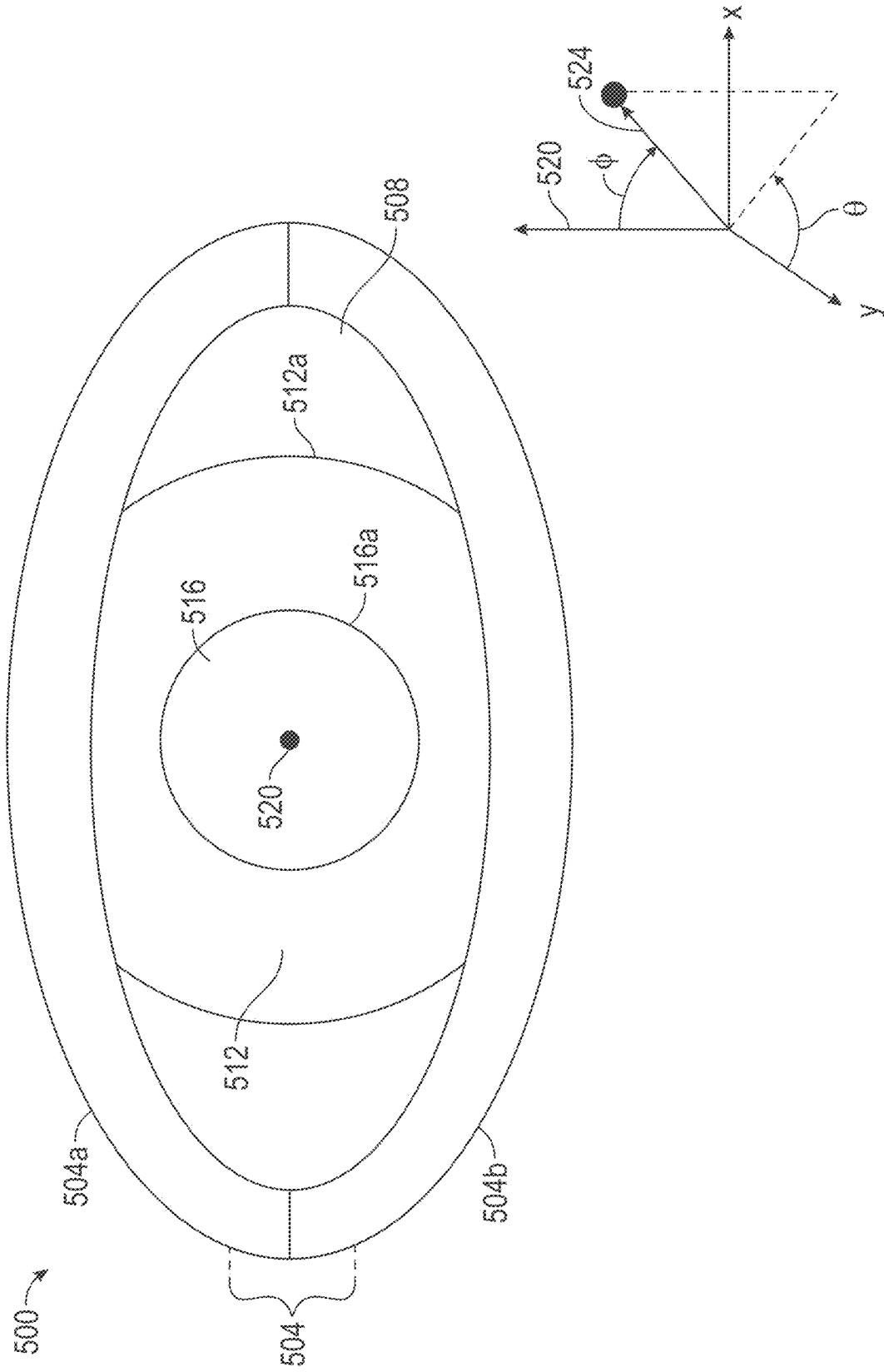


FIG. 5

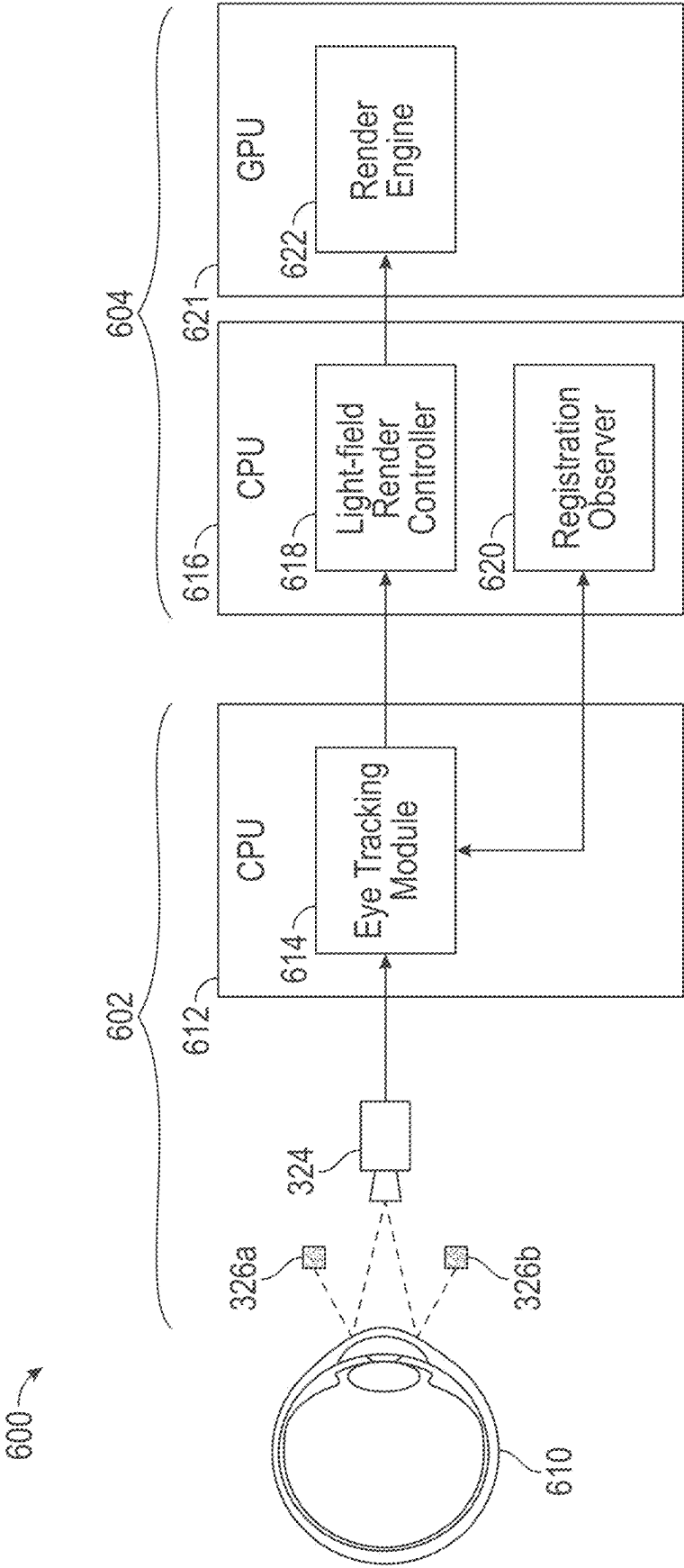


FIG. 6

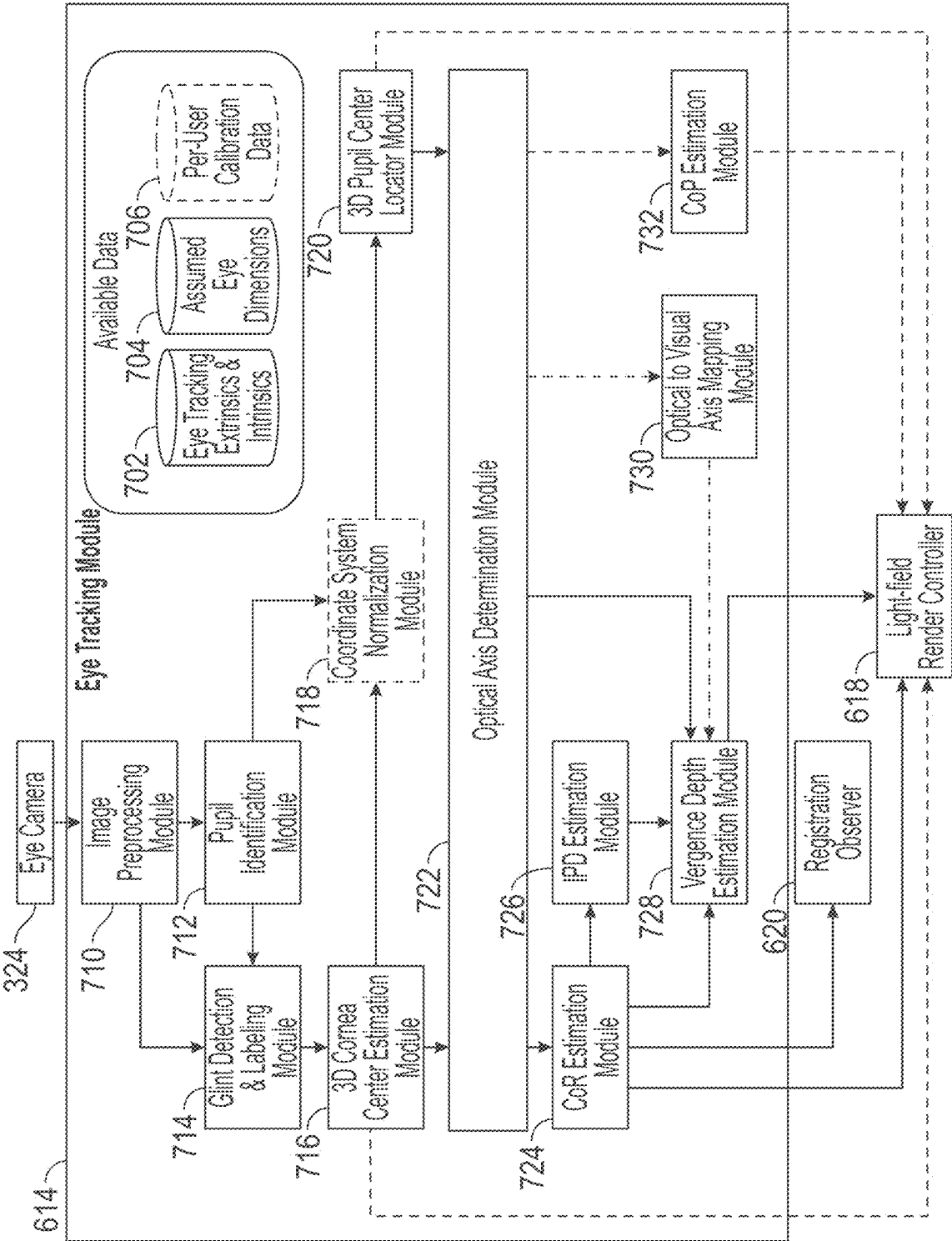


FIG. 7A

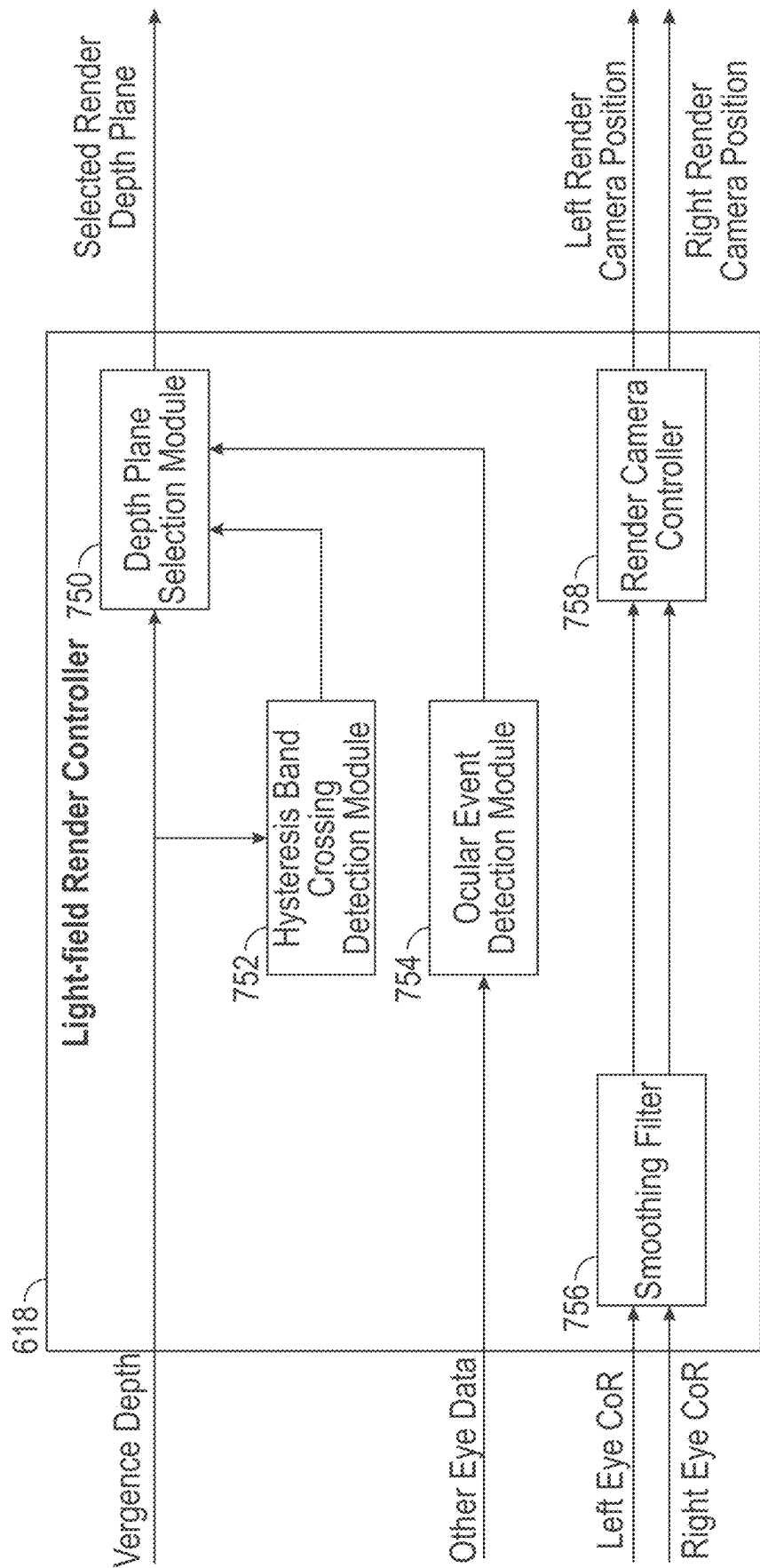


FIG. 7B

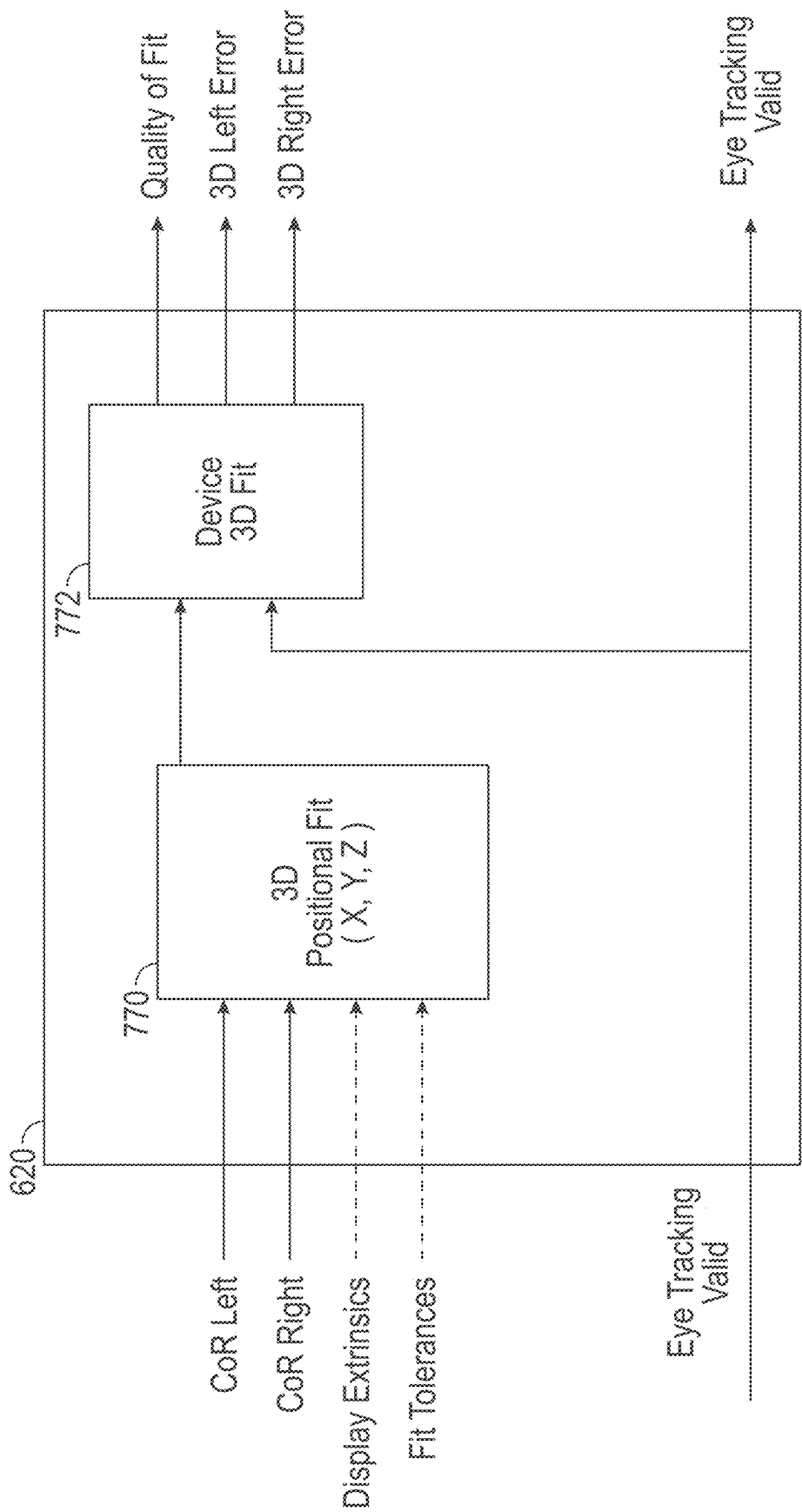


FIG. 7C

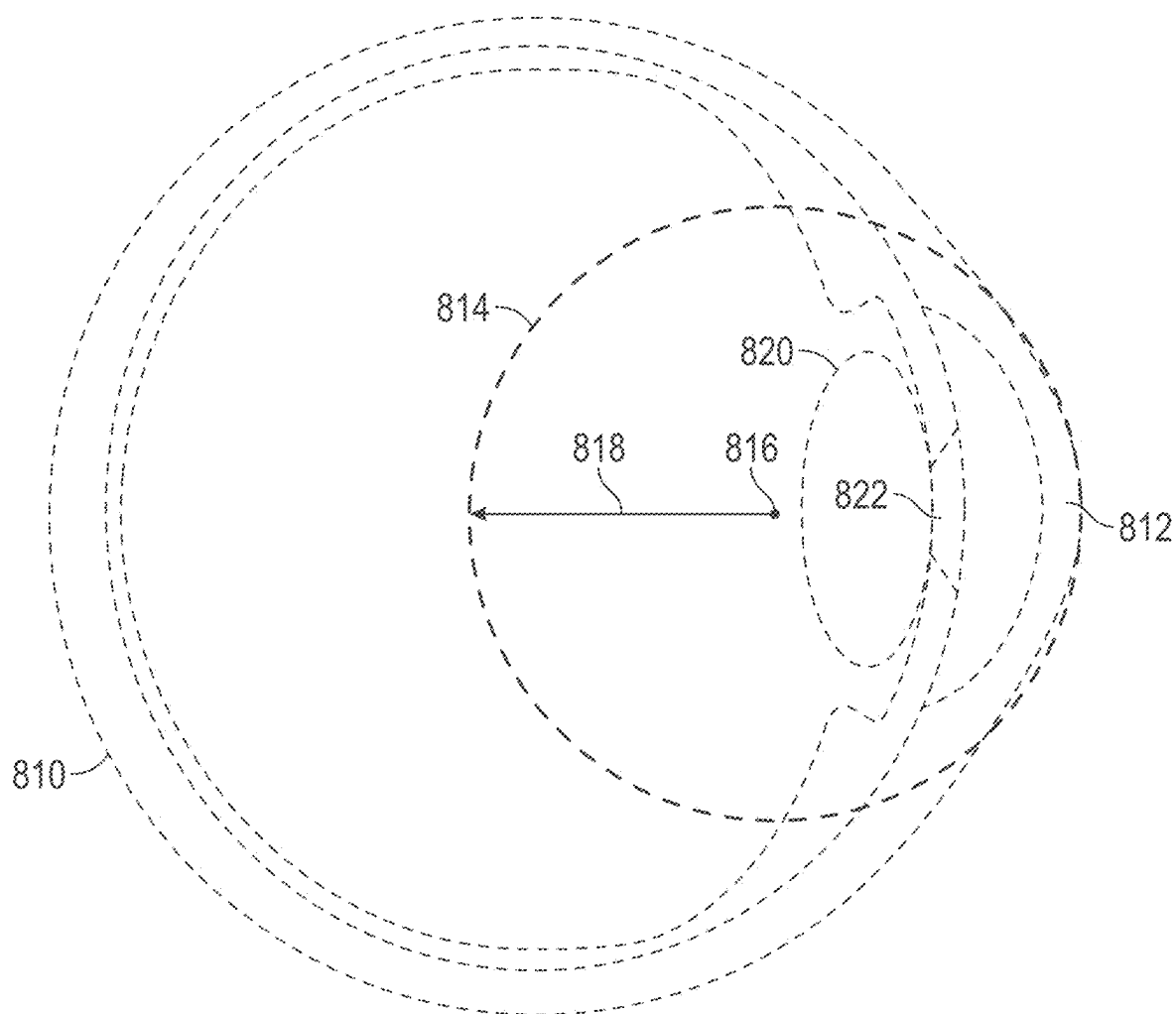


FIG. 8A

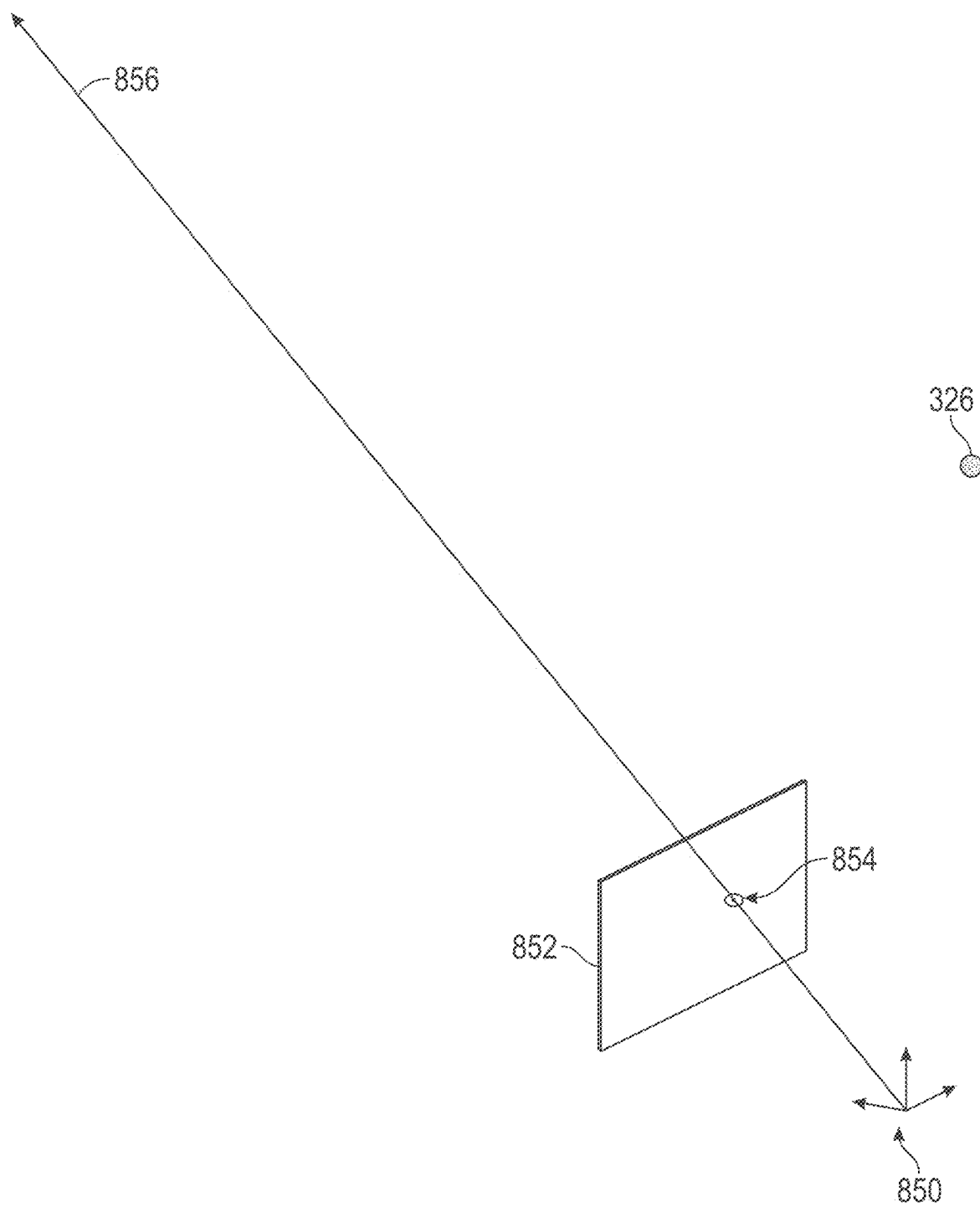
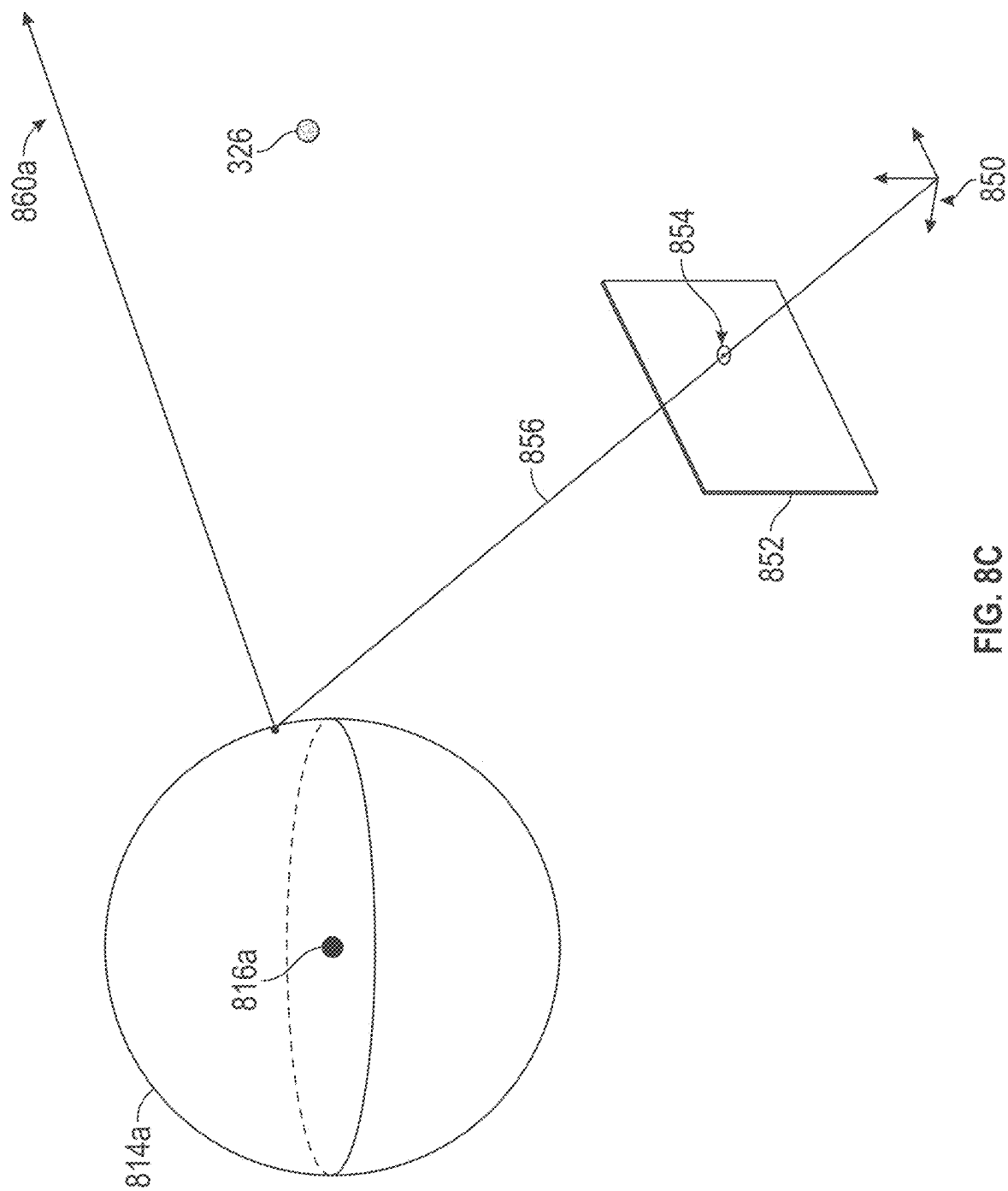


FIG. 8B



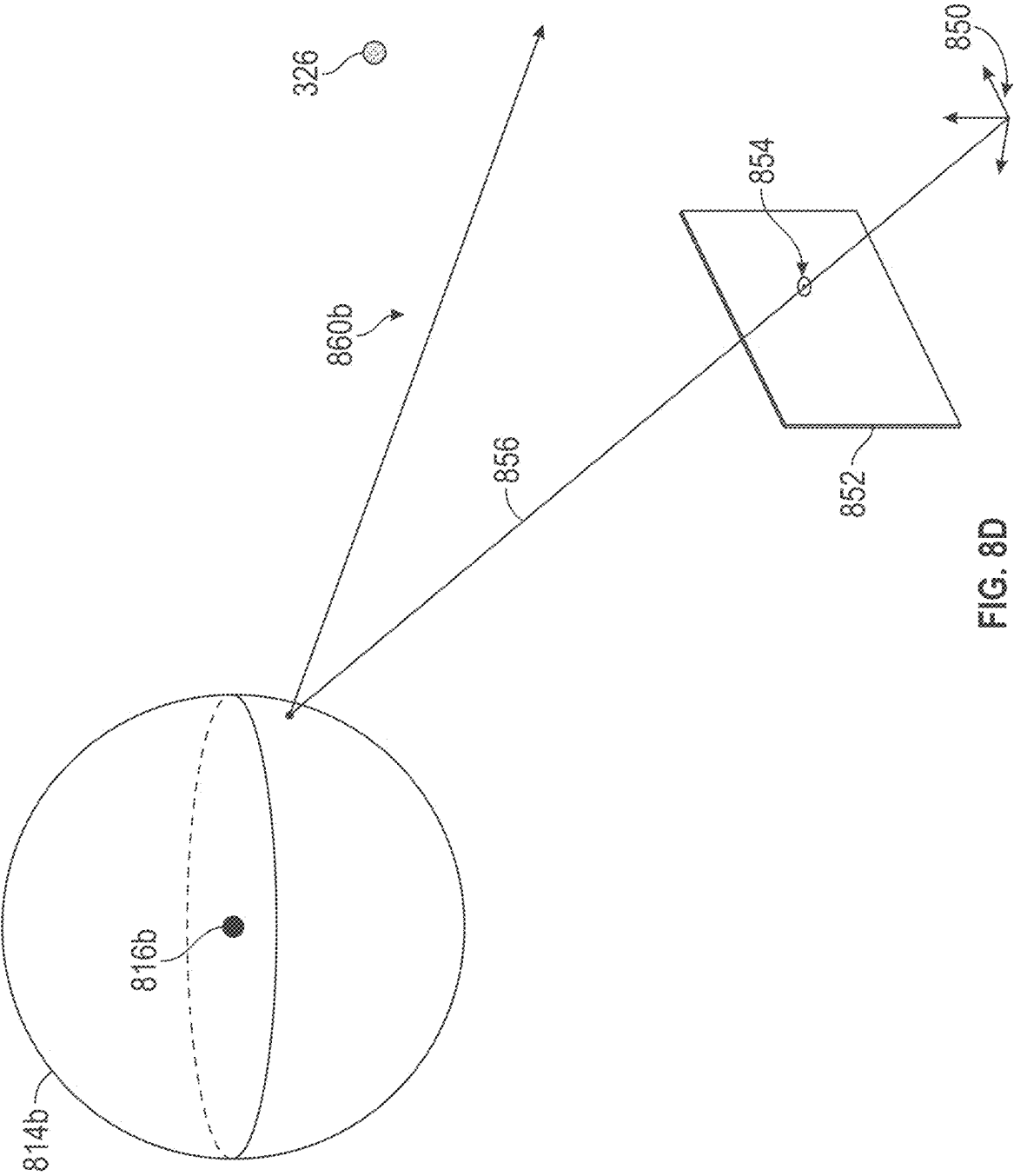
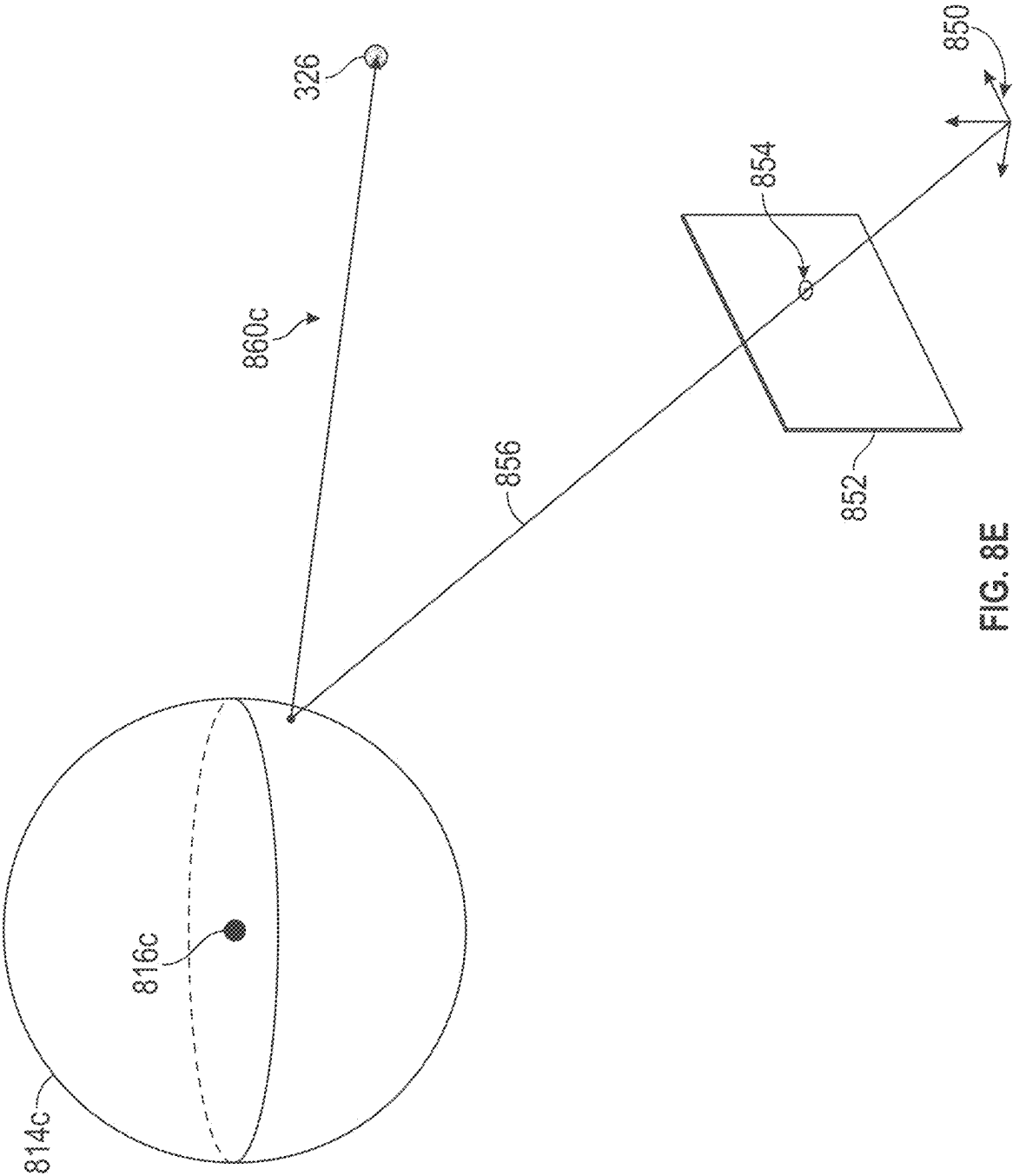


FIG. 8D



900

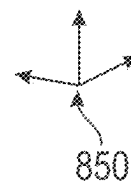
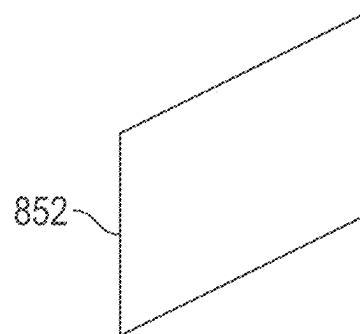


FIG. 9A

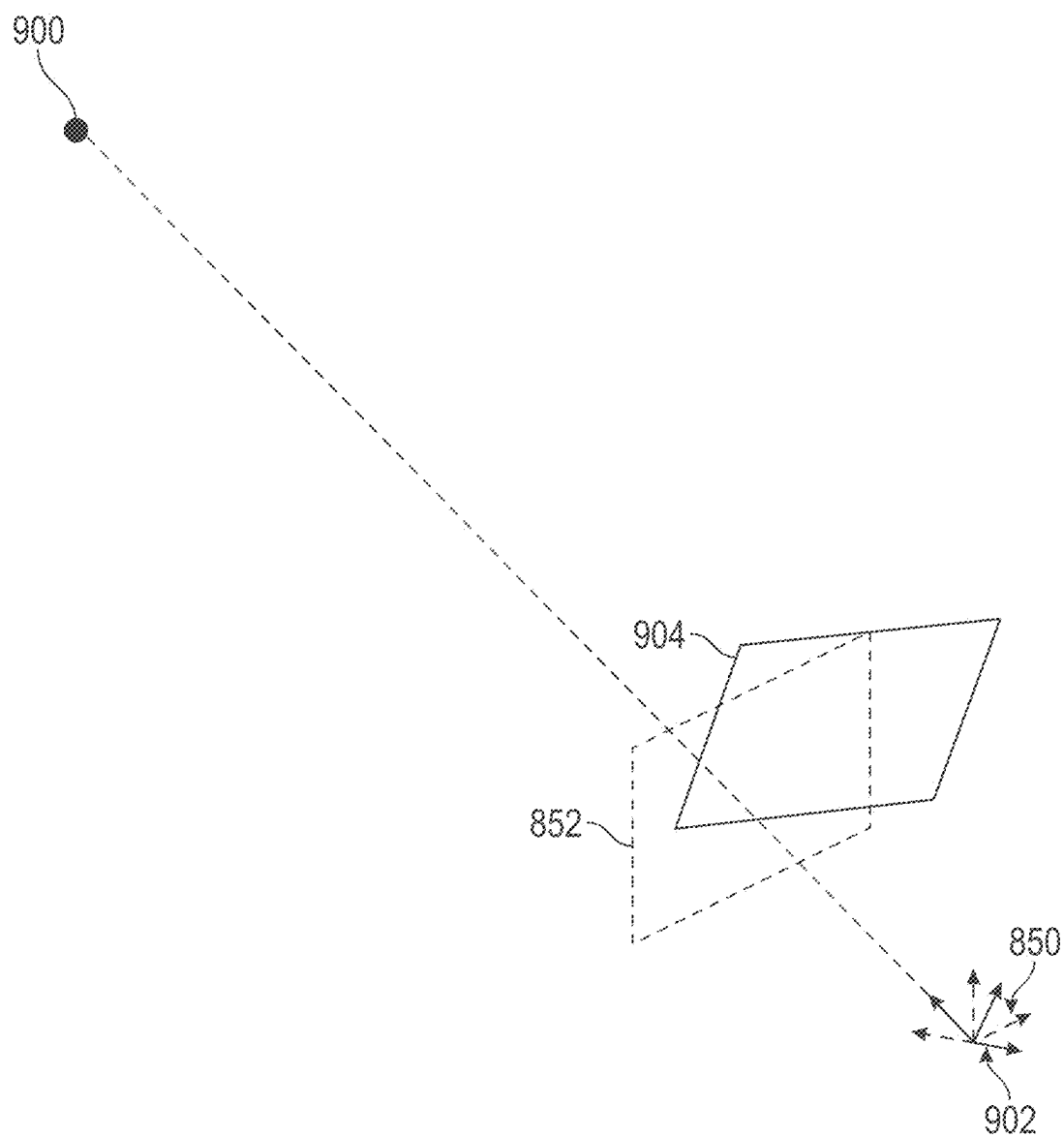


FIG. 9B

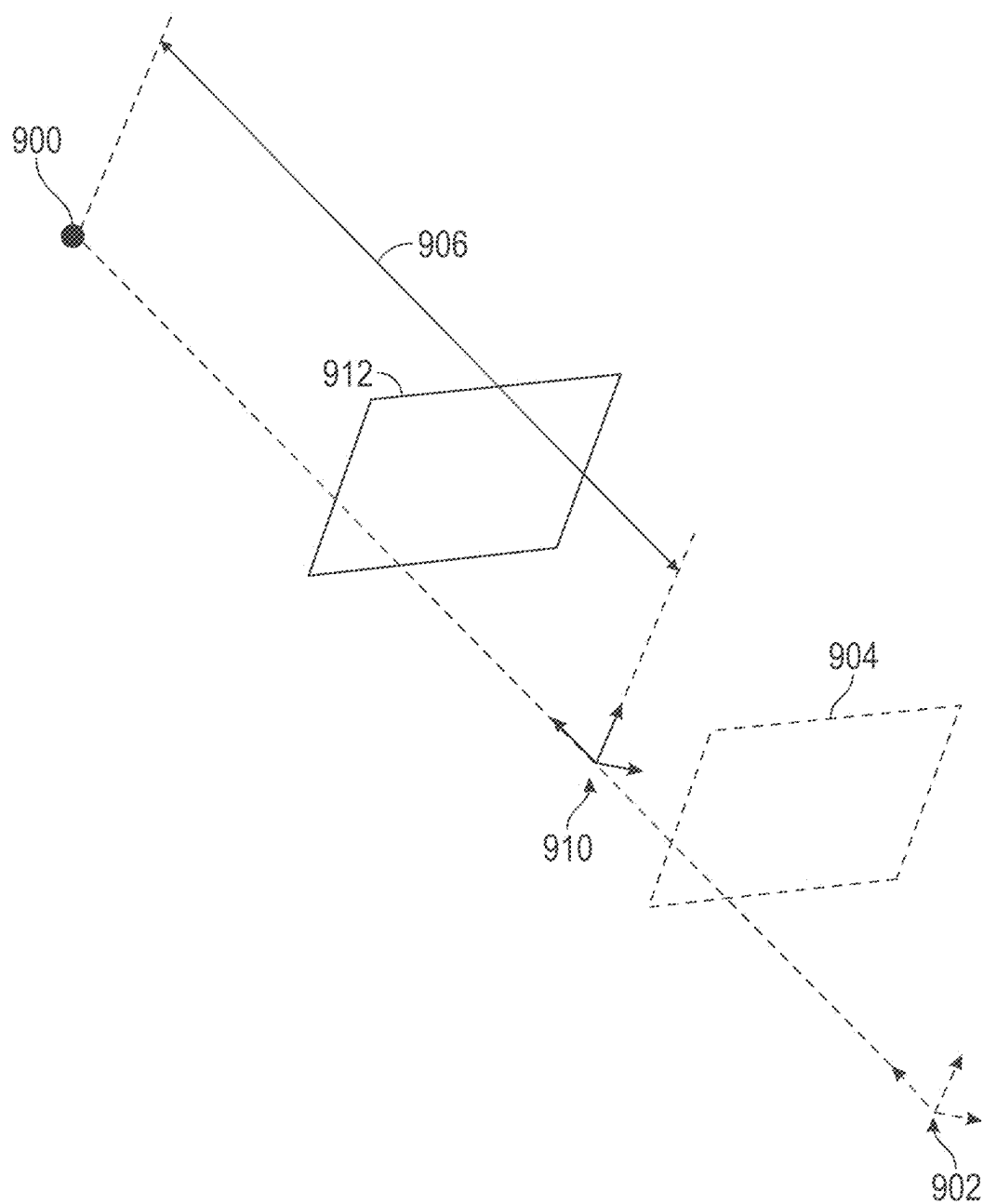


FIG. 9C

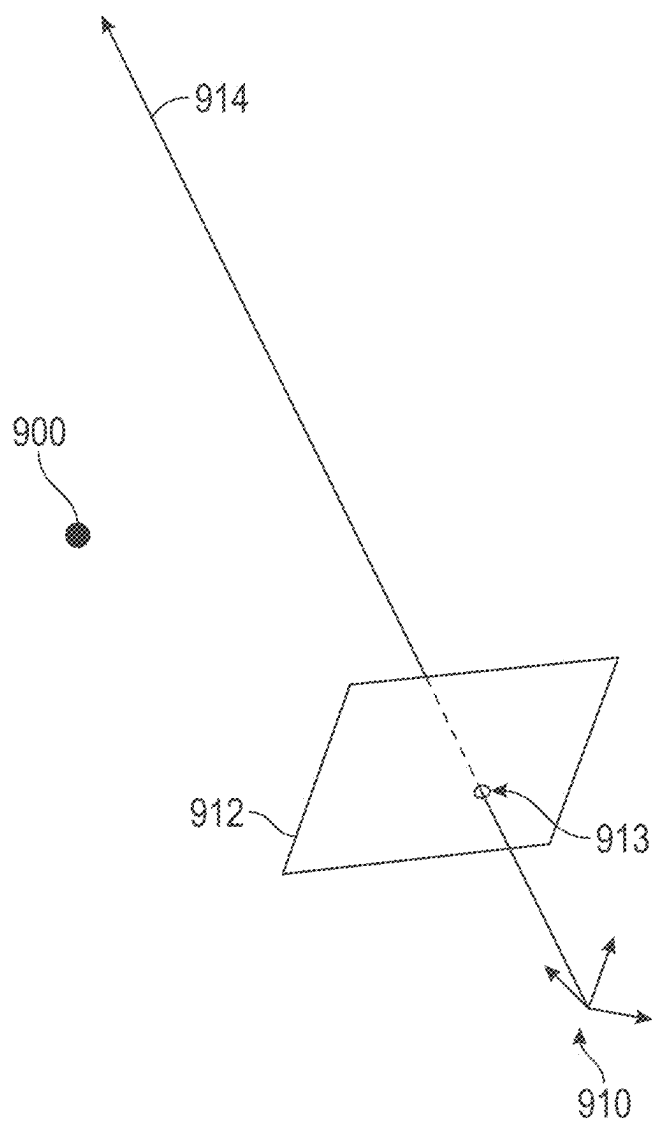


FIG. 9D

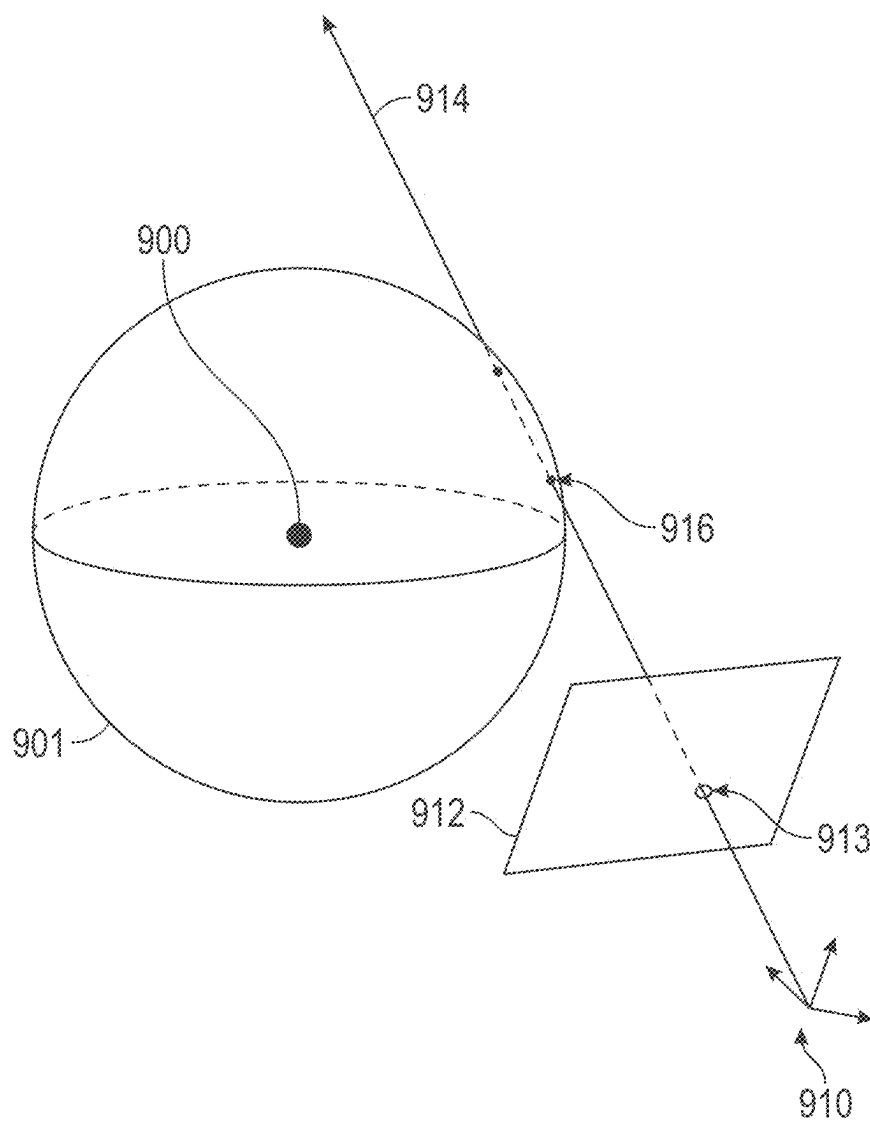


FIG. 9E

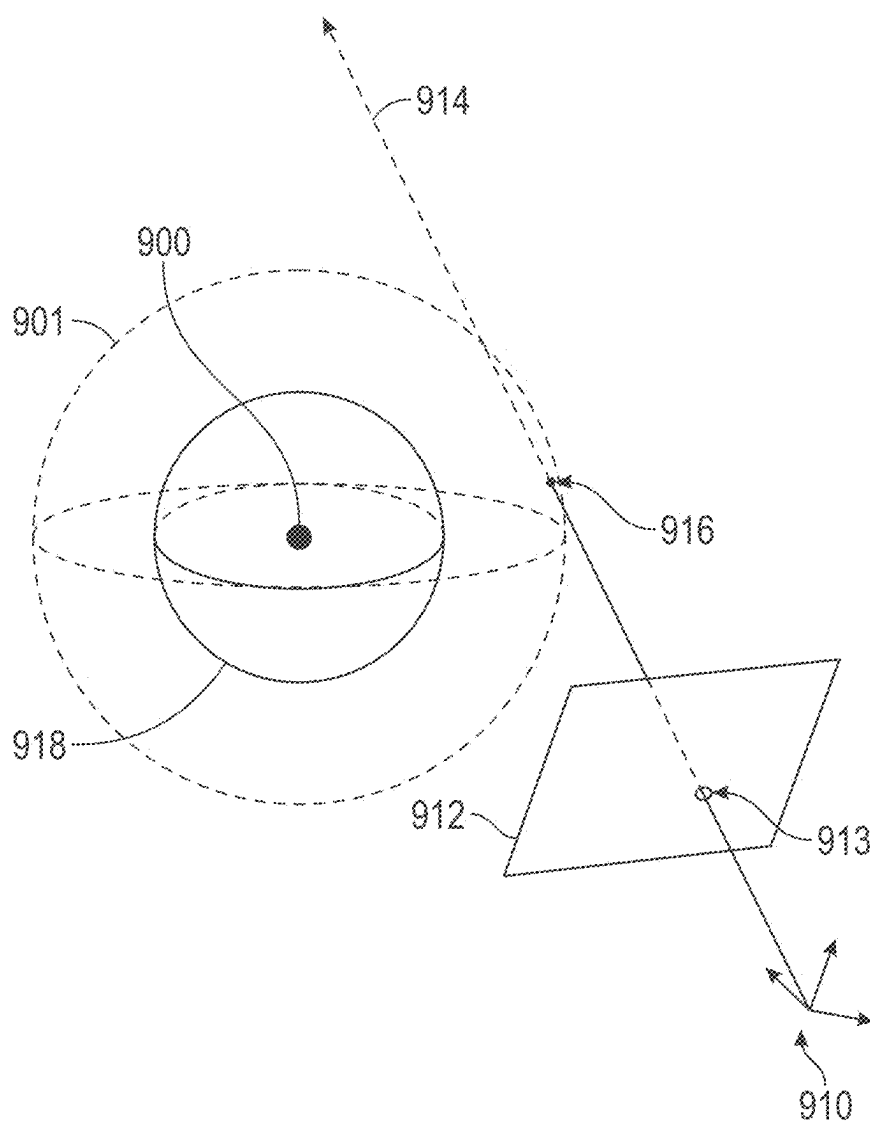


FIG. 9F

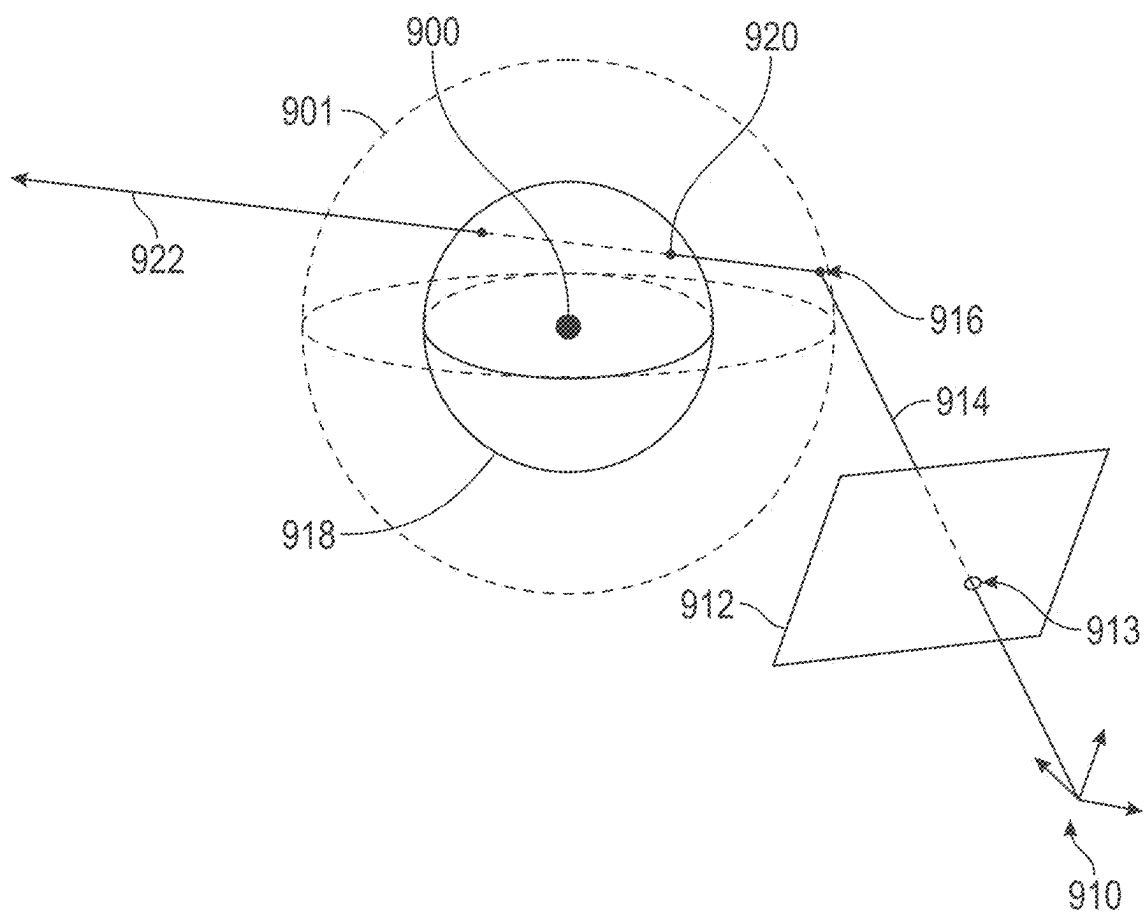
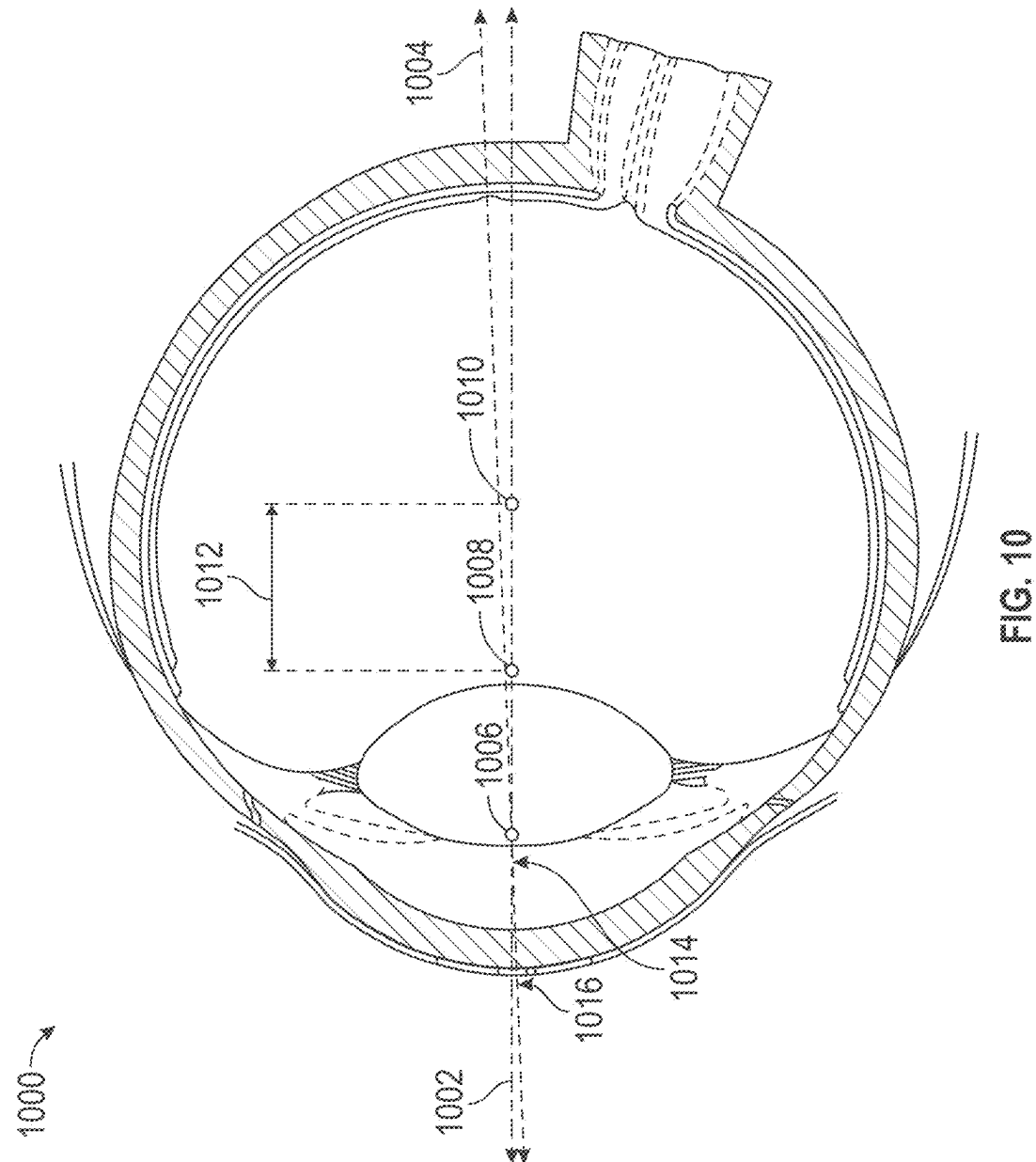


FIG. 9G



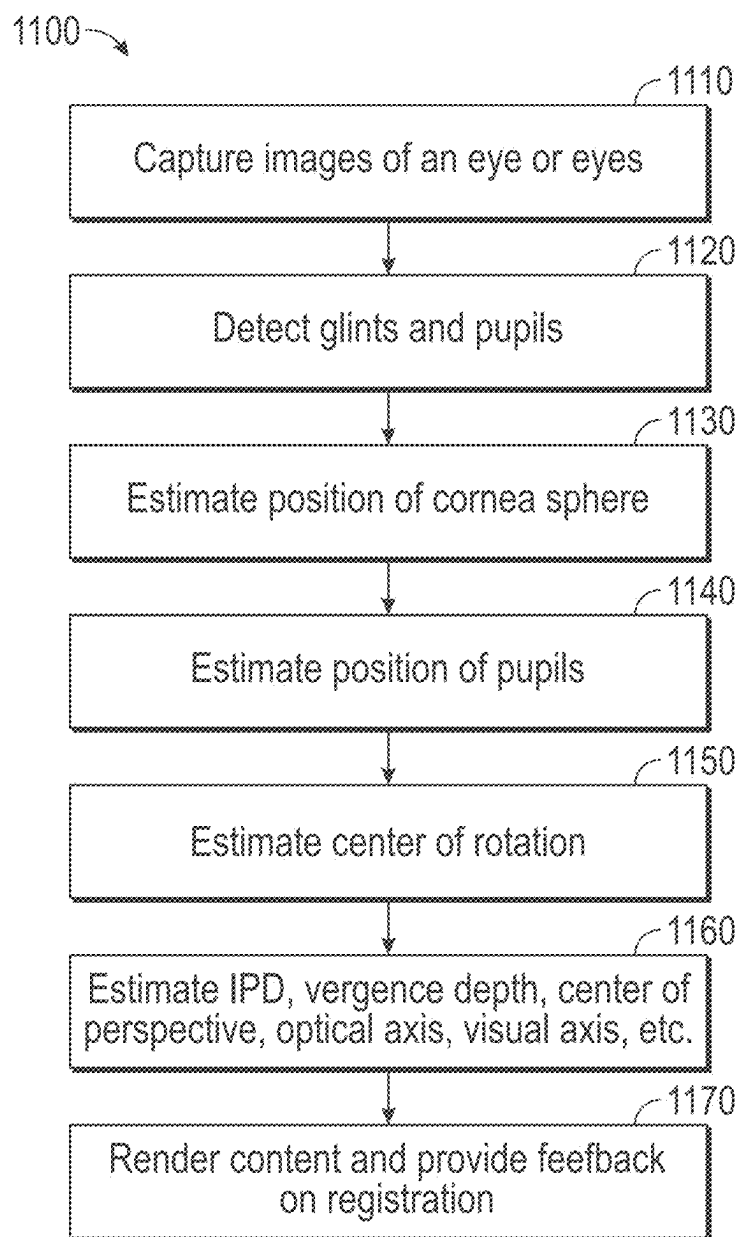


FIG. 11

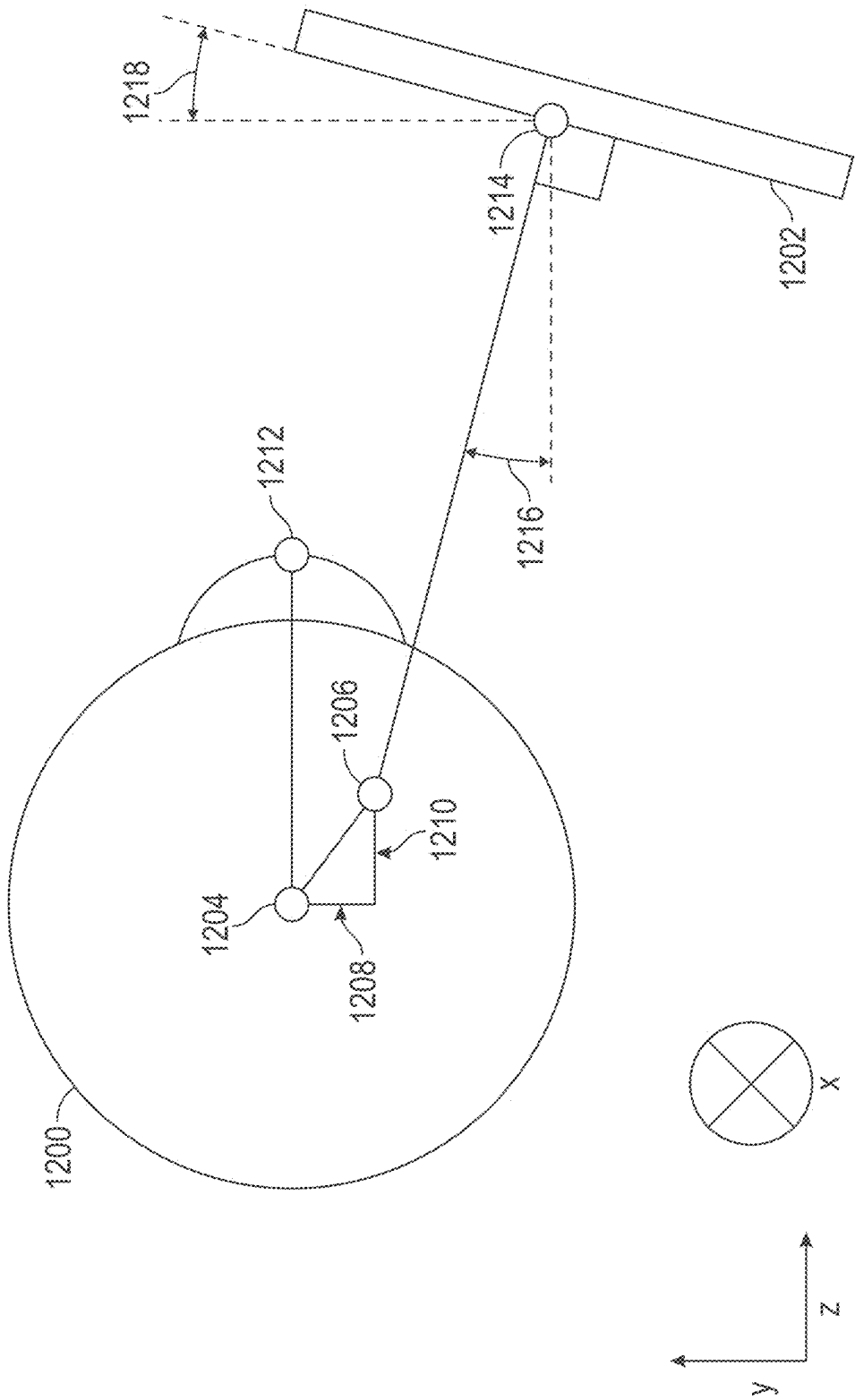


FIG. 12A

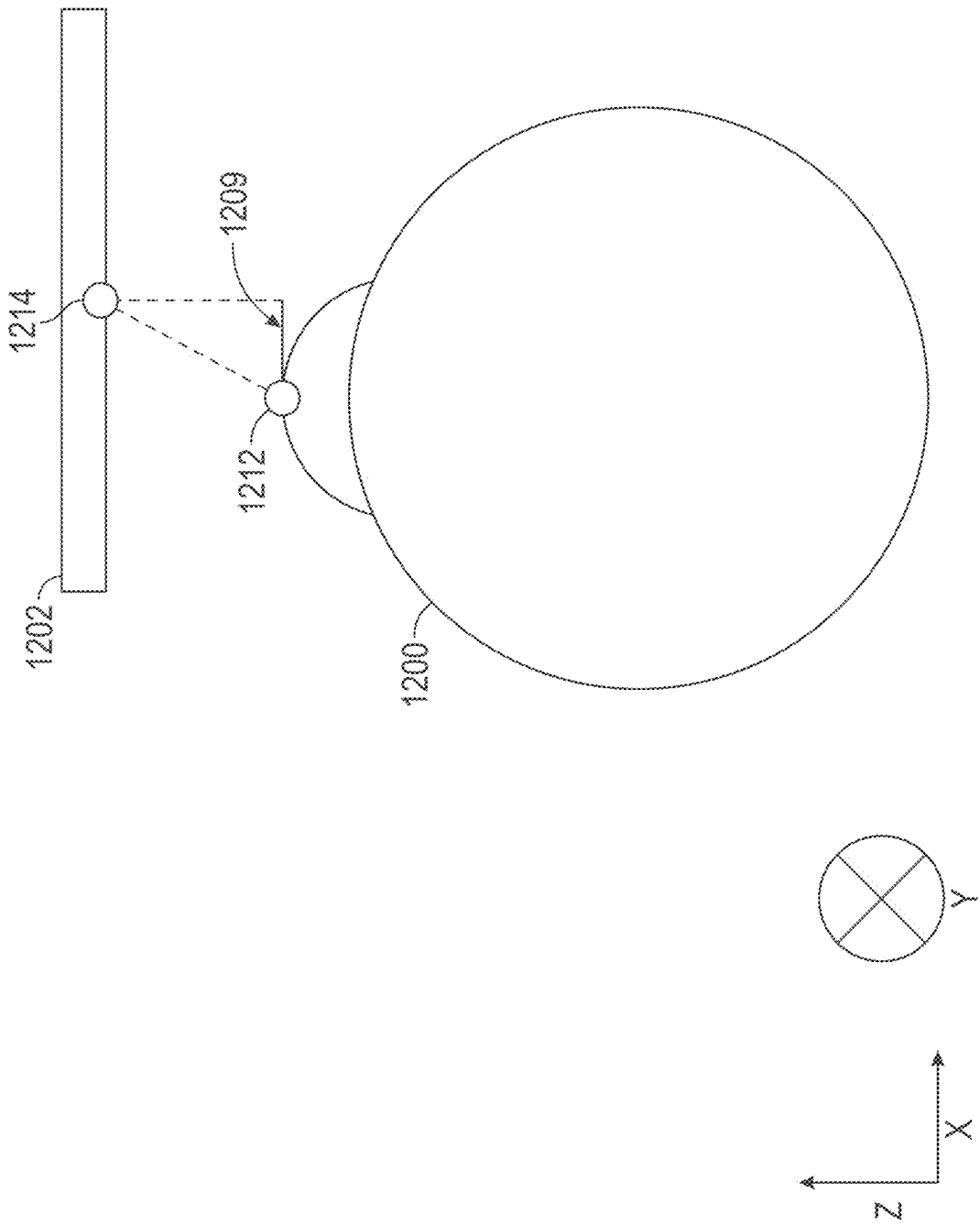


FIG. 12B

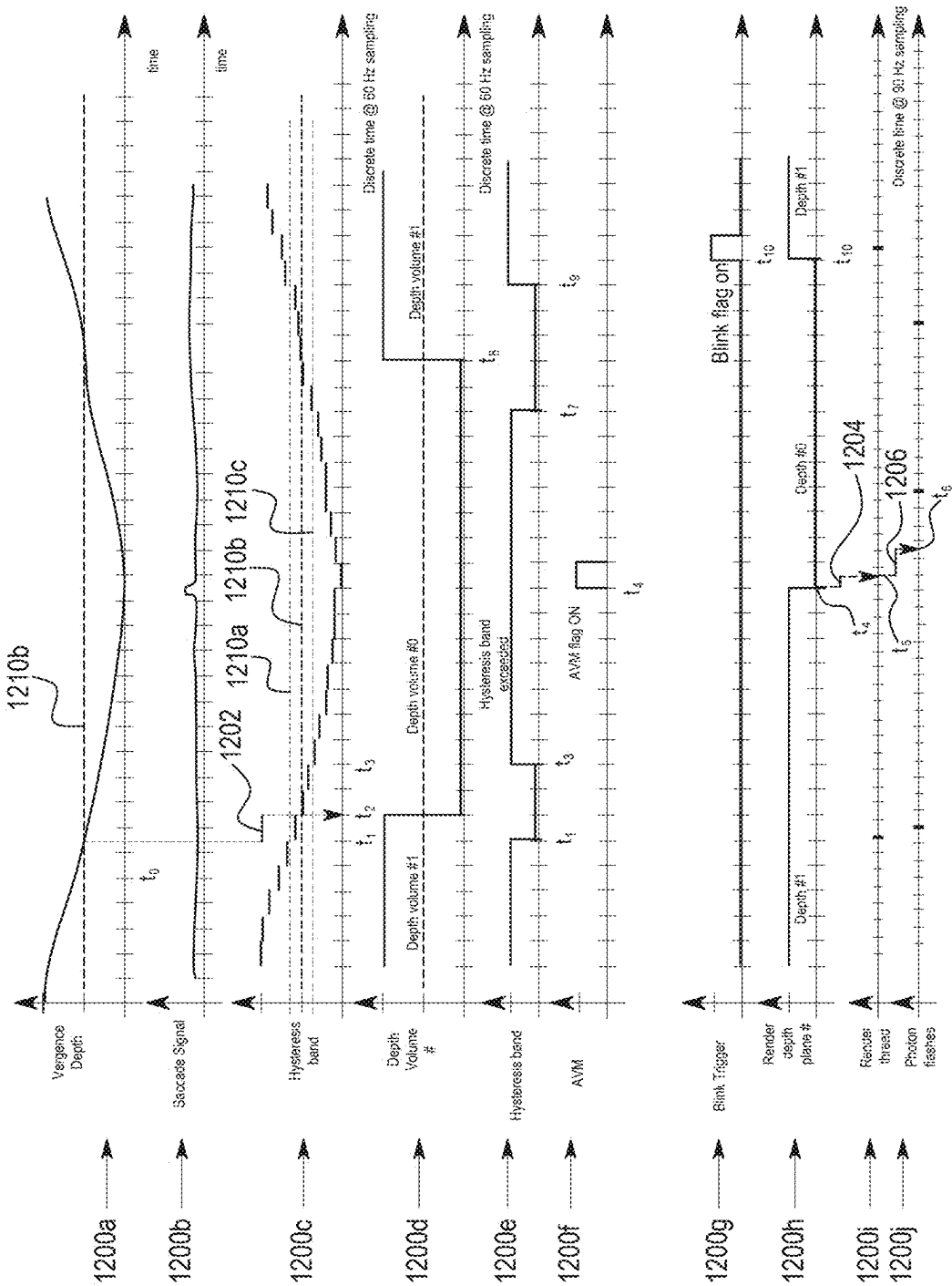


FIG. 13

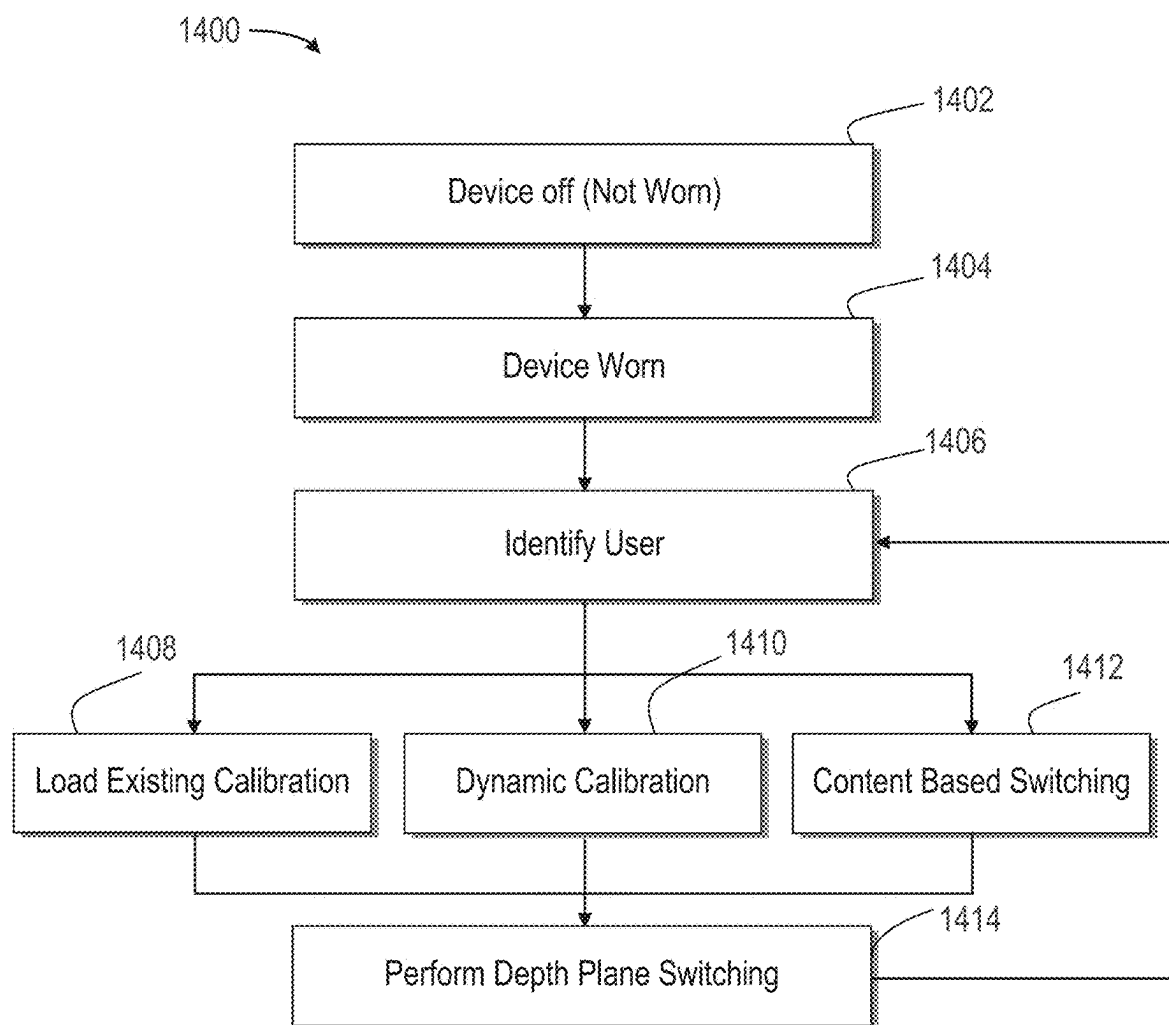


FIG. 14

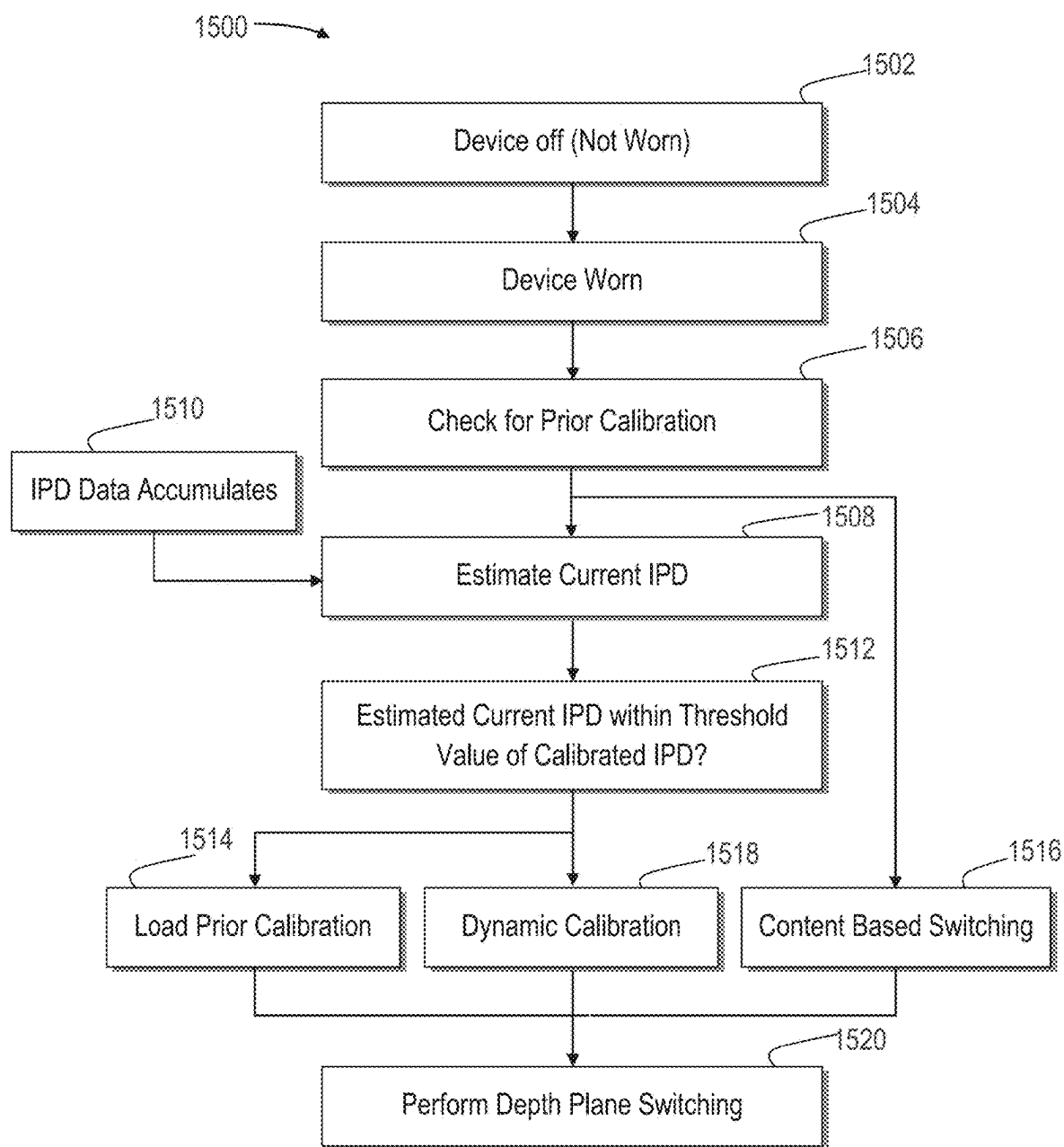


FIG. 15

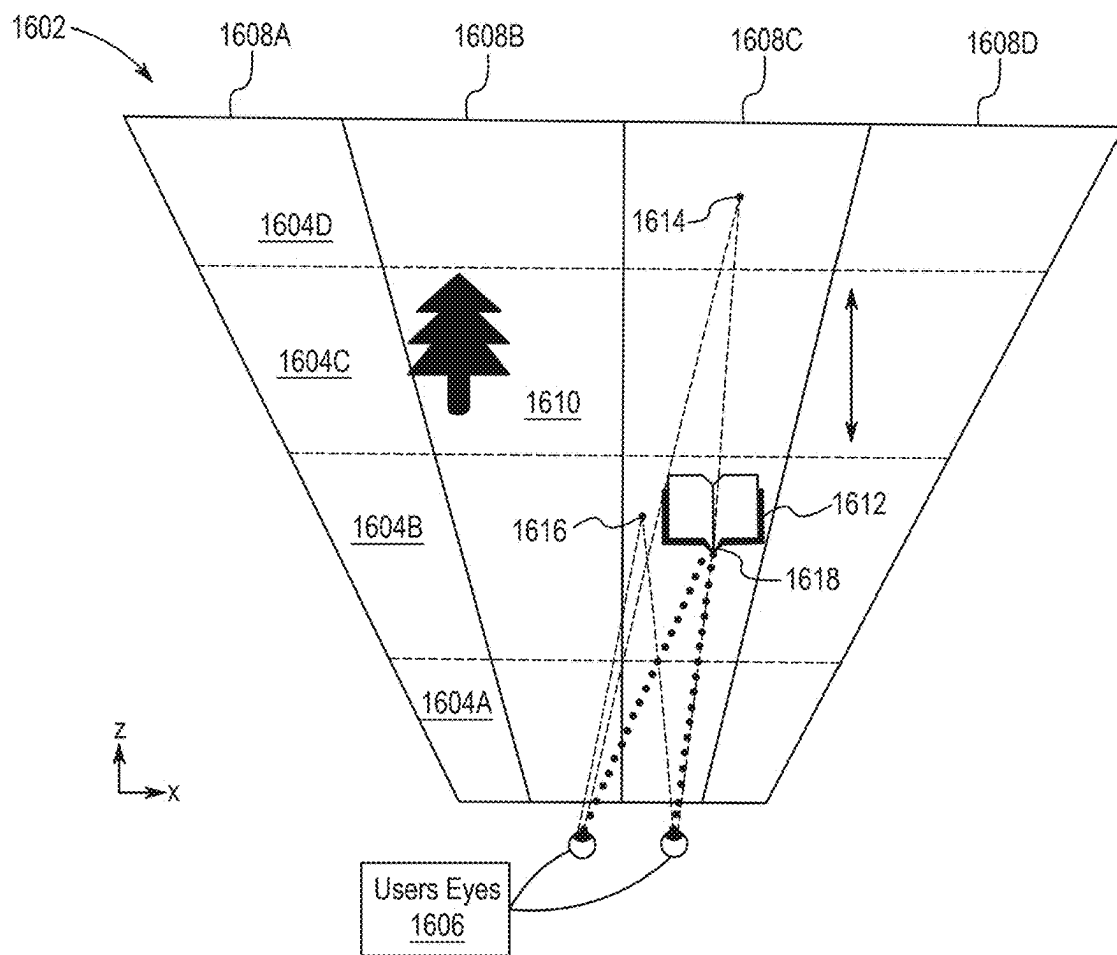


FIG. 16A

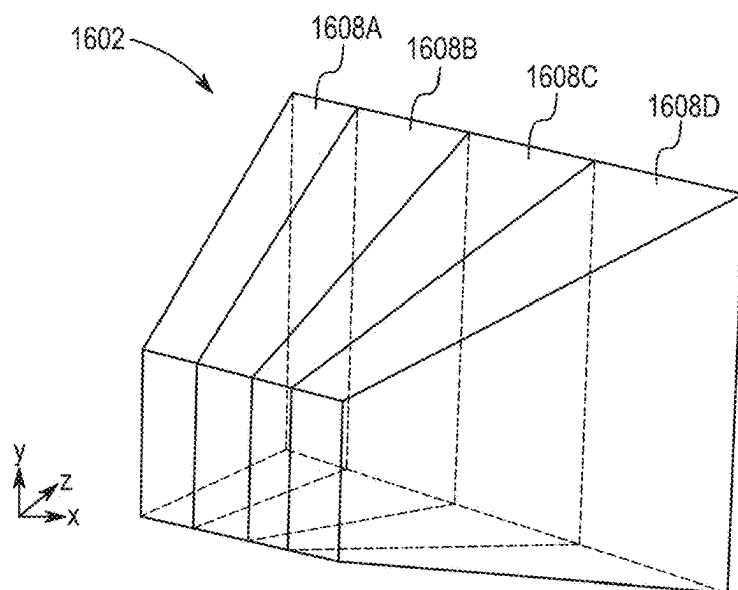


FIG. 16B

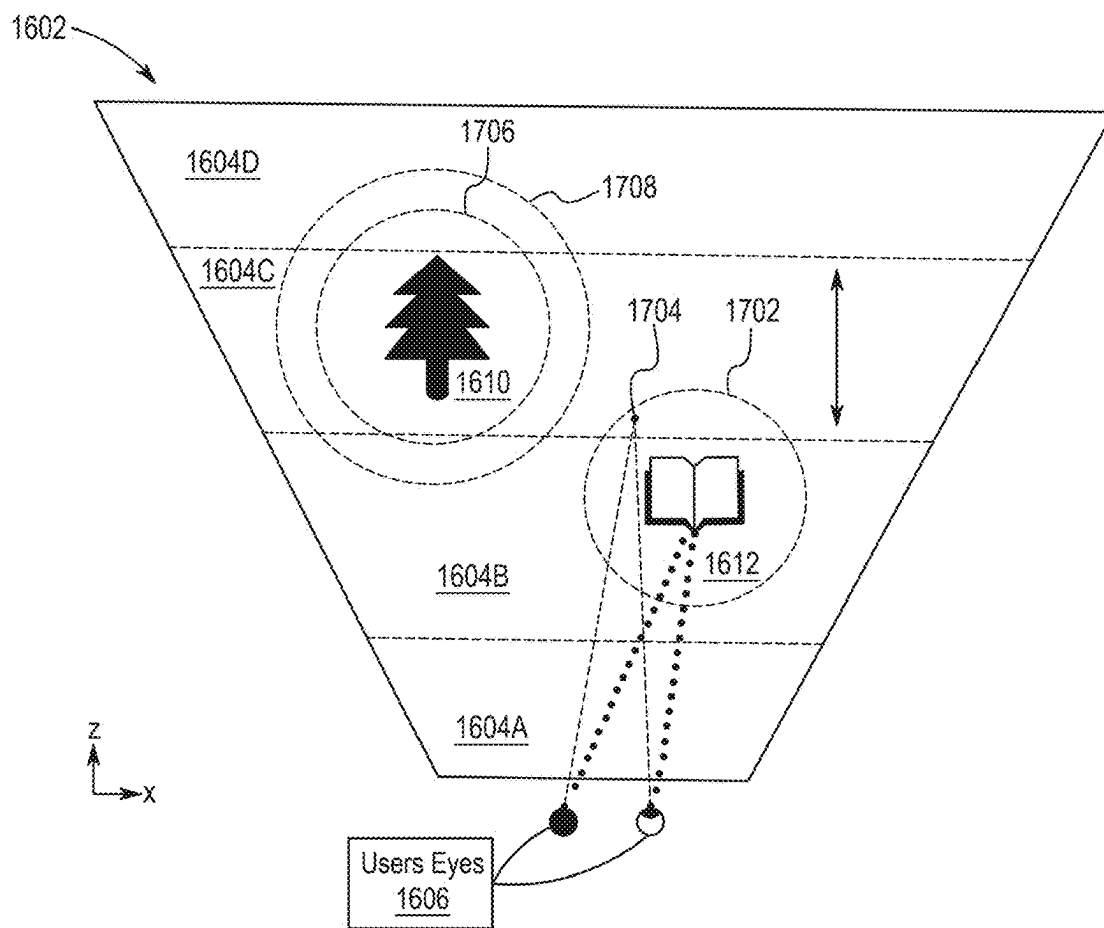


FIG. 17A

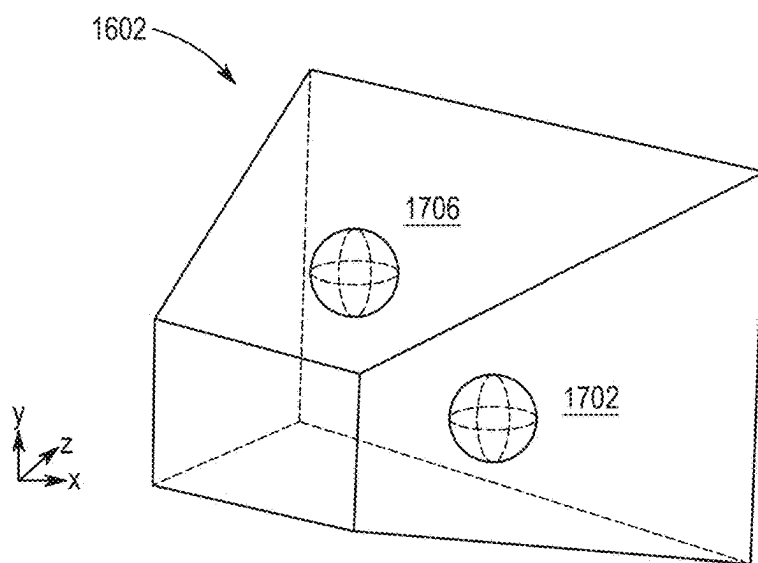


FIG. 17B

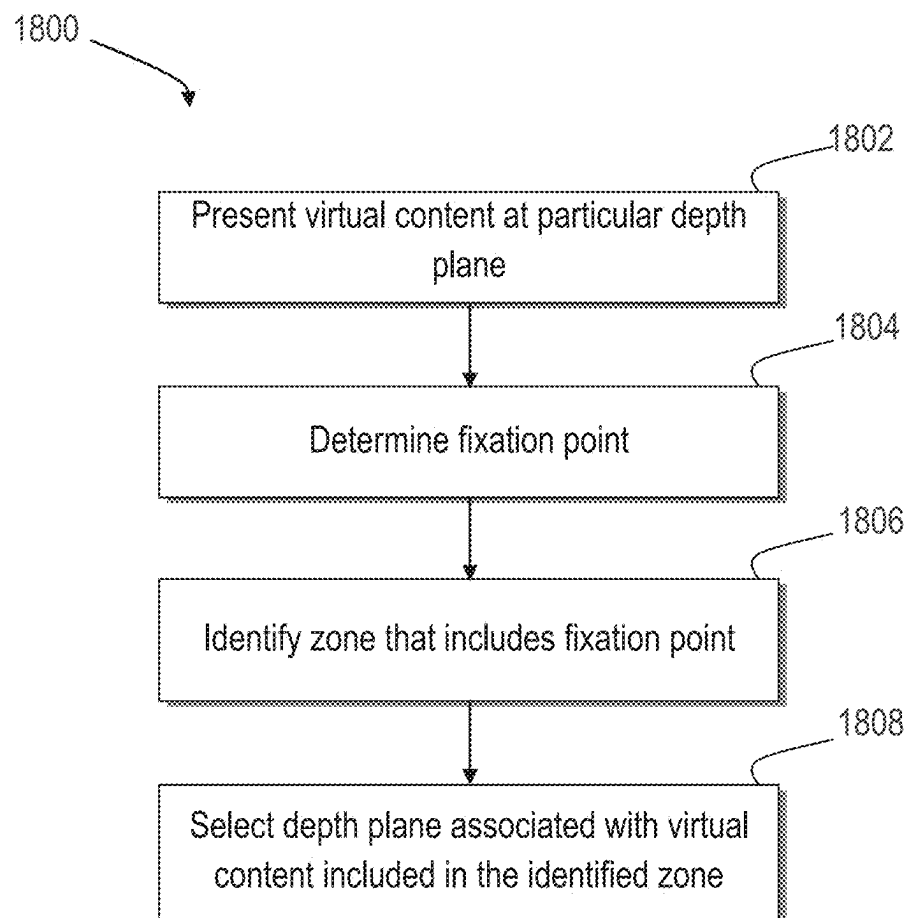


FIG. 18

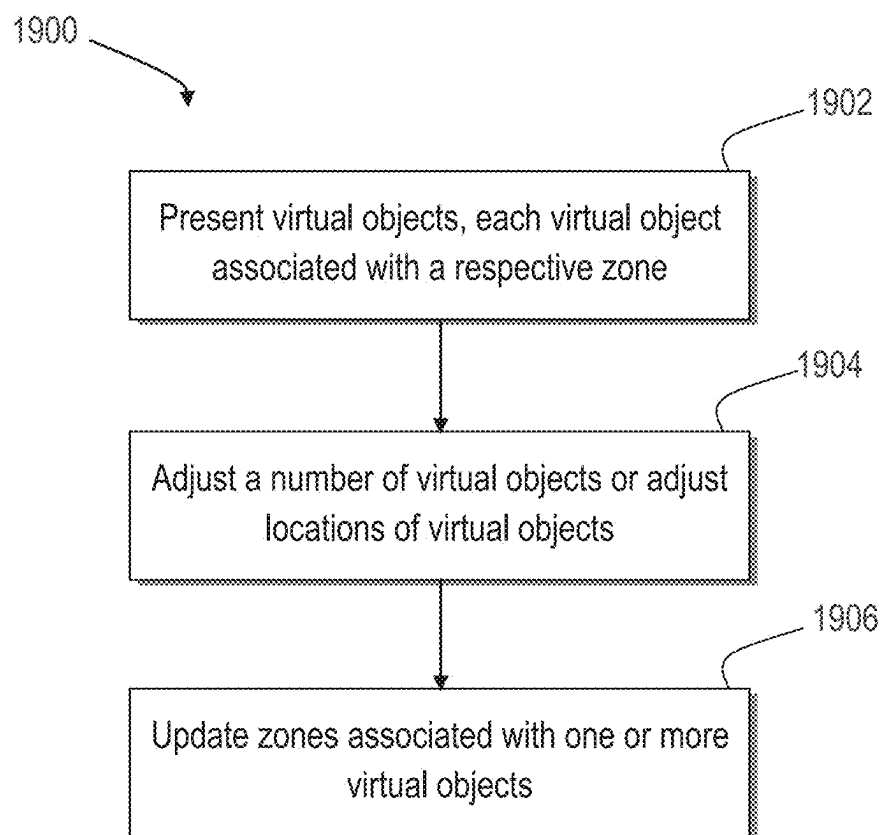


FIG. 19

DEPTH PLANE SELECTION FOR MULTI-DEPTH PLANE DISPLAY SYSTEMS BY USER CATEGORIZATION

PRIORITY CLAIM

[0001] This application is a continuation of U.S. application Ser. No. 18/658,715, filed May 8, 2024, titled “DEPTH PLANE SELECTION FOR MULTI-DEPTH PLANE DISPLAY SYSTEMS BY USER CATEGORIZATION,” which is a continuation of U.S. application Ser. No. 17/962,289, filed Oct. 7, 2022, titled “DEPTH PLANE SELECTION FOR MULTI-DEPTH PLANE DISPLAY SYSTEMS BY USER CATEGORIZATION,” which is a continuation of U.S. application Ser. No. 16/530,904, filed Aug. 2, 2019, titled “DEPTH PLANE SELECTION FOR MULTI-DEPTH PLANE DISPLAY SYSTEMS BY USER CATEGORIZATION,” which claims priority under 35 U.S.C. § 119(e) from U.S. Provisional Application No. 62/714,649, filed Aug. 3, 2018, titled “DEPTH PLANE SELECTION FOR MULTI-DEPTH PLANE DISPLAY SYSTEMS BY DETERMINATION OF INTERPUPILLARY DISTANCE,” and from U.S. Provisional Application No. 62/875,474, filed Jul. 17, 2019, titled “DEPTH PLANE SELECTION FOR MULTI-DEPTH PLANE DISPLAY SYSTEMS BY USER CATEGORIZATION.” The entire contents of each of the above-listed applications are hereby incorporated by reference into this application.

INCORPORATION BY REFERENCE

[0002] This application incorporates by reference the entirety of each of the following patent applications and publications: U.S. application Ser. No. 14/555,585 filed on Nov. 27, 2014, published on Jul. 23, 2015 as U.S. Publication No. 2015/0205126; U.S. application Ser. No. 14/690,401 filed on Apr. 18, 2015, published on Oct. 22, 2015 as U.S. Publication No. 2015/0302652; U.S. application Ser. No. 14/212,961 filed on Mar. 14, 2014, now U.S. Pat. No. 9,417,452 issued on Aug. 16, 2016; U.S. application Ser. No. 14/331,218 filed on Jul. 14, 2014, published on Oct. 29, 2015 as U.S. Publication No. 2015/0309263; U.S. application Ser. No. 15/927,808 filed on Mar. 21, 2018; U.S. application Ser. No. 15/291,929 filed on Oct. 12, 2016, published on Apr. 20, 2017 as U.S. Publication No. 2017/0109580; U.S. application Ser. No. 15/408,197 filed on Jan. 17, 2017, published on Jul. 20, 2017 as U.S. Publication No. 2017/0206412; U.S. application Ser. No. 15/469,369 filed on Mar. 24, 2017, published on Sep. 28, 2017 as U.S. Publication No. 2017/0276948; U.S. Provisional Application No. 62/618,559 filed on Jan. 17, 2018; U.S. application Ser. No. 16/250,931 filed on Jan. 17, 2019; U.S. application Ser. No. 14/705,741 filed on May 6, 2015, published on Apr. 21, 2016 as U.S. Publication No. 2016/0110920; and US patent publication No. 2017/0293145, published Oct. 12, 2017.

FIELD

[0003] The present disclosure relates to display systems, virtual reality, and augmented reality imaging and visualization systems and, more particularly, to depth plane selection based in part on a user’s interpupillary distance.

BACKGROUND

[0004] Modern computing and display technologies have facilitated the development of systems for so called “virtual

reality”, “augmented reality”, or “mixed reality” experiences, wherein digitally reproduced images or portions thereof are presented to a user in a manner wherein they seem to be, or may be perceived as, real. A virtual reality, or “VR”, scenario typically involves presentation of digital or virtual image information without transparency to other actual real-world visual input; an augmented reality, or “AR”, scenario typically involves presentation of digital or virtual image information as an augmentation to visualization of the actual world around the user; a mixed reality, or “MR”, related to merging real and virtual worlds to produce new environments where physical and virtual objects co-exist and interact in real time. As it turns out, the human visual perception system is very complex, and producing a VR, AR, or MR technology that facilitates a comfortable, natural-feeling, rich presentation of virtual image elements amongst other virtual or real-world imagery elements is challenging. Systems and methods disclosed herein address various challenges related to VR, AR and MR technology.

SUMMARY

[0005] Various examples of systems and methods for depth plane selection in display system such as augmented reality display systems, including mixed reality display systems, are disclosed.

[0006] In some embodiments, a display system can be configured to project light to an eye of a wearer to display virtual image content in a field of view of the wearer, who may also be referred as a user. The wearer’s eye may have a cornea, an iris, a pupil, a lens, a retina, and an optical axis extending through the lens, pupil, and cornea. The display system can include a frame configured to be supported on a head of the wearer; a head-mounted display disposed on the frame, the display configured to project light into the wearer’s eyes to display virtual image content to the wearer’s field of view at different amounts of at least one of wavefront divergence and thus the displayed virtual image content may appear to originate from different depths at different periods of time; one or more eye tracking cameras configured to image the wearer’s eyes; and processing electronics in communication with the display and the one or more eye tracking cameras. In some embodiments, the processing electronics may be configured to determine whether the wearer is a calibrated user or a guest user of the display system and, based on that categorization, select a scheme for switching the presentation of virtual content between two or more depth planes. For example, the processing electronics may be configured to obtain an estimate of a wearer’s interpupillary distance based on images of the eye obtained with the one or more eye tracking cameras, determine whether the wearer is a calibrated or guest user, and, for guest users, the display system may be configured to switch between two or more depth planes based in part of the wearer’s estimated interpupillary distance or based on depth plane information associated with particular virtual content.

[0007] In some embodiments, an augmented reality display system comprises a head-mounted display configured to present virtual content by outputting light to a wearer, where the head-mounted display is configured to output light to an eye of the wearer with different amounts of wavefront divergence corresponding to different perceived depths away from the wearer. The display system also comprises at least one processor communicatively coupled to the head-mounted display. The at least one processor is configured to

determine whether the wearer is a calibrated wearer or a guest user. When the wearer is determined to be a calibrated user the at least one processor is configured to load pre-existing user depth plane switching calibration information; and set depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information. When the wearer is determined to be a guest user, the at least one processor is configured to identify a virtual object at which the guest user is most likely to be looking, which has an associated depth plane, and to set depth plane switching parameters for the head-mounted display based upon the associated depth plane.

[0008] In some other embodiments, a method for determining parameters for depth plane switching in a display system is provided. The display system is configured to direct image light to eyes of a user to display virtual image content, and to present the virtual image content on a plurality of depth planes. Each depth plane is associated with image light having a different amount of wavefront divergence, and the display system is configured to switch the virtual image content between different depth planes by changing a wavefront divergence of the image light. The method comprises determining whether the user is a calibrated user or a guest user. When the user is determined to be a calibrated user, pre-existing user depth plane switching calibration information is loaded and depth plane switching parameters for the head-mounted display are set based upon the pre-existing user depth plane switching calibration information. When the user is determined to be a guest user, the virtual object that the guest user is looking at is determined. The virtual object has an associated depth plane; and depth plane switching parameters are set for the head-mounted display based upon the associated depth plane.

[0009] In yet other embodiments, an augmented reality display system comprises a head-mounted display configured to present virtual content by outputting light to a wearer. The head-mounted display comprises a waveguide stack configured to pass light from the world into an eye of the wearer. The waveguide stack comprises a plurality of waveguides and one or more waveguides of the plurality of waveguides are configured to output light to the eye of the wearer with a different amount of wavefront divergence than one or more other waveguides of the plurality of waveguides. Different amounts of wavefront divergence are associated with different accommodation by the eye, and the outputted light with different amounts of wavefront divergence forms virtual objects at different perceived depths away from the wearer. The display system further comprises an imaging device configured to capture images of eyes of the wearer and at least one processor communicatively coupled to the head-mounted display and the imaging device. The at least one processor is configured to determine whether the wearer is a calibrated user or a guest user based at least in part on images of the eyes of the wearer from the imaging device. When the wearer is determined to be a calibrated user, the at least one processor is configured to load pre-existing user depth plane switching calibration information; and set depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information. When the wearer is determined to be a guest user, the at least one processor is configured to determine the interpupillary distance of the guest user; and set depth plane switching

parameters for the head-mounted display based upon the determined interpupillary distance.

[0010] In some other embodiments, a method is provided for determining parameters for depth plane switching in a display system configured to direct image light to eyes of a user to display virtual image content. The eyes are separated by an interpupillary distance and the display system configured to present the virtual image content on a plurality of depth planes. Each depth plane is associated with image light having a different amount of wavefront divergence, and the display system is configured to switch the virtual image content between different depth planes by changing a wavefront divergence of the image light. The method comprises determining whether the user is a calibrated user or a guest user. When the user is determined to be a calibrated user, pre-existing user depth plane switching calibration information is loaded and depth plane switching parameters for the display system are set based upon the pre-existing user depth plane switching calibration information. When the user is determined to be a guest user, the interpupillary distance of the guest user is determined and depth plane switching parameters for the display system are set based upon the determined interpupillary distance.

[0011] Additional examples of embodiments are enumerated below.

[0012] Example 1. An augmented reality display system comprising: a head-mounted display configured to present virtual content by outputting light to a wearer, wherein the head-mounted display is configured to output light to an eye of the wearer with different amounts of wavefront divergence corresponding to different perceived depths away from the wearer; and at least one processor communicatively coupled to the head-mounted display, the at least one processor configured to: determine whether the wearer is a calibrated user or a guest user; when the wearer is determined to be a calibrated user: load pre-existing user depth plane switching calibration information; and set depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information; and when the wearer is determined to be a guest user: identify a virtual object at which the guest user is most likely to be looking, wherein the virtual object has an associated depth plane; and set depth plane switching parameters for the head-mounted display based upon the associated depth plane.

[0013] Example 2. The augmented reality display system of Example 1, wherein the display system is configured to determine whether the wearer is a calibrated user or a guest user by determining the wearer's interpupillary distance.

[0014] Example 3. The augmented reality display system of Example 1, wherein the display system is configured to determine whether the guest user is most likely looking at the virtual object by determining whether the guest user's eyes are fixating within a volume encompassing the virtual object.

[0015] Example 4. The augmented reality display system of Example 3, wherein the display system is configured to: determine an uncertainty associated with determining a position of a fixation point of the wearer; and vary a size of the volume encompassing the virtual object based upon the uncertainty.

[0016] Example 5. The augmented reality display system of Example 1, wherein the display system is configured to transition to dynamic calibration of the guest user if an

uncertainty associated with determining a position of a fixation point of the wearer exceeds a threshold value.

[0017] Example 6. The augmented reality display system of claim 1, wherein the display system is configured to transition to dynamic calibration of the guest user if an uncertainty associated with a location of the virtual object exceeds a threshold value.

[0018] Example 7. The augmented reality display system of Example 1, wherein, upon detecting that the calibrated user is no longer wearing the device after determining that the wearer is a calibrated user, the display system is configured to continue to utilize the calibrated user's depth plane switching calibration information for a predetermined amount of time or for a predetermined number of image frames.

[0019] Example 8. The augmented reality display system of Example 1, wherein the head-mounted display comprises a waveguide stack configured to pass light from the world into an eye of the wearer, wherein the waveguide stack comprises a plurality of waveguides comprising one or more waveguides configured to output light to the eye of the wearer with a different amount of wavefront divergence than one or more other waveguides of the plurality of waveguides.

[0020] Example 9. A method for determining parameters for depth plane switching in a display system configured to direct image light to eyes of a user to display virtual image content, the display system configured to present the virtual image content on a plurality of depth planes, wherein each depth plane is associated with image light having a different amount of wavefront divergence, wherein the display system is configured to switch the virtual image content between different depth planes by changing a wavefront divergence of the image light, the method comprising: determining whether the user is a calibrated user or a guest user; when the user is determined to be a calibrated user: loading pre-existing user depth plane switching calibration information; and setting depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information; and when the user is determined to be a guest user: determining whether the guest user is looking at a virtual object, wherein the virtual object has an associated depth plane; and setting depth plane switching parameters for the head-mounted display based upon the associated depth plane.

[0021] Example 10. The method of Example 9, wherein determining whether the guest user is looking at a virtual object comprises determining whether the guest user's eyes are fixating within a volume encompassing the virtual object.

[0022] Example 11. The method of Example 10, further comprising: determining an uncertainty associated with determining a position of a fixation point of the user; and varying a size of the volume encompassing the virtual object based upon the uncertainty.

[0023] Example 12. The method of Example 9, further comprising transitioning from virtual content-based depth plane switching to dynamic calibration of the guest user if an uncertainty associated with determining a position of a fixation point of the user exceeds a threshold value, wherein virtual content-based depth plane switching comprises setting depth plane switching parameters for the head-mounted display based upon the associated depth plane.

[0024] Example 13. The method of Example 9, further comprising, upon detecting that the calibrated user is no longer wearing a display of the display system after determining that the user is a calibrated user, continue utilizing the calibrated user's depth plane switching calibration information for a predetermined amount of time or for a predetermined number of image frames.

[0025] Example 14. The method of Example 9, wherein the display system comprises a waveguide stack configured to pass light from the world into an eye of the user, wherein the waveguide stack comprises a plurality of waveguides comprising one or more waveguides configured to output light to an eye of the user with a different amount of wavefront divergence than one or more other waveguides of the plurality of waveguides.

[0026] Example 15. An augmented reality display system comprising: a head-mounted display configured to present virtual content by outputting light to a wearer, wherein the head-mounted display comprises: a waveguide stack configured to pass light from the world into an eye of the wearer, wherein the waveguide stack comprises a plurality of waveguides, wherein one or more waveguides of the plurality of waveguides are configured to output light to the eye of the wearer with a different amount of wavefront divergence than one or more other waveguides of the plurality of waveguides, wherein different amounts of wavefront divergence are associated with different accommodation by the eye, and wherein the outputted light with different amounts of wavefront divergence forms virtual objects at different perceived depths away from the wearer; an imaging device configured to capture images of eyes of the wearer; and at least one processor communicatively coupled to the head-mounted display and the imaging device, the at least one processor configured to: determine whether the wearer is a calibrated user or a guest user based at least in part on images of the eyes of the wearer from the imaging device; when the wearer is determined to be a calibrated user: load pre-existing user depth plane switching calibration information; and set depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information; and when the wearer is determined to be a guest user: determine the interpupillary distance of the guest user; and set depth plane switching parameters for the head-mounted display based upon the determined interpupillary distance.

[0027] Example 16. The augmented reality display system of Example 15, wherein the imaging device comprises a left eye tracking system and a right eye tracking system together configured to measure the wearer's interpupillary distance.

[0028] Example 17. The augmented reality display system of Example 15, wherein the processor is configured to determine whether the wearer is a calibrated user or a guest user based on the wearer's interpupillary distance.

[0029] Example 18. The augmented reality display system of Example 15, wherein the processor is configured to determine whether the wearer is a calibrated user or a guest user based on whether the wearer's interpupillary distance is within a predetermined threshold of the interpupillary distance of the calibrated user.

[0030] Example 19. The augmented reality display system of Example 15, wherein the processor is configured to determine whether the wearer is a calibrated user or a guest

user based on whether the wearer's interpupillary distance is within 1.0 mm of the interpupillary distance of the calibrated user.

[0031] Example 20. A method for determining parameters for depth plane switching in a display system configured to direct image light to eyes of a user to display virtual image content, the eyes separated by an interpupillary distance and the display system configured to present the virtual image content on a plurality of depth planes, wherein each depth plane is associated with image light having a different amount of wavefront divergence, wherein the display system is configured to switch the virtual image content between different depth planes by changing a wavefront divergence of the image light, the method comprising: determining whether the user is a calibrated user or a guest user; when the user is determined to be a calibrated user: loading pre-existing user depth plane switching calibration information; and setting depth plane switching parameters for the display system based upon the pre-existing user depth plane switching calibration information; and when the user is determined to be a guest user: determining the interpupillary distance of the guest user; and setting depth plane switching parameters for the display system based upon the determined interpupillary distance.

[0032] Example 21. The method of Example 20, wherein determining the interpupillary distance of the guest user comprises determining an interpupillary distance of eyes of the guest user focused at optical infinity.

[0033] Example 22. The method of Example 20, wherein determining whether the user is the calibrated user or is a guest user comprises determining the interpupillary distance of the user with one or more eye tracking cameras configured to image the eyes of the user.

[0034] Example 23. The method of Example 20, wherein determining whether the user is the calibrated user comprises determining the interpupillary distance of the user with one or more eye tracking cameras configured to image the eyes of the user and determining that the interpupillary distance of the user is within a predetermined range.

[0035] Example 24. The method of Example 23, wherein determining whether the user is the guest user comprises determining the interpupillary distance of the user with the one or more eye tracking cameras and determining that the interpupillary distance of the user is outside of the predetermined range.

[0036] Example 25. The method of Example 20, wherein the pre-existing user depth plane switching calibration information comprises a measured interpupillary distance of the calibrated user, wherein determining whether the user is the calibrated user or is a guest user comprises determining the interpupillary distance of the user, determining that the user is the calibrated user when the user's interpupillary distance is within 1.0 mm of the measured interpupillary distance of the calibrated user, and determining that the user is a guest user when the user's interpupillary distance is not within 1.0 mm of the measured interpupillary distance of the calibrated user.

[0037] Example 26. The method of Example 20, wherein determining whether the user is the calibrated user or is a guest user comprises identifying the user with at least one eye tracking camera.

[0038] Example 27. The method of Example 20, further comprising: with an eye tracking system, determining an optical axis for each of the user's eyes; and determining a

vergence depth of the user based at least on the determined optical axes for the user's eyes and the set depth plane switching parameters.

[0039] Example 28. The method of Example 20, further comprising: determining a vergence distance of the user based at least in part on the set depth plane switching parameters.

[0040] Example 29. The method of Example 20, further comprising: selecting which depth plane to use to present the virtual image content based at least in part on the depth plane switching parameters.

[0041] Details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Neither this summary nor the following detailed description purports to define or limit the scope of the inventive subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0042] FIG. 1 depicts an illustration of a mixed reality scenario with certain virtual reality objects, and certain physical objects viewed by a person.

[0043] FIG. 2 schematically illustrates an example of a wearable system.

[0044] FIG. 3 schematically illustrates example components of a wearable system.

[0045] FIG. 4 schematically illustrates an example of a waveguide stack of a wearable device for outputting image information to a user.

[0046] FIG. 5 schematically illustrates an example of an eye and an example coordinate system for determining an eye pose of an eye.

[0047] FIG. 6 is a schematic diagram of a wearable system that includes an eye tracking system.

[0048] FIG. 7A is a block diagram of a wearable system that may include an eye tracking system.

[0049] FIG. 7B is a block diagram of a render controller in a wearable system.

[0050] FIG. 7C is a block diagram of a registration observer in a head-mounted display system.

[0051] FIG. 8A is a schematic diagram of an eye showing the eye's corneal sphere.

[0052] FIG. 8B illustrates an example corneal glint detected by an eye tracking camera.

[0053] FIGS. 8C-8E illustrate example stages of locating a user's corneal center with an eye tracking module in a wearable system.

[0054] FIGS. 9A-9C illustrate an example normalization of the coordinate system of eye tracking images.

[0055] FIGS. 9D-9G illustrate example stages of locating a user's pupil center with an eye tracking module in a wearable system.

[0056] FIG. 10 illustrates an example of an eye including the eye's optical and visual axes and the eye's center of rotation.

[0057] FIG. 11 is a process flow diagram of an example of a method for using eye tracking in rendering content and providing feedback on registration in a wearable device.

[0058] FIGS. 12A and 12B illustrate a nominal position of a display element relative to a user's eye and illustrate a coordinate system for describing the positions of the display element and the user's eye relative to one another.

[0059] FIG. 13 is a set of example graphs illustrating how the wearable system may switch depth planes in response to the user's eye movements.

[0060] FIG. 14 is a process flow diagram of an example of a method for depth plane selection using an existing calibration, a dynamic calibration, and/or a content-based switching scheme.

[0061] FIG. 15 is a process flow diagram of an example of a method for depth plane selection based at least partly on a user's interpupillary distance.

[0062] FIG. 16A illustrates an example of a top-down view of a representation of a user viewing content presented by a display system configured to switch depth planes by detecting user fixation within one of a plurality of zones that segment the user's field of view along the horizontal axis.

[0063] FIG. 16B illustrates an example of a perspective view of the representation of FIG. 16A.

[0064] FIG. 17A illustrates an example of a top-down view of a representation of a user viewing content presented by a display system configured to switch depth planes by detecting user fixation within discrete marker volumes within the display frustum.

[0065] FIG. 17B illustrates an example of a perspective view of the representation of FIG. 16A.

[0066] FIG. 18 illustrates a flowchart of an example process to select a depth plane based on content-based switching.

[0067] FIG. 19 illustrates a flowchart of another example process to adjust zones based on content-based switching.

[0068] Throughout the drawings, reference numbers may be re-used to indicate correspondence between referenced elements. The drawings are provided to illustrate example embodiments described herein and are not intended to limit the scope of the disclosure.

DETAILED DESCRIPTION

[0069] As described herein, display systems (e.g., augmented reality or virtual reality display systems) may render virtual content for presentation to a user at different perceived depths from the user. In augmented reality display systems, different depth planes may be utilized to project virtual content with each depth plane being associated with a particular perceived depth from the user. For example, a stack of waveguides configured to output light with different wavefront divergences may be utilized, with each depth plane having a corresponding wavefront divergence and being associated with at least one waveguide. As virtual content moves about the user's field of view, the virtual content may be adjusted along three discrete axes. For example, the virtual content may be adjusted along the X, Y, and Z axes such that the virtual content may be presented at different perceived depths from the user. The display system may switch between depth planes as the virtual content is perceived to be moved further from, or closer to, the user. It will be appreciated that switching depth planes may involve changing the wavefront divergence of light forming the virtual content in a discrete step. In a waveguide-based system, in some embodiments, such a depth plane switch may involve switching the waveguide outputting light to form the virtual content.

[0070] In some embodiments, the display system may be configured to monitor the gaze of a user's eyes and determine a three-dimensional fixation point at which the user is fixating. The fixation point may be determined based upon,

among other things, the distance between the user's eyes and the gaze direction of each eye. It will be appreciated that these variables may be understood to form a triangle with the fixation point at one corner of the triangle and the eyes at the other corners. It will also be appreciated that a calibration may be performed to accurately track the orientations of the user's eyes and determine or estimate the gaze of those eyes to determine the fixation point. Thus, after undergoing a full calibration, the display device may have a calibration file or calibration information for the main user of that device. The main user may also be referred to as a calibrated user herein. Further details regarding calibration and eye tracking may be found in, e.g., U.S. patent application Ser. No. 15/993,371, entitled "EYE TRACKING CALIBRATION TECHNIQUES," which is incorporated herein by reference in its entirety.

[0071] The display system may occasionally be used by guest users that have not completed the full calibration. In addition, these guest users may not have the time or the desire to perform a full calibration. However, if the display system does not actually track the fixation point of the guest user, then the depth plane switching may not be appropriate for providing a realistic and comfortable viewing experience for the user.

[0072] Thus, it will be appreciated that the current user of the display system may be categorized as a calibrated user or as a guest user. In some embodiments, the display system may be configured to categorize the current user by determining whether or not the current user is a calibrated user by performing an identification, or authentication, process, e.g., to determine whether information provided by and/or obtained from the current user matches information associated with the calibrated user. For example, the authentication or identification process may be one or more of: asking for and verifying a username and/or password, conducting iris scanning (e.g., by comparing a current image of the user's iris with a reference image), conducting voice recognition (e.g., by comparing a current sample of the user's voice with a reference voice file), and IPD matching. In some embodiments, IPD matching may include determining whether the current user's IPD matches the calibrated user's IPD. If there is a match, the current user may be assumed to be the calibrated user in some embodiments. If there is no match, the current user may be assumed to not be the calibrated user (e.g., to be a guest user) in some embodiments. In some other embodiments, multiple authentication processes may be conducted for a current user to increase the accuracy of the determination of whether the current user is a calibrated user.

[0073] In some embodiments, if the current user is determined to not be a calibrated user (e.g., a guest user), the display system may use content-based depth plane switching; for example, rather than determining the fixation point of the user's eyes and switch depth planes based on the depth plane in which the fixation point is location, the display system may be configured to display content with the appropriate amount of wavefront divergence for where, in 3D space, that content (e.g., a virtual object) is specified to be placed. It will be appreciated that virtual objects may have associated locations or coordinates in the three-dimensional volume around the user, and the display system may be configured to present the object using light with the

amount of wavefront divergence appropriate for the object's depth, relative to the user, within that three-dimensional volume.

[0074] In cases where multiple virtual objects at different depths are to be displayed, content-based depth plane switching may involve making a rough determination as to which virtual object is being fixated on and then using the location of that virtual object to establish the plane to which depth plane switching should switch; for example, upon determining that the user is roughly fixating on a particular virtual object, the display system may be configured to output light with wavefront divergence corresponding to the depth plane associated with that virtual object. In some embodiments, this rough determination of whether the user is fixating on an object may involve determining whether the fixation point is within a display system-defined volume unique to the object and, if yes, switching to the depth plane associated with that object, irrespective of the depth of the determined fixation point (so that, if the volume extends across multiple depth planes, the display system will switch to the depth plane associated with the object).

[0075] In some embodiments, if the current user is determined to not be a calibrated user (e.g., a guest user) the display system may perform a rough calibration by measuring the guest user's interpupillary distance (IPD). This rough calibration may also be referred to as a dynamic calibration. Preferably, the IPD is measured while the user's eyes are directed to, or focused on an object at, optical infinity. In some embodiments, this IPD value may be understood to be the maximum IPD value. The maximum IPD value may be used or a lesser value selected within a distribution of sampled values (e.g., the value at the 95th percentile of sampled IPD values) may be utilized as a reference value for determining the fixation point. For example, this IPD value may constitute one side (e.g., the base) of the imaginary triangle of which the fixation point forms a corner (e.g., the apex).

[0076] Thus, in some embodiments, the display system may be configured to monitor for whether the user is a main or guest user. If the user is a main user, then the calibration file may be accessed. If the user is a guest user, then content-based depth plane switching may be utilized, and/or a rough calibration involving determining IPD may be performed to establish a reference IPD value. The display system may be configured to use this reference IPD value to determine or estimate the guest user's fixation point and, thus, make decisions regarding when to switch depth planes (e.g., when to switch the wavefront divergence of light used to form virtual content).

[0077] In some embodiments, the display system may transition from performing content-based depth plane switching to performing depth plane switching based on a dynamic calibration of the current user. For example, such a transition may occur in situations where data obtained in connection with the content-based depth plane switching scheme is deemed to be unreliable (e.g., where values for the virtual object that the user is roughly fixating on have high levels of uncertainty or variability), or where content is provided at a range of different depths spanning multiple depth planes (e.g., wherein the virtual content spans greater than a threshold number of depth planes).

[0078] Reference will now be made to the drawings, in which like reference numerals refer to like parts throughout.

Unless indicated otherwise, the drawings are schematic not and necessarily drawn to scale.

Examples of 3D Display of a Wearable System

[0079] A wearable system (also referred to herein as an augmented reality (AR) system) can be configured to present 2D or 3D virtual images to a user. The images may be still images, frames of a video, or a video, in combination or the like. At least a portion of the wearable system can be implemented on a wearable device that can present a VR, AR, or MR environment, alone or in combination, for user interaction. The wearable device can be used interchangeably as an AR device (ARD). Further, for the purpose of the present disclosure, the term "AR" is used interchangeably with the term "MR".

[0080] FIG. 1 depicts an illustration of a mixed reality scenario with certain virtual reality objects, and certain physical objects viewed by a person. In FIG. 1, an MR scene **100** is depicted wherein a user of an MR technology sees a real-world park-like setting **110** featuring people, trees, buildings in the background, and a concrete platform **120**. In addition to these items, the user of the MR technology also perceives that he "sees" a robot statue **130** standing upon the real-world platform **120**, and a cartoon-like avatar character **140** flying by which seems to be a personification of a bumble bee, even though these elements do not exist in the real world.

[0081] In order for the 3D display to produce a true sensation of depth, and more specifically, a simulated sensation of surface depth, it may be desirable for each point in the display's visual field to generate an accommodative response corresponding to its virtual depth. If the accommodative response to a display point does not correspond to the virtual depth of that point, as determined by the binocular depth cues of convergence and stereopsis, the human eye may experience an accommodation conflict, resulting in unstable imaging, harmful eye strain, headaches, and, in the absence of accommodation information, almost a complete lack of surface depth.

[0082] VR, AR, and MR experiences can be provided by display systems having displays in which images corresponding to a plurality of depth planes are provided to a viewer. The images may be different for each depth plane (e.g., provide slightly different presentations of a scene or object) and may be separately focused by the viewer's eyes, thereby helping to provide the user with depth cues based on the accommodation of the eye required to bring into focus different image features for the scene located on different depth plane or based on observing different image features on different depth planes being out of focus. As discussed elsewhere herein, such depth cues provide credible perceptions of depth.

[0083] FIG. 2 illustrates an example of wearable system **200** which can be configured to provide an AR/VR/MR scene. The wearable system **200** can also be referred to as the AR system **200**. The wearable system **200** includes a display **220**, and various mechanical and electronic modules and systems to support the functioning of display **220**. The display **220** may be coupled to a frame **230**, which is wearable by a user, wearer, or viewer **210**. The display **220** can be positioned in front of the eyes of the user **210**. The display **220** can present AR/VR/MR content to a user. The display **220** can comprise a head mounted display (HMD) that is worn on the head of the user.

[0084] In some embodiments, a speaker 240 is coupled to the frame 230 and positioned adjacent the ear canal of the user (in some embodiments, another speaker, not shown, is positioned adjacent the other ear canal of the user to provide for stereo/shapeable sound control). The display 220 can include an audio sensor (e.g., a microphone) 232 for detecting an audio stream from the environment and capture ambient sound. In some embodiments, one or more other audio sensors, not shown, are positioned to provide stereo sound reception. Stereo sound reception can be used to determine the location of a sound source. The wearable system 200 can perform voice or speech recognition on the audio stream.

[0085] The wearable system 200 can include an outward-facing imaging system 464 (shown in FIG. 4) which observes the world in the environment around the user. The wearable system 200 can also include an inward-facing imaging system 462 (shown in FIG. 4) which can track the eye movements of the user. The inward-facing imaging system may track either one eye's movements or both eyes' movements. The inward-facing imaging system 462 may be attached to the frame 230 and may be in electrical communication with the processing modules 260 or 270, which may process image information acquired by the inward-facing imaging system to determine, e.g., the pupil diameters or orientations of the eyes, eye movements or eye pose of the user 210. The inward-facing imaging system 462 may include one or more cameras. For example, at least one camera may be used to image each eye. The images acquired by the cameras may be used to determine pupil size or eye pose for each eye separately, thereby allowing presentation of image information to each eye to be dynamically tailored to that eye.

[0086] As an example, the wearable system 200 can use the outward-facing imaging system 464 or the inward-facing imaging system 462 to acquire images of a pose of the user. The images may be still images, frames of a video, or a video.

[0087] The display 220 can be operatively coupled 250, such as by a wired lead or wireless connectivity, to a local data processing module 260 which may be mounted in a variety of configurations, such as fixedly attached to the frame 230, fixedly attached to a helmet or hat worn by the user, embedded in headphones, or otherwise removably attached to the user 210 (e.g., in a backpack-style configuration, in a belt-coupling style configuration).

[0088] The local processing and data module 260 may comprise a hardware processor, as well as digital memory, such as non-volatile memory (e.g., flash memory), both of which may be utilized to assist in the processing, caching, and storage of data. The data may include data a) captured from sensors (which may be, e.g., operatively coupled to the frame 230 or otherwise attached to the user 210), such as image capture devices (e.g., cameras in the inward-facing imaging system or the outward-facing imaging system), audio sensors (e.g., microphones), inertial measurement units (IMUs), accelerometers, compasses, global positioning system (GPS) units, radio devices, or gyroscopes; or b) acquired or processed using remote processing module 270 or remote data repository 280, possibly for passage to the display 220 after such processing or retrieval. The local processing and data module 260 may be operatively coupled by communication links 262 or 264, such as via wired or wireless communication links, to the remote processing

module 270 or remote data repository 280 such that these remote modules are available as resources to the local processing and data module 260. In addition, remote processing module 280 and remote data repository 280 may be operatively coupled to each other.

[0089] In some embodiments, the remote processing module 270 may comprise one or more processors configured to analyze and process data or image information. In some embodiments, the remote data repository 280 may comprise a digital data storage facility, which may be available through the internet or other networking configuration in a "cloud" resource configuration. In some embodiments, all data is stored and all computations are performed in the local processing and data module, allowing fully autonomous use from a remote module.

Example Components of a Wearable System

[0090] FIG. 3 schematically illustrates example components of a wearable system. FIG. 3 shows a wearable system 200 which can include a display 220 and a frame 230. A blown-up view 202 schematically illustrates various components of the wearable system 200. In certain implementations, one or more of the components illustrated in FIG. 3 can be part of the display 220. The various components alone or in combination can collect a variety of data (such as e.g., audio or visual data) associated with the user of the wearable system 200 or the user's environment. It should be appreciated that other embodiments may have additional or fewer components depending on the application for which the wearable system is used. Nevertheless, FIG. 3 provides a basic idea of some of the various components and types of data that may be collected, analyzed, and stored through the wearable system.

[0091] FIG. 3 shows an example wearable system 200 which can include the display 220. The display 220 can comprise a display lens 226 that may be mounted to a user's head or a housing or frame 230, which corresponds to the frame 230. The display lens 226 may comprise one or more transparent mirrors positioned by the housing 230 in front of the user's eyes 302, 304 and may be configured to bounce projected light 338 into the eyes 302, 304 and facilitate beam shaping, while also allowing for transmission of at least some light from the local environment. The wavefront of the projected light beam 338 may be bent or focused to coincide with a desired focal distance of the projected light. As illustrated, two wide-field-of-view machine vision cameras 316 (also referred to as world cameras) can be coupled to the housing 230 to image the environment around the user. These cameras 316 can be dual capture visible light/non-visible (e.g., infrared) light cameras. The cameras 316 may be part of the outward-facing imaging system 464 shown in FIG. 4. Image acquired by the world cameras 316 can be processed by the pose processor 336. For example, the pose processor 336 can implement one or more object recognizers 708 (e.g., shown in FIG. 7) to identify a pose of a user or another person in the user's environment or to identify a physical object in the user's environment.

[0092] With continued reference to FIG. 3, a pair of scanned-laser shaped-wavefront (e.g., for depth) light projector modules with display mirrors and optics configured to project light 338 into the eyes 302, 304 are shown. The depicted view also shows two miniature infrared cameras 324 paired with infrared light sources 326 (such as light emitting diodes "LED"s), which are configured to be able to

track the eyes **302**, **304** of the user to support rendering and user input. The cameras **324** may be part of the inward-facing imaging system **462** shown in FIG. 4. The wearable system **200** can further feature a sensor assembly **339**, which may comprise X, Y, and Z axis accelerometer capability as well as a magnetic compass and X, Y, and Z axis gyro capability, preferably providing data at a relatively high frequency, such as 200 Hz. The sensor assembly **339** may be part of the IMU described with reference to FIG. 2A. The depicted system **200** can also comprise a head pose processor **336**, such as an ASIC (application specific integrated circuit), FPGA (field programmable gate array), or ARM processor (advanced reduced-instruction-set machine), which may be configured to calculate real or near-real time user head pose from wide field of view image information output from the capture devices **316**. The head pose processor **336** can be a hardware processor and can be implemented as part of the local processing and data module **260** shown in FIG. 2A.

[0093] The wearable system can also include one or more depth sensors **234**. The depth sensor **234** can be configured to measure the distance between an object in an environment to a wearable device. The depth sensor **234** may include a laser scanner (e.g., a lidar), an ultrasonic depth sensor, or a depth sensing camera. In certain implementations, where the cameras **316** have depth sensing ability, the cameras **316** may also be considered as depth sensors **234**.

[0094] Also shown is a processor **332** configured to execute digital or analog processing to derive pose from the gyro, compass, or accelerometer data from the sensor assembly **339**. The processor **332** may be part of the local processing and data module **260** shown in FIG. 2. The wearable system **200** as shown in FIG. 3 can also include a position system such as, e.g., a GPS **337** (global positioning system) to assist with pose and positioning analyses. In addition, the GPS may further provide remotely-based (e.g., cloud-based) information about the user's environment. This information may be used for recognizing objects or information in user's environment.

[0095] The wearable system may combine data acquired by the GPS **337** and a remote computing system (such as, e.g., the remote processing module **270**, another user's ARD, etc.) which can provide more information about the user's environment. As one example, the wearable system can determine the user's location based on GPS data and retrieve a world map (e.g., by communicating with a remote processing module **270**) including virtual objects associated with the user's location. As another example, the wearable system **200** can monitor the environment using the world cameras **316** (which may be part of the outward-facing imaging system **464** shown in FIG. 4). Based on the images acquired by the world cameras **316**, the wearable system **200** can detect objects in the environment (e.g., by using one or more object recognizers **708** shown in FIG. 7). The wearable system can further use data acquired by the GPS **337** to interpret the characters.

[0096] The wearable system **200** may also comprise a rendering engine **334** which can be configured to provide rendering information that is local to the user to facilitate operation of the scanners and imaging into the eyes of the user, for the user's view of the world. The rendering engine **334** may be implemented by a hardware processor (such as, e.g., a central processing unit or a graphics processing unit). In some embodiments, the rendering engine is part of the

local processing and data module **260**. The rendering engine **334** can be communicatively coupled (e.g., via wired or wireless links) to other components of the wearable system **200**. For example, the rendering engine **334**, can be coupled to the eye cameras **324** via communication link **274**, and be coupled to a projecting subsystem **318** (which can project light into user's eyes **302**, **304** via a scanned laser arrangement in a manner similar to a retinal scanning display) via the communication link **272**. The rendering engine **334** can also be in communication with other processing units such as, e.g., the sensor pose processor **332** and the image pose processor **336** via links **276** and **294** respectively.

[0097] The cameras **324** (e.g., mini infrared cameras) may be utilized to track the eye pose to support rendering and user input. Some example eye poses may include where the user is looking or at what depth he or she is focusing (which may be estimated with eye vergence). The GPS **337**, gyros, compass, and accelerometers **339** may be utilized to provide coarse or fast pose estimates. One or more of the cameras **316** can acquire images and pose, which in conjunction with data from an associated cloud computing resource, may be utilized to map the local environment and share user views with others.

[0098] The example components depicted in FIG. 3 are for illustration purposes only. Multiple sensors and other functional modules are shown together for ease of illustration and description. Some embodiments may include only one or a subset of these sensors or modules. Further, the locations of these components are not limited to the positions depicted in FIG. 3. Some components may be mounted to or housed within other components, such as a belt-mounted component, a hand-held component, or a helmet component. As one example, the image pose processor **336**, sensor pose processor **332**, and rendering engine **334** may be positioned in a backpack and configured to communicate with other components of the wearable system via wireless communication, such as ultra-wideband, Wi-Fi, Bluetooth, etc., or via wired communication. The depicted housing **230** preferably is head-mountable and wearable by the user. However, some components of the wearable system **200** may be worn to other portions of the user's body. For example, the speaker **240** may be inserted into the ears of a user to provide sound to the user.

[0099] Regarding the projection of light **338** into the eyes **302**, **304** of the user, in some embodiment, the cameras **324** may be utilized to measure where the centers of a user's eyes are geometrically verged to, which, in general, coincides with a position of focus, or "depth of focus", of the eyes. A 3-dimensional surface of all points the eyes verge to can be referred to as the "horopter". The focal distance may take on a finite number of depths, or may be infinitely varying. Light projected from the vergence distance appears to be focused to the subject eye **302**, **304**, while light in front of or behind the vergence distance is blurred. Examples of wearable devices and other display systems of the present disclosure are also described in U.S. Patent Publication No. 2016/0270656, which is incorporated by reference herein in its entirety.

[0100] The human visual system is complicated and providing a realistic perception of depth is challenging. Viewers of an object may perceive the object as being three-dimensional due to a combination of vergence and accommodation. Vergence movements (e.g., rolling movements of the pupils toward or away from each other to converge the lines

of sight of the eyes to fixate upon an object) of the two eyes relative to each other are closely associated with focusing (or “accommodation”) of the lenses of the eyes. Under normal conditions, changing the focus of the lenses of the eyes, or accommodating the eyes, to change focus from one object to another object at a different distance will automatically cause a matching change in vergence to the same distance, under a relationship known as the “accommodation-vergence reflex.” Likewise, a change in vergence will trigger a matching change in accommodation, under normal conditions. Display systems that provide a better match between accommodation and vergence may form more realistic and comfortable simulations of three-dimensional imagery.

[0101] Further spatially coherent light with a beam diameter of less than about 0.7 millimeters can be correctly resolved by the human eye regardless of where the eye focuses. Thus, to create an illusion of proper focal depth, the eye vergence may be tracked with the cameras **324**, and the rendering engine **334** and projection subsystem **318** may be utilized to render all objects on or close to the horopter in focus, and all other objects at varying degrees of defocus (e.g., using intentionally-created blurring). Preferably, the system **220** renders to the user at a frame rate of about 60 frames per second or greater. As described above, preferably, the cameras **324** may be utilized for eye tracking, and software may be configured to pick up not only vergence geometry but also focus location cues to serve as user inputs. Preferably, such a display system is configured with brightness and contrast suitable for day or night use.

[0102] In some embodiments, the display system preferably has latency of less than about 20 milliseconds for visual object alignment, less than about 0.1 degree of angular alignment, and about 1 arc minute of resolution, which, without being limited by theory, is believed to be approximately the limit of the human eye. The display system **220** may be integrated with a localization system, which may involve GPS elements, optical tracking, compass, accelerometers, or other data sources, to assist with position and pose determination; localization information may be utilized to facilitate accurate rendering in the user’s view of the pertinent world (e.g., such information would facilitate the glasses to know where they are with respect to the real world).

[0103] In some embodiments, the wearable system **200** is configured to display one or more virtual images based on the accommodation of the user’s eyes. Unlike prior 3D display approaches that force the user to focus where the images are being projected, in some embodiments, the wearable system is configured to automatically vary the focus of projected virtual content to allow for a more comfortable viewing of one or more images presented to the user. For example, if the user’s eyes have a current focus of 1 m, the image may be projected to coincide with the user’s focus. If the user shifts focus to 3 m, the image is projected to coincide with the new focus. Thus, rather than forcing the user to a predetermined focus, the wearable system **200** of some embodiments allows the user’s eye to a function in a more natural manner.

[0104] Such a wearable system **200** may eliminate or reduce the incidences of eye strain, headaches, and other physiological symptoms typically observed with respect to virtual reality devices. To achieve this, various embodiments of the wearable system **200** are configured to project virtual

images at varying focal distances, through one or more variable focus elements (VFEs). In one or more embodiments, 3D perception may be achieved through a multi-plane focus system that projects images at fixed focal planes away from the user. Other embodiments employ variable plane focus, wherein the focal plane is moved back and forth in the z-direction to coincide with the user’s present state of focus.

[0105] In both the multi-plane focus systems and variable plane focus systems, wearable system **200** may employ eye tracking to determine a vergence of the user’s eyes, determine the user’s current focus, and project the virtual image at the determined focus. In other embodiments, wearable system **200** comprises a light modulator that variably projects, through a fiber scanner, or other light generating source, light beams of varying focus in a raster pattern across the retina. Thus, the ability of the display of the wearable system **200** to project images at varying focal distances not only eases accommodation for the user to view objects in 3D, but may also be used to compensate for user ocular anomalies, as further described in U.S. Patent Publication No. 2016/0270656, which is incorporated by reference herein in its entirety. In some other embodiments, a spatial light modulator may project the images to the user through various optical components. For example, as described further below, the spatial light modulator may project the images onto one or more waveguides, which then transmit the images to the user.

Waveguide Stack Assembly

[0106] FIG. 4 illustrates an example of a waveguide stack for outputting image information to a user. A wearable system **400** includes a stack of waveguides, or stacked waveguide assembly **480** that may be utilized to provide three-dimensional perception to the eye/brain using a plurality of waveguides **432b**, **434b**, **436b**, **438b**, **440b**. In some embodiments, the wearable system **400** may correspond to wearable system **200** of FIG. 2, with FIG. 4 schematically showing some parts of that wearable system **200** in greater detail. For example, in some embodiments, the waveguide assembly **480** may be integrated into the display **220** of FIG. 2.

[0107] With continued reference to FIG. 4, the waveguide assembly **480** may also include a plurality of features **458**, **456**, **454**, **452** between the waveguides. In some embodiments, the features **458**, **456**, **454**, **452** may be lenses. In some embodiments, the lens may be variable focus elements (VFEs). For example, in some embodiments, the waveguide assembly **480** may simply include two variable focus elements and one or more waveguides in between those two variable focus elements. Examples of waveguide assemblies with VFEs are disclosed in US patent publication No. 2017/0293145, published Oct. 12, 2017, the entire disclosure of which is incorporated herein by reference. In other embodiments, the features **458**, **456**, **454**, **452** may not be lenses. Rather, they may simply be spacers (e.g., cladding layers or structures for forming air gaps).

[0108] The waveguides **432b**, **434b**, **436b**, **438b**, **440b** or the plurality of lenses **458**, **456**, **454**, **452** may be configured to send image information to the eye with various levels of wavefront curvature or light ray divergence. Each waveguide level may be associated with a particular depth plane and may be configured to output image information corresponding to that depth plane. Image injection devices **420**,

422, 424, 426, 428 may be utilized to inject image information into the waveguides **440b, 438b, 436b, 434b, 432b**, each of which may be configured to distribute incoming light across each respective waveguide, for output toward the eye **410**. Light exits an output surface of the image injection devices **420, 422, 424, 426, 428** and is injected into a corresponding input edge of the waveguides **440b, 438b, 436b, 434b, 432b**. In some embodiments, a single beam of light (e.g., a collimated beam) may be injected into each waveguide to output an entire field of cloned collimated beams that are directed toward the eye **410** at particular angles (and amounts of divergence) corresponding to the depth plane associated with a particular waveguide.

[0109] In some embodiments, the image injection devices **420, 422, 424, 426, 428** are discrete displays that each produce image information for injection into a corresponding waveguide **440b, 438b, 436b, 434b, 432b**, respectively. In some other embodiments, the image injection devices **420, 422, 424, 426, 428** are the output ends of a single multiplexed display which may, e.g., pipe image information via one or more optical conduits (such as fiber optic cables) to each of the image injection devices **420, 422, 424, 426, 428**.

[0110] A controller **460** controls the operation of the stacked waveguide assembly **480** and the image injection devices **420, 422, 424, 426, 428**. The controller **460** includes programming (e.g., instructions in a non-transitory computer-readable medium) that regulates the timing and provision of image information to the waveguides **440b, 438b, 436b, 434b, 432b**. In some embodiments, the controller **460** may be a single integral device, or a distributed system connected by wired or wireless communication channels. The controller **460** may be part of the processing modules **260** or **270** (illustrated in FIG. 2) in some embodiments.

[0111] The waveguides **440b, 438b, 436b, 434b, 432b** may be configured to propagate light within each respective waveguide by total internal reflection (TIR). The waveguides **440b, 438b, 436b, 434b, 432b** may each be planar or have another shape (e.g., curved), with major top and bottom surfaces and edges extending between those major top and bottom surfaces. In the illustrated configuration, the waveguides **440b, 438b, 436b, 434b, 432b** may each include light extracting optical elements **440a, 438a, 436a, 434a, 432a** that are configured to extract light out of a waveguide by redirecting the light, propagating within each respective waveguide, out of the waveguide to output image information to the eye **410**. Extracted light may also be referred to as outcoupled light, and light extracting optical elements may also be referred to as outcoupling optical elements. An extracted beam of light is outputted by the waveguide at locations at which the light propagating in the waveguide strikes a light redirecting element. The light extracting optical elements (**440a, 438a, 436a, 434a, 432a**) may, for example, be reflective or diffractive optical features. While illustrated disposed at the bottom major surfaces of the waveguides **440b, 438b, 436b, 434b, 432b** for ease of description and drawing clarity, in some embodiments, the light extracting optical elements **440a, 438a, 436a, 434a, 432a** may be disposed at the top or bottom major surfaces, or may be disposed directly in the volume of the waveguides **440b, 438b, 436b, 434b, 432b**. In some embodiments, the light extracting optical elements **440a, 438a, 436a, 434a, 432a** may be formed in a layer of material that is attached to a transparent substrate to form the waveguides **440b,**

438b, 436b, 434b, 432b. In some other embodiments, the waveguides **440b, 438b, 436b, 434b, 432b** may be a monolithic piece of material and the light extracting optical elements **440a, 438a, 436a, 434a, 432a** may be formed on a surface or in the interior of that piece of material.

[0112] With continued reference to FIG. 4, as discussed herein, each waveguide **440b, 438b, 436b, 434b, 432b** is configured to output light to form an image corresponding to a particular depth plane. For example, the waveguide **432b** nearest the eye may be configured to deliver collimated light, as injected into such waveguide **432b**, to the eye **410**. The collimated light may be representative of the optical infinity focal plane. The next waveguide up **434b** may be configured to send out collimated light which passes through the first lens **452** (e.g., a negative lens) before it can reach the eye **410**. First lens **452** may be configured to create a slight convex wavefront curvature so that the eye/brain interprets light coming from that next waveguide up **434b** as coming from a first focal plane closer inward toward the eye **410** from optical infinity. Similarly, the third up waveguide **436b** passes its output light through both the first lens **452** and second lens **454** before reaching the eye **410**. The combined optical power of the first and second lenses **452** and **454** may be configured to create another incremental amount of wavefront curvature so that the eye/brain interprets light coming from the third waveguide **436b** as coming from a second focal plane that is even closer inward toward the person from optical infinity than was light from the next waveguide up **434b**.

[0113] The other waveguide layers (e.g., waveguides **438b, 440b**) and lenses (e.g., lenses **456, 458**) are similarly configured, with the highest waveguide **440b** in the stack sending its output through all of the lenses between it and the eye for an aggregate focal power representative of the closest focal plane to the person. To compensate for the stack of lenses **458, 456, 454, 452** when viewing/interpreting light coming from the world **470** on the other side of the stacked waveguide assembly **480**, a compensating lens layer **430** may be disposed at the top of the stack to compensate for the aggregate power of the lens stack **458, 456, 454, 452** below. (Compensating lens layer **430** and the stacked waveguide assembly **480** as a whole may be configured such that light coming from the world **470** is conveyed to the eye **410** at substantially the same level of divergence (or collimation) as the light had when it was initially received by the stacked waveguide assembly **480**.) Such a configuration provides as many perceived focal planes as there are available waveguide/lens pairings. Both the light extracting optical elements of the waveguides and the focusing aspects of the lenses may be static (e.g., not dynamic or electro-active). In some alternative embodiments, either or both may be dynamic using electro-active features.

[0114] With continued reference to FIG. 4, the light extracting optical elements **440a, 438a, 436a, 434a, 432a** may be configured to both redirect light out of their respective waveguides and to output this light with the appropriate amount of divergence or collimation for a particular depth plane associated with the waveguide. As a result, waveguides having different associated depth planes may have different configurations of light extracting optical elements, which output light with a different amount of divergence depending on the associated depth plane. In some embodiments, as discussed herein, the light extracting optical elements **440a, 438a, 436a, 434a, 432a** may be volumetric

or surface features, which may be configured to output light at specific angles. For example, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** may be volume holograms, surface holograms, and/or diffraction gratings. Light extracting optical elements, such as diffraction gratings, are described in U.S. Patent Publication No. 2015/0178939, published Jun. 25, 2015, which is incorporated by reference herein in its entirety.

[0115] In some embodiments, the light extracting optical elements **440a**, **438a**, **436a**, **434a**, **432a** are diffractive features that form a diffraction pattern, or “diffractive optical element” (also referred to herein as a “DOE”). Preferably, the DOE has a relatively low diffraction efficiency so that only a portion of the light of the beam is deflected away toward the eye **410** with each intersection of the DOE, while the rest continues to move through a waveguide via total internal reflection. The light carrying the image information can thus be divided into a number of related exit beams that exit the waveguide at a multiplicity of locations and the result is a fairly uniform pattern of exit emission toward the eye **304** for this particular collimated beam bouncing around within a waveguide.

[0116] In some embodiments, one or more DOEs may be switchable between “on” state in which they actively diffract, and “off” state in which they do not significantly diffract. For instance, a switchable DOE may comprise a layer of polymer dispersed liquid crystal, in which microdroplets comprise a diffraction pattern in a host medium, and the refractive index of the microdroplets can be switched to substantially match the refractive index of the host material (in which case the pattern does not appreciably diffract incident light) or the microdroplet can be switched to an index that does not match that of the host medium (in which case the pattern actively diffracts incident light).

[0117] In some embodiments, the number and distribution of depth planes or depth of field may be varied dynamically based on the pupil sizes or orientations of the eyes of the viewer. Depth of field may change inversely with a viewer's pupil size. As a result, as the sizes of the pupils of the viewer's eyes decrease, the depth of field increases such that one plane that is not discernible because the location of that plane is beyond the depth of focus of the eye may become discernible and appear more in focus with reduction of pupil size and commensurate with the increase in depth of field. Likewise, the number of spaced apart depth planes used to present different images to the viewer may be decreased with the decreased pupil size. For example, a viewer may not be able to clearly perceive the details of both a first depth plane and a second depth plane at one pupil size without adjusting the accommodation of the eye away from one depth plane and to the other depth plane. These two depth planes may, however, be sufficiently in focus at the same time to the user at another pupil size without changing accommodation.

[0118] In some embodiments, the display system may vary the number of waveguides receiving image information based upon determinations of pupil size or orientation, or upon receiving electrical signals indicative of particular pupil size or orientation. For example, if the user's eyes are unable to distinguish between two depth planes associated with two waveguides, then the controller **460** (which may be an embodiment of the local processing and data module **260**) can be configured or programmed to cease providing image information to one of these waveguides. Advantageously, this may reduce the processing burden on the system,

thereby increasing the responsiveness of the system. In embodiments in which the DOEs for a waveguide are switchable between the on and off states, the DOEs may be switched to the off state when the waveguide does receive image information.

[0119] In some embodiments, it may be desirable to have an exit beam meet the condition of having a diameter that is less than the diameter of the eye of a viewer. However, meeting this condition may be challenging in view of the variability in size of the viewer's pupils. In some embodiments, this condition is met over a wide range of pupil sizes by varying the size of the exit beam in response to determinations of the size of the viewer's pupil. For example, as the pupil size decreases, the size of the exit beam may also decrease. In some embodiments, the exit beam size may be varied using a variable aperture.

[0120] The wearable system **400** can include an outward-facing imaging system **464** (e.g., a digital camera) that images a portion of the world **470**. This portion of the world **470** may be referred to as the field of view (FOV) of a world camera and the imaging system **464** is sometimes referred to as an FOV camera. The FOV of the world camera may or may not be the same as the FOV of a viewer **210** which encompasses a portion of the world **470** the viewer **210** perceives at a given time. For example, in some situations, the FOV of the world camera may be larger than the viewer **210** of the viewer **210** of the wearable system **400**. The entire region available for viewing or imaging by a viewer may be referred to as the field of regard (FOR). The FOR may include 4π steradians of solid angle surrounding the wearable system **400** because the wearer can move his body, head, or eyes to perceive substantially any direction in space. In other contexts, the wearer's movements may be more constricted, and accordingly the wearer's FOR may subtend a smaller solid angle. Images obtained from the outward-facing imaging system **464** can be used to track gestures made by the user (e.g., hand or finger gestures), detect objects in the world **470** in front of the user, and so forth.

[0121] The wearable system **400** can include an audio sensor **232**, e.g., a microphone, to capture ambient sound. As described above, in some embodiments, one or more other audio sensors can be positioned to provide stereo sound reception useful to the determination of location of a speech source. The audio sensor **232** can comprise a directional microphone, as another example, which can also provide such useful directional information as to where the audio source is located. The wearable system **400** can use information from both the outward-facing imaging system **464** and the audio sensor **230** in locating a source of speech, or to determine an active speaker at a particular moment in time, etc. For example, the wearable system **400** can use the voice recognition alone or in combination with a reflected image of the speaker (e.g., as seen in a mirror) to determine the identity of the speaker. As another example, the wearable system **400** can determine a position of the speaker in an environment based on sound acquired from directional microphones. The wearable system **400** can parse the sound coming from the speaker's position with speech recognition algorithms to determine the content of the speech and use voice recognition techniques to determine the identity (e.g., name or other demographic information) of the speaker.

[0122] The wearable system **400** can also include an inward-facing imaging system **466** (e.g., a digital camera),

which observes the movements of the user, such as the eye movements and the facial movements. The inward-facing imaging system 466 may be used to capture images of the eye 410 to determine the size and/or orientation of the pupil of the eye 304. The inward-facing imaging system 466 can be used to obtain images for use in determining the direction the user is looking (e.g., eye pose) or for biometric identification of the user (e.g., via iris identification). In some embodiments, at least one camera may be utilized for each eye, to separately determine the pupil size or eye pose of each eye independently, thereby allowing the presentation of image information to each eye to be dynamically tailored to that eye. In some other embodiments, the pupil diameter or orientation of only a single eye 410 (e.g., using only a single camera per pair of eyes) is determined and assumed to be similar for both eyes of the user. The images obtained by the inward-facing imaging system 466 may be analyzed to determine the user's eye pose or mood, which can be used by the wearable system 400 to decide which audio or visual content should be presented to the user. The wearable system 400 may also determine head pose (e.g., head position or head orientation) using sensors such as IMUs, accelerometers, gyroscopes, etc.

[0123] The wearable system 400 can include a user input device 466 by which the user can input commands to the controller 460 to interact with the wearable system 400. For example, the user input device 466 can include a trackpad, a touchscreen, a joystick, a multiple degree-of-freedom (DOF) controller, a capacitive sensing device, a game controller, a keyboard, a mouse, a directional pad (D-pad), a wand, a haptic device, a totem (e.g., functioning as a virtual user input device), and so forth. A multi-DOF controller can sense user input in some or all possible translations (e.g., left/right, forward/backward, or up/down) or rotations (e.g., yaw, pitch, or roll) of the controller. A multi-DOF controller which supports the translation movements may be referred to as a 3DOF while a multi-DOF controller which supports the translations and rotations may be referred to as 6DOF. In some cases, the user may use a finger (e.g., a thumb) to press or swipe on a touch-sensitive input device to provide input to the wearable system 400 (e.g., to provide user input to a user interface provided by the wearable system 400). The user input device 466 may be held by the user's hand during the use of the wearable system 400. The user input device 466 can be in wired or wireless communication with the wearable system 400.

Other Components of the Wearable System

[0124] In many implementations, the wearable system may include other components in addition or in alternative to the components of the wearable system described above. The wearable system may, for example, include one or more haptic devices or components. The haptic devices or components may be operable to provide a tactile sensation to a user. For example, the haptic devices or components may provide a tactile sensation of pressure or texture when touching virtual content (e.g., virtual objects, virtual tools, other virtual constructs). The tactile sensation may replicate a feel of a physical object which a virtual object represents, or may replicate a feel of an imagined object or character (e.g., a dragon) which the virtual content represents. In some implementations, haptic devices or components may be

worn by the user (e.g., a user wearable glove). In some implementations, haptic devices or components may be held by the user.

[0125] The wearable system may, for example, include one or more physical objects which are manipulable by the user to allow input or interaction with the wearable system. These physical objects may be referred to herein as totems. Some totems may take the form of inanimate objects, such as for example, a piece of metal or plastic, a wall, a surface of table. In certain implementations, the totems may not actually have any physical input structures (e.g., keys, triggers, joystick, trackball, rocker switch). Instead, the totem may simply provide a physical surface, and the wearable system may render a user interface so as to appear to a user to be on one or more surfaces of the totem. For example, the wearable system may render an image of a computer keyboard and trackpad to appear to reside on one or more surfaces of a totem. For example, the wearable system may render a virtual computer keyboard and virtual trackpad to appear on a surface of a thin rectangular plate of aluminum which serves as a totem. The rectangular plate does not itself have any physical keys or trackpad or sensors. However, the wearable system may detect user manipulation or interaction or touches with the rectangular plate as selections or inputs made via the virtual keyboard or virtual trackpad. The user input device 466 (shown in FIG. 4) may be an embodiment of a totem, which may include a trackpad, a touchpad, a trigger, a joystick, a trackball, a rocker or virtual switch, a mouse, a keyboard, a multi-degree-of-freedom controller, or another physical input device. A user may use the totem, alone or in combination with poses, to interact with the wearable system or other users.

[0126] Examples of haptic devices and totems usable with the wearable devices, HMD, and display systems of the present disclosure are described in U.S. Patent Publication No. 2015/0016777, which is incorporated by reference herein in its entirety.

Example of an Eye Image

[0127] FIG. 5 illustrates an image of an eye 500 with eyelids 504, sclera 508 (the "white" of the eye), iris 512, and pupil 516. Curve 516a shows the pupillary boundary between the pupil 516 and the iris 512, and curve 512a shows the limbic boundary between the iris 512 and the sclera 508. The eyelids 504 include an upper eyelid 504a and a lower eyelid 504b. The eye 500 is illustrated in a natural resting pose (e.g., in which the user's face and gaze are both oriented as they would be toward a distant object directly ahead of the user). The natural resting pose of the eye 500 can be indicated by a natural resting direction 520, which is a direction orthogonal to the surface of the eye 500 when in the natural resting pose (e.g., directly out of the plane for the eye 500 shown in FIG. 5) and in this example, centered within the pupil 516.

[0128] As the eye 500 moves to look toward different objects, the eye pose will change relative to the natural resting direction 520. The current eye pose can be determined with reference to an eye pose direction 524, which is a direction orthogonal to the surface of the eye (and centered in within the pupil 516) but oriented toward the object at which the eye is currently directed. With reference to an example coordinate system shown in FIG. 5, the pose of the eye 500 can be expressed as two angular parameters indicating an azimuthal deflection and a zenithal deflection of

the eye pose direction **524** of the eye, both relative to the natural resting direction **520** of the eye. For purposes of illustration, these angular parameters can be represented as θ (azimuthal deflection, determined from a fiducial azimuth) and ϕ (zenithal deflection, sometimes also referred to as a polar deflection). In some implementations, angular roll of the eye around the eye pose direction **524** can be included in the determination of eye pose, and angular roll can be included in the following analysis. In other implementations, other techniques for determining the eye pose can be used, for example, a pitch, yaw, and optionally roll system.

[0129] An eye image can be obtained from a video using any appropriate process, for example, using a video processing algorithm that can extract an image from one or more sequential frames. The pose of the eye can be determined from the eye image using a variety of eye-tracking techniques. For example, an eye pose can be determined by considering the lensing effects of the cornea on light sources that are provided. Any suitable eye tracking technique can be used for determining eye pose in the eyelid shape estimation techniques described herein. Example of an Eye Tracking System

[0130] FIG. 6 illustrates a schematic diagram of a wearable system **600** that includes an eye tracking system. The wearable system **600** may, in at least some embodiments, include components located in a head-mounted unit **602** and components located in a non-head-mounted unit **604**. Non-head mounted unit **604** may be, as examples, a belt-mounted component, a hand-held component, a component in a backpack, a remote component, etc. Incorporating some of the components of the wearable system **600** in non-head-mounted unit **604** may help to reduce the size, weight, complexity, and cost of the head-mounted unit **602**. In some implementations, some or all of the functionality described as being performed by one or more components of head-mounted unit **602** and/or non-head mounted **604** may be provided by way of one or more components included elsewhere in the wearable system **600**. For example, some or all of the functionality described below in association with a CPU **612** of head-mounted unit **602** may be provided by way of a CPU **616** of non-head mounted unit **604**, and vice versa. In some examples, some or all of such functionality may be provided by way of peripheral devices of wearable system **600**. Furthermore, in some implementations, some or all of such functionality may be provided by way of one or more cloud computing devices or other remotely-located computing devices in a manner similar to that which has been described above with reference to FIG. 2.

[0131] As shown in FIG. 6, wearable system **600** can include an eye tracking system including a camera **324** that captures images of a user's eye **610**. If desired, the eye tracking system may also include light sources **326a** and **326b** (such as light emitting diodes "LED"s). The light sources **326a** and **326b** may generate glints (i.e., reflections off of the user's eyes that appear in images of the eye captured by camera **324**). The positions of the light sources **326a** and **326b** relative to the camera **324** may be known and, as a consequence, the positions of the glints within images captured by camera **324** may be used in tracking the user's eyes (as will be discussed in more detail below in connection with FIGS. 7-11). In at least one embodiment, there may be one light source **326** and one camera **324** associated with a single one of the user's eyes **610**. In another embodiment, there may be one light source **326** and

one camera **324** associated with each of a user's eyes **610**. In yet other embodiments, there may be one or more cameras **324** and one or more light sources **326** associated with one or each of a user's eyes **610**. As a specific example, there may be two light sources **326a** and **326b** and one or more cameras **324** associated with each of a user's eyes **610**. As another example, there may be three or more light sources such as light sources **326a** and **326b** and one or more cameras **324** associated with each of a user's eyes **610**.

[0132] Eye tracking module **614** may receive images from eye tracking camera(s) **324** and may analyze the images to extract various pieces of information. As examples, the eye tracking module **614** may detect the user's eye poses, a three-dimensional position of the user's eye relative to the eye tracking camera **324** (and to the head-mounted unit **602**), the direction one or both of the user's eyes **610** are focused on, the user's vergence depth (i.e., the depth from the user at which the user is focusing on), the positions of the user's pupils, the positions of the user's cornea and cornea sphere, the center of rotation of each of the user's eyes, and the center of perspective of each of the user's eyes. The eye tracking module **614** may extract such information using techniques described below in connection with FIGS. 7-11. As shown in FIG. 6, eye tracking module **614** may be a software module implemented using a CPU **612** in a head-mounted unit **602**. Further details discussing the creation, adjustment, and use of eye tracking module components are provided in U.S. patent application Ser. No. 15/993,371, entitled "EYE TRACKING CALIBRATION TECHNIQUES," which is incorporated herein by reference in its entirety.

[0133] Data from eye tracking module **614** may be provided to other components in the wearable system. As an example, such data may be transmitted to components in a non-head-mounted unit **604** such as CPU **616** including software modules for a light-field render controller **618** and a registration observer **620**.

[0134] Render controller **618** may use information from eye tracking module **614** to adjust images displayed to the user by render engine **622** (e.g., a render engine that may be a software module in GPU **620** and that may provide images to display **220**). As an example, the render controller **618** may adjust images displayed to the user based on the user's center of rotation or center of perspective. In particular, the render controller **618** may use information on the user's center of perspective to simulate a render camera (i.e., to simulate collecting images from the user's perspective) and may adjust images displayed to the user based on the simulated render camera.

[0135] A "render camera," which is sometimes also referred to as a "pinhole perspective camera" (or simply "perspective camera") or "virtual pinhole camera" (or simply "virtual camera"), is a simulated camera for use in rendering virtual image content possibly from a database of objects in a virtual world. The objects may have locations and orientations relative to the user or wearer and possibly relative to real objects in the environment surrounding the user or wearer. In other words, the render camera may represent a perspective within render space from which the user or wearer is to view 3D virtual contents of the render space (e.g., virtual objects). The render camera may be managed by a render engine to render virtual images based on the database of virtual objects to be presented to the eye. The virtual images may be rendered as if taken from the

perspective of the user or wearer. For example, the virtual images may be rendered as if captured by a pinhole camera (corresponding to the “render camera”) having a specific set of intrinsic parameters (e.g., focal length, camera pixel size, principal point coordinates, skew/distortion parameters, etc.), and a specific set of extrinsic parameters (e.g., translational components and rotational components relative to the virtual world). The virtual images are taken from the perspective of such a camera having a position and orientation of the render camera (e.g., extrinsic parameters of the render camera). It follows that the system may define and/or adjust intrinsic and extrinsic render camera parameters. For example, the system may define a particular set of extrinsic render camera parameters such that virtual images may be rendered as if captured from the perspective of a camera having a specific location with respect to the user’s or wearer’s eye so as to provide images that appear to be from the perspective of the user or wearer. The system may later dynamically adjust extrinsic render camera parameters on-the-fly so as to maintain registration with the specific location. Similarly, intrinsic render camera parameters may be defined and dynamically adjusted over time. In some implementations, the images are rendered as if captured from the perspective of a camera having an aperture (e.g., pinhole) at a specific location with respect to the user’s or wearer’s eye (such as the center of perspective or center of rotation, or elsewhere).

[0136] In some embodiments, the system may create or dynamically reposition and/or reorient one render camera for the user’s left eye, and another render camera for the user’s right eye, as the user’s eyes are physically separated from one another and thus consistently positioned at different locations. It follows that, in at least some implementations, virtual content rendered from the perspective of a render camera associated with the viewer’s left eye may be presented to the user through an eyepiece on the left side of a head-mounted display (e.g., head-mounted unit **602**), and that virtual content rendered from the perspective of a render camera associated with the user’s right eye may be presented to the user through an eyepiece on the right side of such a head-mounted display. Further details discussing the creation, adjustment, and use of render cameras in rendering processes are provided in U.S. patent application Ser. No. 15/274,823, entitled “METHODS AND SYSTEMS FOR DETECTING AND COMBINING STRUCTURAL FEATURES IN 3D RECONSTRUCTION,” which is expressly incorporated herein by reference in its entirety for all purposes.

[0137] In some examples, one or more modules (or components) of the system **600** (e.g., light-field render controller **618**, render engine **620**, etc.) may determine the position and orientation of the render camera within render space based on the position and orientation of the user’s head and eyes (e.g., as determined based on head pose and eye tracking data, respectively). For example, the system **600** may effectively map the position and orientation of the user’s head and eyes to particular locations and angular positions within a 3D virtual environment, place and orient render cameras at the particular locations and angular positions within the 3D virtual environment, and render virtual content for the user as it would be captured by the render camera. Further details discussing real world to virtual world mapping processes are provided in U.S. patent application Ser. No. 15/296,869, entitled “SELECTING VIRTUAL OBJECTS IN A THREE-

DIMENSIONAL SPACE,” which is expressly incorporated herein by reference in its entirety for all purposes. As an example, the render controller **618** may adjust the depths at which images are displayed by selecting which depth plane (or depth planes) are utilized at any given time to display the images. In some implementations, such a depth plane switch may be carried out through an adjustment of one or more intrinsic render camera parameters.

[0138] Registration observer **620** may use information from eye tracking module **614** to identify whether the head-mounted unit **602** is properly positioned on a user’s head. As an example, the eye tracking module **614** may provide eye location information, such as the positions of the centers of rotation of the user’s eyes, indicative of the three-dimensional position of the user’s eyes relative to camera **324** and head-mounted unit **602** and the eye tracking module **614** may use the location information to determine if display **220** is properly aligned in the user’s field of view, or if the head-mounted unit **602** (or headset) has slipped or is otherwise misaligned with the user’s eyes. As examples, the registration observer **620** may be able to determine if the head-mounted unit **602** has slipped down the user’s nose bridge, thus moving display **220** away and down from the user’s eyes (which may be undesirable), if the head-mounted unit **602** has been moved up the user’s nose bridge, thus moving display **220** closer and up from the user’s eyes, if the head-mounted unit **602** has been shifted left or right relative to the user’s nose bridge, if the head-mounted unit **602** has been lifted above the user’s nose bridge, or if the head-mounted unit **602** has been moved in these or other ways away from a desired position or range of positions. In general, registration observer **620** may be able to determine if head-mounted unit **602**, in general, and displays **220**, in particular, are properly positioned in front of the user’s eyes. In other words, the registration observer **620** may determine if a left display in display system **220** is appropriately aligned with the user’s left eye and a right display in display system **220** is appropriately aligned with the user’s right eye. The registration observer **620** may determine if the head-mounted unit **602** is properly positioned by determining if the head-mounted unit **602** is positioned and oriented within a desired range of positions and/or orientations relative to the user’s eyes.

[0139] In at least some embodiments, registration observer **620** may generate user feedback in the form of alerts, messages, or other content. Such feedback may be provided to the user to inform the user of any misalignment of the head-mounted unit **602**, along with optional feedback on how to correct the misalignment (such as a suggestion to adjust the head-mounted unit **602** in a particular manner).

[0140] Example registration observation and feedback techniques, which may be utilized by registration observer **620**, are described in U.S. patent application Ser. No. 15/717,747, filed Sep. 27, 2017 (Attorney Docket No. MLEAP.052A2), which is incorporated by reference herein in its entirety.

Example of an Eye Tracking Module

[0141] A detailed block diagram of an example eye tracking module **614** is shown in FIG. 7A. As shown in FIG. 7A, eye tracking module **614** may include a variety of different submodules, may provide a variety of different outputs, and may utilize a variety of available data in tracking the user’s eyes. As examples, eye tracking module **614** may utilize

available data including eye tracking extrinsics and intrinsics, such as the geometric arrangements of the eye tracking camera 324 relative to the light sources 326 and the head-mounted-unit 602; assumed eye dimensions 704 such as a typical distance of approximately 4.7 mm between a user's center of cornea curvature and the average center of rotation of the user's eye or typical distances between a user's center of rotation and center of perspective; and per-user calibration data 706 such as a particular user's interpupillary distance. Additional examples of extrinsics, intrinsics, and other information that may be employed by the eye tracking module 614 are described in U.S. patent application Ser. No. 15/497,726, filed Apr. 26, 2017 (Attorney Docket No. MLEAP.023A7), which is incorporated by reference herein in its entirety.

[0142] Image preprocessing module 710 may receive images from an eye camera such as eye camera 324 and may perform one or more preprocessing (i.e., conditioning) operations on the received images. As examples, image preprocessing module 710 may apply a Gaussian blur to the images, may down sample the images to a lower resolution, may applying an unsharp mask, may apply an edge sharpening algorithm, or may apply other suitable filters that assist with the later detection, localization, and labelling of glints, a pupil, or other features in the images from eye camera 324. The image preprocessing module 710 may apply a low-pass filter or a morphological filter such as an open filter, which can remove high-frequency noise such as from the pupillary boundary 516a (see FIG. 5), thereby removing noise that can hinder pupil and glint determination. The image preprocessing module 710 may output preprocessed images to the pupil identification module 712 and to the glint detection and labeling module 714.

[0143] Pupil identification module 712 may receive preprocessed images from the image preprocessing module 710 and may identify regions of those images that include the user's pupil. The pupil identification module 712 may, in some embodiments, determine the coordinates of the position, or coordinates, of the center, or centroid, of the user's pupil in the eye tracking images from camera 324. In at least some embodiments, pupil identification module 712 may identify contours in eye tracking images (e.g., contours of pupil iris boundary), identify contour moments (i.e., centers of mass), apply a starburst pupil detection and/or a canny edge detection algorithm, reject outliers based on intensity values, identify sub-pixel boundary points, correct for eye-camera distortion (i.e., distortion in images captured by eye camera 324), apply a random sample consensus (RANSAC) iterative algorithm to fit an ellipse to boundaries in the eye tracking images, apply a tracking filter to the images, and identify sub-pixel image coordinates of the user's pupil centroid. The pupil identification module 712 may output pupil identification data, which may indicate which regions of the preprocessing images module 712 identified as showing the user's pupil, to glint detection and labeling module 714. The pupil identification module 712 may provide the 2D coordinates of the user's pupil (i.e., the 2D coordinates of the centroid of the user's pupil) within each eye tracking image to glint detection module 714. In at least some embodiments, pupil identification module 712 may also provide pupil identification data of the same sort to coordinate system normalization module 718.

[0144] Pupil detection techniques, which may be utilized by pupil identification module 712, are described in U.S.

Patent Publication No. 2017/0053165, published Feb. 23, 2017 and in U.S. Patent Publication No. 2017/0053166, published Feb. 23, 2017, each of which is incorporated by reference herein in its entirety.

[0145] Glint detection and labeling module 714 may receive preprocessed images from module 710 and pupil identification data from module 712. Glint detection module 714 may use this data to detect and/or identify glints (i.e., reflections off of the user's eye of the light from light sources 326) within regions of the preprocessed images that show the user's pupil. As an example, the glint detection module 714 may search for bright regions within the eye tracking image, sometimes referred to herein as "blobs" or local intensity maxima, that are in the vicinity of the user's pupil. In at least some embodiments, the glint detection module 714 may rescale (e.g., enlarge) the pupil ellipse to encompass additional glints. The glint detection module 714 may filter glints by size and/or by intensity. The glint detection module 714 may also determine the 2D positions of each of the glints within the eye tracking image. In at least some examples, the glint detection module 714 may determine the 2D positions of the glints relative to the user's pupil, which may also be referred to as the pupil-glint vectors. Glint detection and labeling module 714 may label the glints and output the preprocessing images with labeled glints to the 3D cornea center estimation module 716. Glint detection and labeling module 714 may also pass along data such as preprocessed images from module 710 and pupil identification data from module 712.

[0146] Pupil and glint detection, as performed by modules such as modules 712 and 714, can use any suitable techniques. As examples, edge detection can be applied to the eye image to identify glints and pupils. Edge detection can be applied by various edge detectors, edge detection algorithms, or filters. For example, a Canny Edge detector can be applied to the image to detect edges such as in lines of the image. Edges may include points located along a line that correspond to the local maximum derivative. For example, the pupillary boundary 516a (see FIG. 5) can be located using a Canny edge detector. With the location of the pupil determined, various image processing techniques can be used to detect the "pose" of the pupil 116. Determining an eye pose of an eye image can also be referred to as detecting an eye pose of the eye image. The pose can also be referred to as the gaze, pointing direction, or the orientation of the eye. For example, the pupil may be looking leftwards towards an object, and the pose of the pupil could be classified as a leftwards pose. Other methods can be used to detect the location of the pupil or glints. For example, a concentric ring can be located in an eye image using a Canny Edge detector. As another example, an integro-differential operator can be used to find the pupillary or limbus boundaries of the iris. For example, the Daugman integro-differential operator, the Hough transform, or other iris segmentation techniques can be used to return a curve that estimates the boundary of the pupil or the iris.

[0147] 3D cornea center estimation module 716 may receive preprocessed images including detected glint data and pupil identification data from modules 710, 712, 714. 3D cornea center estimation module 716 may use these data to estimate the 3D position of the user's cornea. In some embodiments, the 3D cornea center estimation module 716 may estimate the 3D position of an eye's center of cornea curvature or a user's corneal sphere, i.e., the center of an

imaginary sphere having a surface portion generally coextensive with the user's cornea. The 3D cornea center estimation module **716** may provide data indicating the estimated 3D coordinates of the corneal sphere and/or user's cornea to the coordinate system normalization module **718**, the optical axis determination module **722**, and/or the light-field render controller **618**. Further details of the operation of the 3D cornea center estimation module **716** are provided herein in connection with FIGS. **8A-8E**. Techniques for estimating the positions of eye features such as a cornea or corneal sphere, which may be utilized by 3D cornea center estimation module **716** and other modules in the wearable systems of the present disclosure are discussed in U.S. patent application Ser. No. 15/497,726, filed Apr. 26, 2017 (Attorney Docket No. MLEAP.023A7), which is incorporated by reference herein in its entirety.

[0148] Coordinate system normalization module **718** may optionally (as indicated by its dashed outline) be included in eye tracking module **614**. Coordinate system normalization module **718** may receive data indicating the estimated 3D coordinates of the center of the user's cornea (and/or the center of the user's corneal sphere) from the 3D cornea center estimation module **716** and may also receive data from other modules. Coordinate system normalization module **718** may normalize the eye camera coordinate system, which may help to compensate for slippages of the wearable device (e.g., slippages of the head-mounted component from its normal resting position on the user's head, which may be identified by registration observer **620**). Coordinate system normalization module **718** may rotate the coordinate system to align the z-axis (i.e., the vergence depth axis) of the coordinate system with the cornea center (e.g., as indicated by the 3D cornea center estimation module **716**) and may translate the camera center (i.e., the origin of the coordinate system) to a predetermined distance away from the cornea center such as 30 mm (i.e., module **718** may enlarge or shrink the eye tracking image depending on whether the eye camera **324** was determined to be nearer or further than the predetermined distance). With this normalization process, the eye tracking module **614** may be able to establish a consistent orientation and distance in the eye tracking data, relatively independent of variations of headset positioning on the user's head. Coordinate system normalization module **718** may provide 3D coordinates of the center of the cornea (and/or corneal sphere), pupil identification data, and pre-processed eye tracking images to the 3D pupil center locator module **720**. Further details of the operation of the coordinate system normalization module **718** are provided herein in connection with FIGS. **9A-9C**.

[0149] 3D pupil center locator module **720** may receive data, in the normalized or the unnormalized coordinate system, including the 3D coordinates of the center of the user's cornea (and/or corneal sphere), pupil location data, and preprocessed eye tracking images. 3D pupil center locator module **720** may analyze such data to determine the 3D coordinates of the center of the user's pupil in the normalized or unnormalized eye camera coordinate system. The 3D pupil center locator module **720** may determine the location of the user's pupil in three-dimensions based on the 2D position of the pupil centroid (as determined by module **712**), the 3D position of the cornea center (as determined by module **716**), assumed eye dimensions **704** such as the size of the a typical user's corneal sphere and the typical distance from the cornea center to the pupil center, and optical

properties of eyes such as the index of refraction of the cornea (relative to the index of refraction of air) or any combination of these. Further details of the operation of the 3D pupil center locator module **720** are provided herein in connection with FIGS. **9D-9G**. Techniques for estimating the positions of eye features such as a pupil, which may be utilized by 3D pupil center locator module **720** and other modules in the wearable systems of the present disclosure are discussed in U.S. patent application Ser. No. 15/497,726, filed Apr. 26, 2017 (Attorney Docket No. MLEAP.023A7), which is incorporated by reference herein in its entirety.

[0150] Optical axis determination module **722** may receive data from modules **716** and **720** indicating the 3D coordinates of the center of the user's cornea and the user's pupil. Based on such data, the optical axis determination module **722** may identify a vector from the position of the cornea center (i.e., from the center of the corneal sphere) to the center of the user's pupil, which may define the optical axis of the user's eye. Optical axis determination module **722** may provide outputs specifying the user's optical axis to modules **724**, **728**, **730**, and **732**, as examples.

[0151] Center of rotation (CoR) estimation module **724** may receive data from module **722** including parameters of the optical axis of the user's eye (i.e., data indicating the direction of the optical axis in a coordinate system with a known relation to the head-mounted unit **602**). CoR estimation module **724** may estimate the center of rotation of a user's eye (i.e., the point around which the user's eye rotates when the user eye rotates left, right, up, and/or down). While eyes may not rotate perfectly around a singular point, assuming a singular point may be sufficient. In at least some embodiments, CoR estimation module **724** may estimate an eye's center of rotation by moving from the center of the pupil (identified by module **720**) or the center of curvature of the cornea (as identified by module **716**) toward the retina along the optical axis (identified by module **722**) a particular distance. This particular distance may be an assumed eye dimension **704**. As one example, the particular distance between the center of curvature of the cornea and the CoR may be approximately 4.7 mm. This distance may be varied for a particular user based on any relevant data including the user's age, sex, vision prescription, other relevant characteristics, etc.

[0152] In at least some embodiments, the CoR estimation module **724** may refine its estimate of the center of rotation of each of the user's eyes over time. As an example, as time passes, the user will eventually rotate their eyes (to look somewhere else, at something closer, further, or sometime left, right, up, or down) causing a shift in the optical axis of each of their eyes. CoR estimation module **724** may then analyze two (or more) optical axes identified by module **722** and locate the 3D point of intersection of those optical axes. The CoR estimation module **724** may then determine the center of rotation lies at that 3D point of intersection. Such a technique may provide for an estimate of the center of rotation, with an accuracy that improves over time. Various techniques may be employed to increase the accuracy of the CoR estimation module **724** and the determined CoR positions of the left and right eyes. As an example, the CoR estimation module **724** may estimate the CoR by finding the average point of intersection of optical axes determined for various different eye poses over time. As additional examples, module **724** may filter or average estimated CoR positions over time, may calculate a moving average of

estimated CoR positions over time, and/or may apply a Kalman filter and known dynamics of the eyes and eye tracking system to estimate the CoR positions over time. As a specific example, module 724 may calculate a weighted average of determined points of optical axes intersection and assumed CoR positions (such as 4.7 mm from an eye's center of cornea curvature), such that the determined CoR may slowly drift from an assumed CoR position (i.e., 4.7 mm behind an eye's center of cornea curvature) to a slightly different location within the user's eye over time as eye tracking data for the user is obtained and thereby enables per-user refinement of the CoR position.

[0153] Interpupillary distance (IPD) estimation module 726 may receive data from CoR estimation module 724 indicating the estimated 3D positions of the centers of rotation of the user's left and right eyes. IPD estimation module 726 may then estimate a user's IPD by measuring the 3D distance between the centers of rotation of the user's left and right eyes. In general, the distance between the estimated CoR of the user's left eye and the estimated CoR of the user's right eye may be roughly equal to the distance between the centers of a user's pupils, when the user is looking at optical infinity (i.e., the optical axes of the user's eyes are substantially parallel to one another), which is the typical definition of interpupillary distance (IPD). A user's IPD may be used by various components and modules in the wearable system. As example, a user's IPD may be provided to registration observer 620 and used in assessing how well the wearable device is aligned with the user's eyes (e.g., whether the left and right display lenses are properly spaced in accordance with the user's IPD). As another example, a user's IPD may be provided to vergence depth estimation module 728 and be used in determining a user's vergence depth. Module 726 may employ various techniques, such as those discussed in connection with CoR estimation module 724, to increase the accuracy of the estimated IPD. As examples, IPD estimation module 724 may apply filtering, averaging over time, weighted averaging including assumed IPD distances, Kalman filters, etc. as part of estimating a user's IPD in an accurate manner.

[0154] In some embodiments, IPD estimation module 726 may receive data from 3D pupil center locator module and/or 3D cornea center estimation module 716 indicating the estimated 3D positions of the user's pupils and/or corneas. IPD estimation module 726 may then estimate a user's IPD by reference to the distances between the pupils and corneas. In general, these distances will vary over time as a user rotates their eyes and changes the depth of their vergence. In some cases, the IPD estimation module 726 may look for the largest measured distance between the pupils and/or corneas, which should occur while the user is looking near optical infinity and should generally correspond to the user's interpupillary distance. In other cases, the IPD estimation module 726 may fit the measured distances between the user's pupils (and/or corneas) to a mathematical relationship of how a person's interpupillary distance changes as a function of their vergence depth. In some embodiments, using these or other similar techniques, the IPD estimation module 726 may be able to estimate the user's IPD even without an observation of the user looking at optical infinity (e.g., by extrapolating out from one or more observations in which the user was verging at distances closer than optical infinity).

[0155] Vergence depth estimation module 728 may receive data from various modules and submodules in the eye tracking module 614 (as shown in connection with FIG. 7A). In particular, vergence depth estimation module 728 may employ data indicating estimated 3D positions of pupil centers (e.g., as provided by module 720 described above), one or more determined parameters of optical axes (e.g., as provided by module 722 described above), estimated 3D positions of centers of rotation (e.g., as provided by module 724 described above), estimated IPD (e.g., Euclidean distance(s) between estimated 3D positions of centers of rotations) (e.g., as provided by module 726 described above), and/or one or more determined parameters of optical and/or visual axes (e.g., as provided by module 722 and/or module 730 described below). Vergence depth estimation module 728 may detect or otherwise obtain a measure of a user's vergence depth, which may be the distance from the user at which the user's eyes are focused. As examples, when the user is looking at an object three feet in front of them, the user's left and right eyes have a vergence depth of three feet; and, while when the user is looking at a distant landscape (i.e., the optical axes of the user's eyes are substantially parallel to one another such that the distance between the centers of the user's pupils may be roughly equal to the distance between the centers of rotation of the user's left and right eyes), the user's left and right eyes have a vergence depth of infinity. In some implementations, the vergence depth estimation module 728 may utilize data indicating the estimated centers of the user's pupils (e.g., as provided by module 720) to determine the 3D distance between the estimated centers of the user's pupils. The vergence depth estimation module 728 may obtain a measure of vergence depth by comparing such a determined 3D distance between pupil centers to estimated IPD (e.g., Euclidean distance(s) between estimated 3D positions of centers of rotations) (e.g., as indicated by module 726 described above). In addition to the 3D distance between pupil centers and estimated IPD, the vergence depth estimation module 728 may utilize known, assumed, estimated, and/or determined geometries to calculate vergence depth. As an example, module 728 may combine 3D distance between pupil centers, estimated IPD, and 3D CoR positions in a trigonometric calculation to estimate (i.e., determine) a user's vergence depth. Indeed, an evaluation of such a determined 3D distance between pupil centers against estimated IPD may serve to indicate a measure of the user's current vergence depth relative to optical infinity. In some examples, the vergence depth estimation module 728 may simply receive or access data indicating an estimated 3D distance between the estimated centers of the user's pupils for purposes of obtaining such a measure of vergence depth. In some embodiments, the vergence depth estimation module 728 may estimate vergence depth by comparing a user's left and right optical axis. In particular, vergence depth estimation module 728 may estimate vergence depth by locating the distance from a user at which the user's left and right optical axes intersect (or where projections of the user's left and right optical axes on a plane such as a horizontal plane intersect). Module 728 may utilize a user's IPD in this calculation, by setting the zero depth to be the depth at which the user's left and right optical axes are separated by the user's IPD. In at least some embodiments, vergence depth estimation module 728 may determine vergence depth by triangulating eye tracking data together with known or derived spatial relationships.

[0156] In some embodiments, vergence depth estimation module 728 may estimate a user's vergence depth based on the intersection of the user's visual axes (instead of their optical axes), which may provide a more accurate indication of the distance at which the user is focused on. In at least some embodiments, eye tracking module 614 may include optical to visual axis mapping module 730. As discussed in further detail in connection with FIG. 10, a user's optical and visual axis are generally not aligned. A visual axis is the axis along which a person is looking, while an optical axis is defined by the center of that person's lens and pupil, and may go through the center of the person's retina. In particular, a user's visual axis is generally defined by the location of the user's fovea, which may be offset from the center of a user's retina, thereby resulting in different optical and visual axis. In at least some of these embodiments, eye tracking module 614 may include optical to visual axis mapping module 730. Optical to visual axis mapping module 730 may correct for the differences between a user's optical and visual axis and provide information on the user's visual axis to other components in the wearable system, such as vergence depth estimation module 728 and light-field render controller 618. In some examples, module 730 may use assumed eye dimensions 704 including a typical offset of approximately 5.2° inwards (nasally, towards a user's nose) between an optical axis and a visual axis. In other words, module 730 may shift a user's left optical axis (nasally) rightwards by 5.2° towards the nose and a user's right optical axis (nasally) leftwards by 5.2° towards the nose in order to estimate the directions of the user's left and right optical axes. In other examples, module 730 may utilize per-user calibration data 706 in mapping optical axes (e.g., as indicated by module 722 described above) to visual axes. As additional examples, module 730 may shift a user's optical axes nasally by between 4.0° and 6.5°, by between 4.5° and 6.0°, by between 5.0° and 5.4°, etc., or any ranges formed by any of these values. In some arrangements, the module 730 may apply a shift based at least in part upon characteristics of a particular user such as their age, sex, vision prescription, or other relevant characteristics and/or may apply a shift based at least in part upon a calibration process for a particular user (i.e., to determine a particular user's optical-visual axis offset). In at least some embodiments, module 730 may also shift the origins of the left and right optical axes to correspond with the user's CoP (as determined by module 732) instead of the user's CoR.

[0157] Optional center of perspective (CoP) estimation module 732, when provided, may estimate the location of the user's left and right centers of perspective (CoP). A CoP may be a useful location for the wearable system and, in at least some embodiments, is a position just in front of a pupil. In at least some embodiments, CoP estimation module 732 may estimate the locations of a user's left and right centers of perspective based on the 3D location of a user's pupil center, the 3D location of a user's center of cornea curvature, or such suitable data or any combination thereof. As an example, a user's CoP may be approximately 5.01 mm in front of the center of cornea curvature (i.e., 5.01 mm from the corneal sphere center in a direction that is towards the eye's cornea and that is along the optical axis) and may be approximately 2.97 mm behind the outer surface of a user's cornea, along the optical or visual axis. A user's center of perspective may be just in front of the center of their pupil. As examples, a user's CoP may be less than approximately

2.0 mm from the user's pupil, less than approximately 1.0 mm from the user's pupil, or less than approximately 0.5 mm from the user's pupil or any ranges between any of these values. As another example, the center of perspective may correspond to a location within the anterior chamber of the eye. As other examples, the CoP may be between 1.0 mm and 2.0 mm, about 1.0 mm, between 0.25 mm and 1.0 mm, between 0.5 mm and 1.0 mm, or between 0.25 mm and 0.5 mm.

[0158] The center of perspective described herein (as a potentially desirable position for a pinhole of a render camera and an anatomical position in a user's eye) may be a position that serves to reduce and/or eliminate undesired parallax shifts. In particular, the optical system of a user's eye is very roughly equivalent to theoretical system formed by a pinhole in front of a lens, projecting onto a screen, with the pinhole, lens, and screen roughly corresponding to a user's pupil/iris, lens, and retina, respectively. Moreover, it may be desirable for there to be little or no parallax shift when two point light sources (or objects) at different distances from the user's eye are rigidly rotated about the opening of the pinhole (e.g., rotated along radii of curvature equal to their respective distance from the opening of the pinhole). Thus, it would seem that the CoP should be located at the center of the pupil of an eye (and such a CoP may be used in some embodiments). However, the human eye includes, in addition to the lens and pinhole of the pupil, a cornea that imparts additional optical power to light propagating toward the retina). Thus, the anatomical equivalent of the pinhole in the theoretical system described in this paragraph may be a region of the user's eye positioned between the outer surface of the cornea of the user's eye and the center of the pupil or iris of the user's eye. For instance, the anatomical equivalent of the pinhole may correspond to a region within the anterior chamber of a user's eye. For various reasons discussed herein, it may be desired to set the CoP to such a position within the anterior chamber of the user's eye. The derivation and significance of the CoP are discussed in additional detail in Appendix (Parts I and II), which forms part of this application.

[0159] As discussed above, eye tracking module 614 may provide data, such as estimated 3D positions of left and right eye centers of rotation (CoR), vergence depth, left and right eye optical axis, 3D positions of a user's eye, 3D positions of a user's left and right centers of cornea curvature, 3D positions of a user's left and right pupil centers, 3D positions of a user's left and right center of perspective, a user's IPD, etc., to other components, such as light-field render controller 618 and registration observer 620, in the wearable system. Eye tracking module 614 may also include other submodules that detect and generate data associated with other aspects of a user's eye. As examples, eye tracking module 614 may include a blink detection module that provides a flag or other alert whenever a user blinks and a saccade detection module that provides a flag or other alert whenever a user's eye saccades (i.e., quickly shifts focus to another point). Example of a Render Controller

[0160] A detailed block diagram of an example light-field render controller 618 is shown in FIG. 7B. As shown in FIGS. 6 and 7B, render controller 618 may receive eye tracking information from eye tracking module 614 and may provide outputs to render engine 622, which may generate images to be displayed for viewing by a user of the wearable system. As examples, render controller 618 may receive a

vergence depth, left and right eye centers of rotation (and/or centers of perspective), and other eye data such as blink data, saccade data, etc.

[0161] Depth plane selection module 750 may receive vergence depth information and other eye data and, based on such data, may cause render engine 622 to convey content to a user with a particular depth plane (i.e., at a particular accommodation or focal distance). As discussed in connection with FIG. 4, a wearable system may include a plurality of discrete depth planes formed by a plurality of waveguides, each conveying image information with a varying level of wavefront curvature. In some embodiments, a wearable system may include one or more variable depth planes, such as an optical element that conveys image information with a level of wavefront curvature that varies over time. In these and other embodiments, depth plane selection module 750 may cause render engine 622 to convey content to a user at a selected depth (i.e., cause render engine 622 to direct display 220 to switch depth planes), based in part of the user's vergence depth. In at least some embodiments, depth plane selection module 750 and render engine 622 may render content at different depths and also generate and/or provide depth plane selection data to display hardware such as display 220. Display hardware such as display 220 may perform an electrical depth plane switching in response to depth plane selection data (which may be control signals) generated by and/or provided by modules such as depth plane selection module 750 and render engine 622.

[0162] In general, it may be desirable for depth plane selection module 750 to select a depth plane matching the user's current vergence depth, such that the user is provided with accurate accommodation cues. However, it may also be desirable to switch depth planes in a discreet and unobtrusive manner. As examples, it may be desirable to avoid excessive switching between depth planes and/or it may be desire to switch depth planes at a time when the user is less likely to notice the switch, such as during a blink or eye saccade.

[0163] Hysteresis band crossing detection module 752 may help to avoid excessive switching between depth planes, particularly when a user's vergence depth fluctuates at the midpoint or transition point between two depth planes. In particular, module 752 may cause depth plane selection module 750 to exhibit hysteresis in its selection of depth planes. As an example, modules 752 may cause depth plane selection module 750 to switch from a first farther depth plane to a second closer depth plane only after a user's vergence depth passes a first threshold. Similarly, module 752 may cause depth plane selection module 750 (which may in turn direct displays such as display 220) to switch to the first farther depth plane only after the user's vergence depth passes a second threshold that is farther from the user than the first threshold. In the overlapping region between the first and second thresholds, module 750 may cause depth plane selection module 750 to maintain whichever depth plane is currently select as the selected depth plane, thus avoiding excessive switching between depth planes.

[0164] Ocular event detection module 750 may receive other eye data from the eye tracking module 614 of FIG. 7A and may cause depth plane selection module 750 to delay some depth plane switches until an ocular event occurs. As an example, ocular event detection module 750 may cause depth plane selection module 750 to delay a planned depth

plane switch until a user blink is detected; may receive data from a blink detection component in eye tracking module 614 that indicates when the user is currently blinking; and, in response, may cause depth plane selection module 750 to execute the planned depth plane switch during the blink event (such by causing module 750 to direct display 220 to execute the depth plane switch during the blink event). In at least some embodiments, the wearable system may be able to shift content onto a new depth plane during a blink event such that the user is unlikely to perceive the shift. As another example, ocular event detection module 750 may delay planned depth plane switches until an eye saccade is detected. As discussed in connection with eye blinks, such as an arrangement may facilitate the discretely shifting of depth planes.

[0165] If desired, depth plane selection module 750 may delay planned depth plane switches only for a limited period of time before executing the depth plane switch, even in the absence of an ocular event. Similarly, depth plane selection module 750 may execute a depth plane switch when the user's vergence depth is substantially outside of a currently-selected depth plane (i.e., when the user's vergence depth has exceeded a predetermined threshold beyond the regular threshold for a depth plane switch), even in the absence of an ocular event. These arrangements may help ensure that ocular event detection module 754 does not indefinitely delay depth plane switches and does not delay depth plane switches when a large accommodation error is present. Further details of the operation of depth plane selection module 750, and how the module may time depth plane switches, are provided herein in connection with FIG. 13.

[0166] Render camera controller 758 may provide information to render engine 622 indicating where the user's left and right eyes are. Render engine 622 may then generate content by simulating cameras at the positions of the user's left and right eyes and generating content based on the perspectives of the simulated cameras. As discussed above, the render camera is a simulated camera for use in rendering virtual image content possibly from a database of objects in a virtual world. The objects may have locations and orientations relative to the user or wearer and possibly relative to real objects in the environment surrounding the user or wearer. The render camera may be included in a render engine to render virtual images based on the database of virtual objects to be presented to the eye. The virtual images may be rendered as if taken from the perspective the user or wearer. For example, the virtual images may be rendered as if captured by a camera (corresponding to the "render camera") having an aperture, lens, and detector viewing the objects in the virtual world. The virtual images are taken from the perspective of such a camera having a position of the "render camera." For example, the virtual images may be rendered as if captured from the perspective of a camera having a specific location with respect to the user's or wearer's eye so as to provide images that appear to be from the perspective of the user or wearer. In some implementations, the images are rendered as if captured from the perspective of a camera having an aperture at a specific location with respect to the user's or wearer's eye (such as the center of perspective or center of rotation as discussed herein, or elsewhere).

[0167] Render camera controller 758 may determine the positions of the left and right cameras based on the left and right eye centers of rotation (CoR), determined by CoR

estimation module 724, and/or based on the left and right eye centers of perspective (CoP), determined by CoP estimation module 732. In some embodiments, render camera controller 758 may switch between the CoR and CoP locations based on various factors. As examples, the render camera controller 758 may, in various modes, register the render camera to the CoR locations at all times, register the render camera to the CoP locations at all times, toggle or discretely switch between registering the render camera to the CoR locations and registering the render camera to the CoP locations over time based on various factors, or dynamically register the render camera to any of a range of different positions along the optical (or visual) axis between the CoR and CoP locations over time based on various factors. The CoR and CoP positions may optionally pass through smoothing filter 756 (in any of the aforementioned modes for render camera positioning) which may average the CoR and CoP locations over time to reduce noise in these positions and prevent jitter in the render simulated render cameras.

[0168] In at least some embodiments, the render camera may be simulated as a pinhole camera with the pinhole disposed at the position of the estimated CoR or CoP identified by eye tracking module 614. As the CoP is offset from the CoR, the location of the render camera and its pinhole both shift as the user's eye rotates, whenever the render camera's position is based on a user's CoP. In contrast, whenever the render camera's position is based on a user's CoR, the location of the render camera's pinhole does not move with eye rotations, although the render camera (which is behind the pinhole) may, in some embodiments, move with eye rotation. In other embodiments where the render camera's position is based on a user's CoR, the render camera may not move (i.e., rotate) with a user's eye.

Example of a Registration Observer

[0169] A block diagram of an example registration observer 620 is shown in FIG. 7C. As shown in FIGS. 6, 7A, and 7C, registration observer 620 may receive eye tracking information from eye tracking module 614 (FIGS. 6 and 7A). As examples, registration observer 620 may receive information on a user's left and right eye centers of rotation (e.g., the three-dimensional positions of the user's left and right eye centers of rotations, which may be on a common coordinate system or have a common frame of reference with the head-mounted display system 600). As other examples, registration observer 620 may receive display extrinsics, fit tolerances, and an eye-tracking valid indicator. The display extrinsics may include information on the display (e.g., display 200 of FIG. 2) such as the field of view of the display, the size of one or more display surfaces, and the positions of the display surfaces relative to the head-mounted display system 600. The fit tolerances may include information on display registration volumes, which may indicate how far the user's left and right eyes may move from nominal positions before display performance is impacted. In addition, the fit tolerances may indicate the amount of display performance impact that is expected as a function of the positions of the user's eyes.

[0170] As shown in FIG. 7C, registration observer 620 may include a 3D positional fit module 770. The positional fit module 770 may obtain and analyze various pieces of data including, as examples, a left eye center of rotation 3D position (e.g., CoR Left), a right eye center of rotation 3D

position (e.g., CoR Right), display extrinsics, and fit tolerances. The 3D positional fit module 770 may determine how far the user's left and right eyes are from the respective left and right eye nominal positions (e.g., may calculate 3D left error and 3D right error) and may provide the error distances (e.g., 3D left error and 3D right error) to device 3D fit module 772.

[0171] 3D positional fit module 770 may also compare the error distances to the display extrinsics and the fit tolerances to determine if the user's eye are within a nominal volume, a partially-degraded volume (e.g., a volume in which the performance of display 220 is partially degraded), or in a fully degraded or nearly fully degraded volume (e.g., a volume in which display 220 is substantially unable to provide content to the user's eyes). In at least some embodiments, 3D positional fit module 770 or 3D fit module 772 may provide an output qualitatively describing the fit of the HMD on the user, such as the Quality of Fit output shown in FIG. 7C. As an example, module 770 may provide an output indicating whether the current fit of the HMD on the user is good, marginal, or failed. A good fit may correspond to a fit that enables the user to view at least a certain percentage of the image (such as 90%), a marginal fit may enable the user to view at least a lower percentage of the image (such as 80%), while a failed fit may be a fit in which only an even lower percentage of the image is visible to the user.

[0172] As another example, the 3D positional fit module 770 and/or device 3D fit module 772 may calculate a visible area metric, which may be a percentage of the overall area (or pixels) of images display by display 220 that are visible to the user. Modules 770 and 772 may calculate the visible area metric by evaluating the positions of the user's left and right eyes (e.g., which may be based on the centers of rotation of the user's eyes) relative to display 220 and using one or more models (e.g., a mathematical or geometric model), one or more look-up tables, or other techniques or combinations of these and other techniques to determine what percentage of the images are visible to the user as a function of the positions of the user's eyes. Additionally, modules 770 and 772 may determine which regions or portions of the images display by display 220 are expected to be visible to the user as a function of the positions of the user's eyes.

[0173] Registration observer 620 may also include a device 3D fit module 772. Module 772 may receive data from 3D positional fit module 770 and may also receive an eye tracking valid indicator, which may be provided by eye tracking module 614 and may indicate whether the eye tracking system is currently tracking the positions of the user's eyes or if eye tracking data is unavailable or in an error condition (e.g., determined to be no reliable). Device 3D fit module 772 may, if desired, modify quality of fit data received from 3D positional fit module 770 depending on the state of the eye tracking valid data. For example, if the data from the eye tracking system is indicated to not be available or to have an error, the device 3D fit module 772 may provide a notification that there is an error and/or not provide output to the user regarding fit quality or fit errors.

[0174] In at least some embodiments, registration observer 620 may provide feedback to users on the quality of fit as well as details of the nature and magnitude of the error. As examples, the head-mounted display system may provide feedback to the user during calibration or fitting processes

(e.g., as part of a setup procedure) and may provide feedback during operation (e.g., if the fit degrades due to slippage, the registration observer 620 may prompt the user to readjust the head-mounted display system). In some embodiments, the registration analysis may be performed automatically (e.g., during use of the head-mounted display system) and the feedback may be provided without user input. These are merely illustrative examples.

Example of Locating a User's Cornea with an Eye Tracking System

[0175] FIG. 8A is a schematic diagram of an eye showing the eye's corneal sphere. As shown in FIG. 8A, a user's eye 810 may have a cornea 812, a pupil 822, and a lens 820. The cornea 812 may have an approximately spherical shape, shown by corneal sphere 814. Corneal sphere 814 may have a center point 816, also referred to as a corneal center, and a radius 818. The semispherical cornea of a user's eye may curve around the corneal center 816.

[0176] FIGS. 8B-8E illustrate an example of locating a user's corneal center 816 using 3D cornea center estimation module 716 and eye tracking module 614.

[0177] As shown in FIG. 8B, 3D cornea center estimation module 716 may receive an eye tracking image 852 that includes a corneal glint 854. The 3D cornea center estimation module 716 may then simulate, in an eye camera coordinate system 850, the known 3D positions of the eye camera 324 and light source 326 (which may be based on data in eye tracking extrinsics & intrinsics database 702, assumed eye dimensions database 704, and/or per-user calibration data 706) in order to cast a ray 856 in the eye camera coordinate system. In at least some embodiments, the eye camera coordinate system 850 may have its origin at the 3D position of the eye tracking camera 324.

[0178] In FIG. 8C, 3D cornea center estimation module 716 simulates a corneal sphere 814a (which may be based on assumed eye dimensions from database 704) and corneal curvature center 816a at a first position. The 3D cornea center estimation module 716 may then check to see whether the corneal sphere 814a would properly reflect light from the light source 326 to the glint position 854. As shown in FIG. 8C, the first position is not a match as the ray 860a does not intersect light source 326.

[0179] Similarly in FIG. 8D, 3D cornea center estimation module 716 simulates a corneal sphere 814b and corneal curvature center 816b at a second position. The 3D cornea center estimation module 716 then checks to see whether the corneal sphere 814b properly reflects light from the light source 326 to the glint position 854. As shown in FIG. 8D, the second position is also not a match.

[0180] As shown in FIG. 8E, the 3D cornea center estimation module 716 eventually is able to determine the correct position of the corneal sphere is corneal sphere 814c and corneal curvature center 816c. The 3D cornea center estimation module 716 confirms the illustrated position is correct by checking that light from source 326 will properly reflect off of the corneal sphere and be imaged by camera 324 at the correct location of glint 854 on image 852. With this arrangement and with the known 3D positions of the light source 326, the camera 324, and the optical properties of the camera (focal length, etc.), the 3D cornea center estimation module 716 can determine the 3D location of the cornea's center of curvature 816 (relative to the wearable system).

[0181] The processes described herein in connection with at least FIGS. 8C-8E may effectively be an iterative, repetitive, or optimization process to identify the 3D position of the user's cornea center. As such, any of a plurality of techniques (e.g., iterative, optimization techniques, etc.) may be used to efficiently and quickly prune or reduce the search space of possible positions. Moreover, in some embodiments, the system may include two, three, four, or more light sources such as light source 326 and some of all of these light sources may be disposed at different positions, resulting in multiple glints such as glint 854 located at different positions on image 852 and multiple rays such as ray 856 having different origins and directions. Such embodiments may enhance the accuracy of the 3D cornea center estimation module 716, as the module 716 may seek to identify a cornea position that results in some or all of the glints & rays being properly reflected between their respective light sources and their respective positions on image 852. In other words and in these embodiments, the positions of some or all of the light sources may be relied upon in the 3D cornea position determination (e.g., iterative, optimization techniques, etc.) processes of FIGS. 8B-8E.

Example of Normalizing the Coordinate System of Eye Tracking Images

[0182] FIGS. 9A-9C illustrate an example normalization of the coordinate system of eye tracking images, by a component in the wearable system such as coordinate system normalization module 718 of FIG. 7A. Normalizing the coordinate system of eye tracking images relative to a user's pupil location may compensate for slippage of the wearable system relative to a user's face (i.e., headset slippage) and such normalization may establish a consistent orientation and distance between eye tracking images and a user's eyes.

[0183] As shown in FIG. 9A, coordinate system normalization module 718 may receive estimated 3D coordinates 900 of a user's center of corneal rotation and may receive un-normalized eye tracking images such as image 852. Eye tracking image 852 and coordinates 900 may be in an un-normalized coordinate system 850 that is based on the location of eye tracking camera 324, as an example.

[0184] As a first normalization step, coordinate system normalization module 718 may rotate coordinate system 850 into rotated coordinate system 902, such that the z-axis (i.e., the vergence depth axis) of the coordinate system may be aligned with a vector between the origin of the coordinate system and cornea center of curvature coordinates 900, as shown in FIG. 9B. In particular, coordinate system normalization module 718 may rotate eye tracking image 850 into rotated eye tracking image 904, until the coordinates 900 of the user's corneal center of curvature are normal to the plane of the rotated image 904.

[0185] As a second normalization step, coordinate system normalization module 718 may translate rotated coordinate system 902 into normalized coordinate system 910, such that cornea center of curvature coordinates 900 are a standard, normalized distance 906 from the origin of normalized coordinate system 910, as shown in FIG. 9C. In particular, coordinate system normalization module 718 may translate rotated eye tracking image 904 into normalized eye tracking image 912. In at least some embodiments, the standard, normalized distance 906 may be approximately 30 millimeters. If desired, the second normalization step may be performed prior to the first normalization step.

Example of Locating a User's Pupil Centroid with an Eye Tracking System

[0186] FIGS. 9D-9G illustrate an example of locating a user's pupil center (i.e., the center of a user's pupil 822 as shown in FIG. 8A) using 3D pupil center locator module 720 and eye tracking module 614.

[0187] As shown in FIG. 9D, 3D pupil center locator module 720 may receive a normalized eye tracking image 912 that includes a pupil centroid 913 (i.e., a center of a user's pupil as identified by pupil identification module 712). The 3D pupil center locator module 720 may then simulate the normalized 3D position 910 of eye camera 324 to cast a ray 914 in the normalized coordinate system 910, through the pupil centroid 913.

[0188] In FIG. 9E, 3D pupil center locator module 720 may simulate a corneal sphere such as corneal sphere 901 having center of curvature 900 based on data from 3D cornea center estimation module 716 (and as discussed in more detail in connection with FIGS. 8B-8E). As an example, the corneal sphere 901 may be positioned in the normalized coordinate system 910 based on the location of the center of curvature 816c identified in connection with FIG. 8E and based on the normalization processes of FIGS. 9A-9C. Additionally, 3D pupil center locator module 720 may identify a first intersection 916 between ray 914 (i.e., a ray between the origin of normalized coordinate system 910 and the normalized location of a user's pupil) and the simulated cornea, as shown in FIG. 9E.

[0189] As shown in FIG. 9F, 3D pupil center locator module 720 may determine pupil sphere 918 based on corneal sphere 901. Pupil sphere 918 may share a common center of curvature with corneal sphere 901, but have a small radius. 3D pupil center locator module 720 may determine a distance between cornea center 900 and pupil sphere 918 (i.e., a radius of pupil sphere 918) based on a distance between the corneal center and the pupil center. In some embodiments, the distance between a pupil center and a corneal center of curvature may be determined from assumed eye dimensions 704 of FIG. 7A, from eye tracking extrinsics and intrinsics database 702, and/or from per-user calibration data 706. In other embodiments, the distance between a pupil center and a corneal center of curvature may be determined from per-user calibration data 706 of FIG. 7A.

[0190] As shown in FIG. 9G, 3D pupil center locator module 720 may locate the 3D coordinates of a user's pupil center based on variety of inputs. As examples, the 3D pupil center locator module 720 may utilize the 3D coordinates and radius of the pupil sphere 918, the 3D coordinates of the intersection 916 between a simulated cornea sphere 901 and a ray 914 associated with a pupil centroid 913 in a normalized eye tracking image 912, information on the index of refraction of a cornea, and other relevant information such as the index of refraction of air (which may be stored in eye tracking extrinsics & intrinsics database 702) to determine the 3D coordinates of the center of a user's pupil. In particular, the 3D pupil center locator module 720 may, in simulation, bend ray 916 into refracted ray 922 based on refraction difference between air (at a first index of refraction of approximately 1.00) and corneal material (at a second index of refraction of approximately 1.38). After taking into account refraction caused by the cornea, 3D pupil center locator module 720 may determine the 3D coordinates of the first intersection 920 between refracted ray 922 and pupil

sphere 918. 3D pupil center locator module 720 may determine that a user's pupil center 920 is located at approximately the first intersection 920 between refracted ray 922 and pupil sphere 918. With this arrangement, the 3D pupil center locator module 720 can determine the 3D location of the pupil center 920 (relative to the wearable system), in the normalized coordinate system 910. If desired, the wearable system can un-normalize the coordinates of the pupil center 920 into the original eye camera coordinate system 850. The pupil center 920 may be used together with the corneal curvature center 900 to determine, among other things, a user's optical axis using optical axis determination module 722 and a user's vergence depth by vergence depth estimation module 728.

Example of Differences Between Optical and Visual Axes

[0191] As discussed in connection with optical to visual mapping module 730 of FIG. 7A, a user's optical and visual axes are generally not aligned, due in part to a user's visual axis being defined by their fovea and that foveae are not generally in the center of a person's retina. Thus, when a person desires to concentrate on a particular object, the person aligns their visual axis with that object to ensure that light from the object falls on their fovea while their optical axis (defined by the center of their pupil and center of curvature of their cornea) is actually slightly offset from that object. FIG. 10 is an example of an eye 1000 illustrating the eye's optical axis 1002, the eye's visual axis 1004, and the offset between these axes. Additionally, FIG. 10 illustrates the eye's pupil center 1006, the eye's center of cornea curvature 1008, and the eye's average center of rotation (CoR) 1010. In at least some populations, the eye's center of cornea curvature 1008 may lie approximately 4.7 mm in front, as indicated by dimension 1012, of the eye's average center of rotation (CoR) 1010. Additionally, the eye's center of perspective 1014 may lie approximately 5.01 mm in front of the eye's center of cornea curvature 1008, about 2.97 mm behind the outer surface 1016 of the user's cornea, and/or just in front of the user's pupil center 1006 (e.g., corresponding to a location within the anterior chamber of eye 1000). As additional examples, dimension 1012 may be between 3.0 mm and 7.0 mm, between 4.0 and 6.0 mm, between 4.5 and 5.0 mm, or between 4.6 and 4.8 mm or any ranges between any values and any values in any of these ranges. The eye's center of perspective (CoP) 1014 may be a useful location for the wearable system as, in at least some embodiments, registering a render camera at the CoP may help to reduce or eliminate parallax artifacts.

[0192] FIG. 10 also illustrates such a within a human eye 1000 with which the pinhole of a render camera can be aligned. As shown in FIG. 10, the pinhole of a render camera may be registered with a location 1014 along the optical axis 1002 or visual axis 1004 of the human eye 1000 closer to the outer surface of the cornea than both (a) the center of the pupil or iris 1006 and (b) the center of cornea curvature 1008 of the human eye 1000. For example, as shown in FIG. 10, the pinhole of a render camera may be registered with a location 1014 along the optical axis 1002 of the human eye 1000 that is about 2.97 millimeters rearward from the outer surface of the cornea 1016 and about 5.01 millimeters forward from the center of cornea curvature 1008. The location 1014 of the pinhole of the render camera and/or the anatomical region of the human eye 1000 to which the location 1014 corresponds can be seen as representing the

center of perspective of the human eye **1000**. The optical axis **1002** of the human eye **1000** as shown in FIG. **10** represents the most direct line through the center of cornea curvature **1008** and the center of the pupil or iris **1006**. The visual axis **1004** of the human eye **1000** differs from the optical axis **1002**, as it represents a line extending from the fovea of the human eye **1000** to the center of the pupil or iris **1006**.

Example Processes of Rendering Content and Checking Registration Based on Eye Tracking

[0193] FIG. **11** is a process flow diagram of an example method **1100** for using eye tracking in rendering content and providing feedback on registration in a wearable device. The method **1100** may be performed by the wearable system described herein. Embodiments of the method **1100** can be used by the wearable system to render content and provide feedback on registration (i.e., fit of the wearable device to the user) based on data from an eye tracking system.

[0194] At block **1110**, the wearable system may capture images of a user's eye or eyes. The wearable system may capture eye images using one or more eye cameras **324**, as shown at least in the example of FIG. **3**. If desired, the wearable system may also include one or more light sources **326** configured to shine IR light on a user's eyes and produce corresponding glints in the eye images captured by eye cameras **324**. As discussed herein, the glints may be used by an eye tracking module **614** to derive various pieces of information about a user's eye including where the eye is looking.

[0195] At block **1120**, the wearable system may detect glints and pupils in the eye images captured in block **1110**. As an example, block **1120** may include processing the eye images by glint detection & labeling module **714** to identify the two-dimensional positions of glints in the eye images and processing the eye images by pupil identification module **712** to identify the two-dimensional positions of pupils in the eye images.

[0196] At block **1130**, the wearable system may estimate the three-dimensional positions of a user's left and right corneas relative to the wearable system. As an example, the wearable system may estimate the positions of the center of curvature of a user's left and right corneas as well as the distances between those centers of curvature and the user's left and right corneas. Block **1130** may involve 3D cornea center estimation module **716** identifying the position of the centers of curvature as described herein at least in connection with FIGS. **7A** and **8A-8E**.

[0197] At block **1140**, the wearable system may estimate the three-dimensional positions of a user's left and right pupil centers relative to the wearable system. As an example, the wearable system and 3D pupil center locator module **720** in particular, may estimate the positions of the user's left and right pupil centers as described at least in connection with FIGS. **7A** and **9D-9G**, as part of block **1140**.

[0198] At block **1150**, the wearable system may estimate the three-dimensional positions of a user's left and right centers or rotation (CoR) relative to the wearable system. As an example, the wearable system and CoR estimation module **724** in particular, may estimate the positions of the CoR for the user's left and right eyes as described at least in connection with FIGS. **7A** and **10**. As a particular example,

the wearable system may find the CoR of an eye by walking back along the optical axis from the center of curvature of a cornea towards the retina.

[0199] At block **1160**, the wearable system may estimate a user's IPD, vergence depth, center of perspective (CoP), optical axis, visual axis, and other desired attributes from eye tracking data. As examples, IPD estimation module **726** may estimate a user's IPD by comparing the 3D positions of the left and right CoRs, vergence depth estimation module **728** may estimate a user's depth by finding an intersection (or near intersection) of the left and right optical axes or an intersection of the left and right visual axes, optical axis determination module **722** may identify the left and right optical axes over time, optical to visual axis mapping module **730** may identify the left and right visual axes over time, and the CoP estimation module **732** may identify the left and right centers of perspective, as part of block **1160**.

[0200] At block **1170**, the wearable system may render content and may, optionally, provide feedback on registration (i.e., fit of the wearable system to the user's head) based in part on the eye tracking data identified in blocks **1120-1160**. As an example, the wearable system may identify a suitable location for a render camera and then generate content for a user based on the render camera's location, as discussed in connection with light-field render controller **618**, FIG. **7B**, and render engine **622**. As another example, the wearable system may determine if it is properly fitted to the user, or has slipped from its proper location relative to the user, and may provide optional feedback to the user indicating whether the fit of the device needs adjustment, as discussed in connection with registration observer **620**. In some embodiments, the wearable system may adjust rendered content based on improper or less than ideal registration in an attempt to reduce, minimize or compensate for the effects of improper or mis-registration.

Example of a Registration Coordinate System

[0201] FIGS. **12A-12B** illustrate an example eye position coordinate system, which may be used for defining three-dimensional positions of a user's left and right eyes relative to the display of the wearable system described herein. As examples, the coordinate system may include axis x, y, and z. Axis z of the coordinate system may correspond to depth, such the distance between the plane a user's eyes lie in and the plane that display **220** lies in (e.g., the direction normal to the plane of the front of a user's face). Axis x of the coordinate system may correspond to a left-right direction, such as the distance between the users left and right eyes. Axis y of the coordinate system may correspond to an up-down direction, which may be a vertical direction when the user is upright.

[0202] FIG. **12A** illustrates a side view of a user's eye **1200** and a display surface **1202** (which may be a part of display **220** of FIG. **2**), while FIG. **12B** illustrates a top down view of the user's eye **1200** and the display surface **1202**. Display surface **1202** may be located in front of the user's eyes and may output image light to the user's eyes. As an example, display surface **1202** may comprise one or more out-coupling light elements, active or pixel display elements, and may be part of a stack of waveguides, such as stacked waveguide assembly **480** of FIG. **4**. In some embodiments, the display surface **1202** may be planar. In some other embodiments, the display surface **1202** may have other topologies (e.g., be curved). It will be appreciated

that the display surface **1202** may be a physical surface of the display, or simply a plane or other imaginary surface from which image light is understood to propagate from the display **220** to the user's eyes.

[0203] As shown in FIG. **12A**, the user's eye **1200** may have an actual position **1204** offset from a nominal position **1206** and the display surface **1202** may be at position **1214**. FIG. **12A** also illustrates the corneal apex **1212** of the user's eye **1200**. The user's line of sight (e.g., their optical and/or visual axis) may be substantially along the line between the actual position **1204** and the corneal apex **1212**. As shown in FIGS. **12A** and **12B**, the actual position **1204** may be offset from the nominal position **1206** by an z-offset **1210**, a y-offset **1208**, and an x-offset **1209**. The nominal position **1206** may represent a preferred position (sometimes referred to as a design position, which may be generally centered within a desired volume) for the user's eye **1200** with respect to the display surface **1202**. As the user's eye **1200** moves away from the nominal position **1206**, the performance of display surface **1202** may be degraded, as discussed herein in connection with FIG. **14** for example.

[0204] It will be appreciated that a point or volume associated with the user's eye **1200** may be used to represent the position of the user's eye in analyses of registration herein. The representative point or volume may be any point or volume associated with the eye **1200**, and preferably is consistently used. For example, the point or volume may be on or in the eye **1200**, or may be disposed away from the eye **1200**. In some embodiments, the point or volume is the center of rotation of the eye **1200**. The center of rotation may be determined as described herein and may have advantages for simplifying the registration analyses, since it is roughly symmetrically disposed on the various axes within the eye **1200** and allows a single display registration volume aligned with the optical axis to be utilized for the analyses.

[0205] FIG. **12A** also illustrates that the display surface **1202** may be centered below the user's horizon (as seen along the y-axis when the user is looking straight ahead, with their optical axis parallel to the ground) and may be tilted (with respect to the y-axis). In particular, the display surface **1202** may be disposed somewhat below the user's horizon such that the user would have to look downward, at approximately the angle **1216**, to look at the center of the display surface **1202**, when the eye **1200** is at position **1206**. This may facilitate a more natural and comfortable interaction with the display surface **1202**, particularly when viewing content rendered at shorter depths (or distances from the user), as users may be more comfortable viewing content below their horizon than above their horizon. Additionally, display surface **1202** may be tilted, such as at angle **1218** (with respect to the y-axis) such that, when the user is looking at the center of the display surface **1202** (e.g., looking slightly below the user's horizon), the display surface **1202** is generally perpendicular to the user's line of sight. In at least some embodiments, the display surface **1202** may also be shifted left or right (e.g., along the x-axis) relative to the nominal position of the user's eye. As an example, a left-eye display surface may be shifted rightwards and a right-eye display surface may be shifted leftwards (e.g., display surfaces **1202** may be shifted towards each other) such that the user's lines of sight hits the centers of the display surfaces when focused at some distance less than infinity, which may increase user comfort during typical usage on the wearable device.

Example Graphs of Rendering Content in Response to User Eye Movements

[0206] FIG. **13** includes a set of example graphs **1200a-1200j** which illustrate how the wearable system may switch depth planes in response to the user's eye movements. As discussed herein in connection with FIGS. **4** and **7**, the wearable system may include multiple depth planes, where the various depth planes are configured to present content to a user at a different simulated depth or with a different accommodation cue (i.e., with various levels of wavefront curvature or light ray divergence). As an example, the wearable system may include a first depth plane configured to simulate a first range of depths and a second depth plane configured to simulate a second range of depths and, while these two ranges may desirably overlap to facilitate hysteresis in switching, the second range of depths may generally extend to greater distances from the user. In such embodiments, the wearable system may track a user's vergence depth, saccade movements, and blinks to switch between the first and second depth planes in a manner that avoid excessive depth plane switching, excessive accommodation-vergence mismatches, and excessive periods of accommodation-vergence mismatch and that seek to reduce the visibility of depth plane switches (i.e., by shifting depth planes during blinks and saccades).

[0207] Graph **1200a** illustrates an example of a user's vergence depth over time. Graph **1200b** illustrates an example of a user's saccade signal or velocity of eye movements over time.

[0208] Graph **1200c** may illustrate vergence depth data generated by eye tracking module **614** and, in particular, data generated by vergence depth estimation module **728**. As shown in graphs **1200c-1200h**, eye tracking data may be sampled within eye tracking module **614** at a rate of approximately 60 Hz. As shown between graphs **1200b** and **1200c**, eye tracking data within eye tracking module **614** may lag behind a user's actual eye movements by a delay **1202**. As an example, at time t_1 a user's vergence depth may cross a hysteresis threshold **1210a**, but the eye tracking module **614** may not recognize the event until time t_2 after delay **1202**.

[0209] Graph **1200c** also illustrates various thresholds **1210a**, **1210b**, **1210c** in a hysteresis band, which may be associated with transitions between first and second depth planes (i.e., depth planes **#1** and **#0** in FIG. **13**). In some embodiments, the wearable system may try to display content with depth plane **#1** whenever a user's vergence depth is greater than threshold **1210b** and to display content with depth plane **#0** whenever a user's vergence depth is less than threshold **1210b**. However, to avoid excessive switching, the wearable system may implement hysteresis, whereby the wearable system will not switch from depth plane **#1** to depth plane **#0** until the user's vergence depth crosses outer threshold **1210c**. Similarly, the wearable system may not switch from depth plane **#0** to depth plane **#1** until the user's vergence depth crosses outer threshold **1210a**.

[0210] Graph **1200d** illustrates an internal flag that may be generated by depth plane selection module **750**, or hysteresis band crossing detection module **752**, indicating whether the user's vergence depth is in the volume generally associated with depth plane **#1** or the volume generally associated with depth plane **#2** (i.e., whether the user's vergence depth is greater or less than threshold **1210b**).

[0211] Graph **1200e** illustrates an internal hysteresis band flag that may be generated by depth plane section module

750, or hysteresis band crossing detection module 752, indicating whether a user's vergence depth has cross an outer threshold such as threshold 1210a or 1210c. In particular, graph 1200e illustrates a flag indicative of whether the user's vergence depth has completely crossed a hysteresis band and into a region outside of the active depth plane's volume (i.e., into a region associated with a depth plane other than an active depth plane), thus potentially leading to undesirable accommodation-vergence mismatch (AVM).

[0212] Graph 1200f illustrates an internal AVM flag that may be generated by depth plane selection module 750, or hysteresis band crossing detection module 752, indicating whether a user's vergence has been in outside of the active depth plane's volume for greater than a predetermined time. The AVM flag may therefore identify when the user may have been subjected to an undesirable accommodation-vergence mismatch for a nearly-excessive or excessive period of time. Additionally or alternatively, the internal AVM flag may also indicate whether a user's vergence has gone a predetermined distance beyond the active depth plane's volume, thus creating a potentially-excessive accommodation-vergence mismatches. In other words, the AVM flag may indicate when a user's vergence has exceeded an additional threshold even further from threshold 1210b than thresholds 1210a and 1210c.

[0213] Graph 1200g illustrates an internal blink flag that may be generated by ocular event detection module 754, which may determine when a user has or is blinking. As noted herein, it may be desired to switch depth planes upon user blink, to reduce the likelihood of the user perceiving the switch in depth planes.

[0214] Graph 1200h illustrates an example output from depth plane selection module 750. In particular, graph 1200h shows that depth plane selection module 750 may output an instruction to utilize a selected depth plane, which may change over time, to a render engine such as render engine 622 (see FIG. 6).

[0215] Graphs 1200i and 1200j illustrate delays that may be present in the wearable system including a delay by render engine 622 to switch depth planes and a delay by the display 220, which may need to provide light associated with a new image frame in a new depth plane to effectuate a change in depth planes.

[0216] Reference will now be made to the events illustrated in graphs 1200a-1200j at various times (t_0 - t_{10}).

[0217] Sometime around time t_0 , a user's vergence depth may cross threshold 1210a, which may be an outer hysteresis threshold. After a delay associated with image capture and signal processing, the wearable system may generate a signal, as indicated in graph 1200e, that indicates that the user's vergence depth lies within the hysteresis band. In the example of graph 1200e, the eye tracking module 614 may present a hysteresis band exceeded flag at approximately time t_1 in connection with the user's vergence depth crossing threshold 1210a.

[0218] The user's vergence depth may continue to decrease from time t_1 to until approximately time t_4 and may thereafter increase.

[0219] At time t_1 , a user's vergence depth may cross threshold 1210b, which may be a midpoint between two depth planes such as depth planes #1 and #0. After processing delay 1202, eye tracking module 614 may alter an internal flag indicating that the user's vergence depth has

moved from a volume generally associated with depth plane #1 into a volume generally associated with depth plane #0, as illustrated in graph 1200d.

[0220] At time t_3 , the eye tracking module 614 may determine that the user's vergence depth, as shown in graph 1200a, has moved entirely through the hysteresis band and cross outer threshold 1210c. As a result, the eye tracking module 614 may generate a signal, as indicated in graph 1200e, that indicates that the user's vergence depth lies outside the hysteresis band. In at least some embodiments, the eye tracking module 614 may switch between first and second depth planes only when a user's vergence depth is outside of the hysteresis band between those two depth planes.

[0221] In at least some embodiments, the eye tracking module 614 may be configured to switch depth planes at time t_3 . In particular, the eye tracking module 614 may be configured to switch depth planes based on a determination that the vergence depth has moved from the volume of the currently selected depth plane (depth plane #1 as indicated by graph 1200h) into the volume of another depth plane (depth plane #0) and entirely crossed a hysteresis band. In other words, the eye tracking module 614 may implement a depth plane switch whenever the hysteresis band is exceeded (graph 1200e is high) and an accommodation-vergence mismatch based on time or magnitude of mismatch is detected (graph 1200f is high). In such embodiments, the eye tracking module 614 may provide a signal to render engine 622 instructing render engine 622 to switch to the other depth plane (depth plane #0). In the example of FIG. 13, however, the eye tracking module 614 may be configured to delay depth plane switches until at least one other condition has been satisfied. These additional conditions may include, as examples, a blink condition, an accommodation-vergence mismatch timeout condition, and an accommodation-vergence magnitude condition.

[0222] At time t_4 and in the example of FIG. 13, the eye tracking module 614 may be configured to switch depth planes. In particular, eye tracking module 614 may determine that the user's vergence has been in the volume associated with depth plane #0 for longer than a predetermined threshold of time (and optionally, also outside of the hysteresis band for that period of time). Examples of predetermined thresholds of time include 5 seconds, 10 seconds, 20 seconds, 30 seconds, 1 minute, and 90 seconds and any range between any of these values. Upon such a determination, the eye tracking module 614 may generate an AVM flag, as indicated in graph 1200f, and direct render engine 622 to switch to depth plane #0, as indicated in graph 1200h. In some embodiments, the eye tracking module 614 may generate an AVM flag and direct render engine 622 to switch depth planes if the user's vergence depth is detected to be more than a threshold distance from the current selected depth volume.

[0223] At time t_5 and after delay 1204, the render engine 622 may start rendering content at the newly-selected depth plane #0. After a delay 1206 associated with rendering and conveying light to a user through the display 220, the display 220 may be fully switched to the newly-selected depth plane #0 by time t_6 .

[0224] Thus, graphs 1200a-j illustrates, between times t_0 and t_6 , how the system may respond to a user's changing vergence and may switch depth planes after the user's vergence has moved away from a prior depth volume for

more than a predetermined period of time. Graphs **1200a-j**, between times t_7 and t_{10} , may illustrate how the system responds to a user's changing vergence and may switch depth planes upon detection of the user blinking, which may be prior to the predetermined period of time.

[0225] At time t_7 , the eye tracking module **614** may detect that the user's vergence depth has entered the hysteresis region between depth planes #0 and #1 (i.e., that the user's vergence depth has crossed outer threshold **1210c**). In response, the eye tracking module **614** may alter a hysteresis flag as shown in graph **1200e**.

[0226] At time t_8 , the eye tracking module **614** may detect that the user's vergence depth has cross threshold **1210b** and moved from the volume generally associated with depth plane #0 into the volume generally associated with depth plane #1. As such, the eye tracking module **614** may alter a depth volume flag, as shown in graph **1200d**.

[0227] At time t_9 , the eye tracking module **614** may detect that the user's vergence depth has crossed threshold **1210a** and moved out of the hysteresis volume into the volume generally associated exclusively with depth plane #1. In response, the eye tracking module **614** may alter a hysteresis flag as shown in graph **1200e**.

[0228] At around time t_{10} , the user may blink and the eye tracking module **614** may detect that blink. As one example, ocular event detection module **754** may detect a user's blink. In response, the eye tracking module **614** may generate a blink flag, as shown in graph **1200h**. In at least some embodiments, the eye tracking module **614** may implement a depth plane switch whenever the hysteresis band is exceeded (graph **1200e** is high) and a blink is detected (graph **1200g** is high). Thus, the eye tracking module **614** may instruct render engine **622** to switch depth planes at time t_{10} .

Example Processes of Calibration of Depth Plane Selection

[0229] As discussed herein, a head-mounted display such as display **220** of FIG. 2 may include a plurality of depth planes, each of which provides a different amount of wavefront divergence to provide a different accommodation cue to a user's eyes. The depth planes may be formed from optical elements such the waveguides **432b**, **434b**, **436b**, and **440b** of FIG. 4, which may be configured to send image information to user's eyes with desired levels of wavefront divergence, as examples.

[0230] In at least some embodiments, the wearable system including display **220** may be configured to display image content having an accommodation cue based on the current fixation point or vergence depth of the user's gaze (e.g., to reduce or minimize accommodation-vergence mismatches). In other words, the wearable system may be configured to identify the vergence depth of the user's gaze (e.g., using vergence depth estimation module **728** of FIG. 7A, as an example) and then display image content on the depth plane that provides an accommodation cue associated with the current vergence depth. Thus, when a user is looking at optical infinity, the wearable system may display image content on a first depth plane that provides an accommodation cue of optical infinity. In contrast, when the user is looking in the near field (e.g., within a meter), the wearable system may display image content on a second depth plane that provides an accommodation cue within or at least closer to the near field.

[0231] As previously noted, vergence depth estimation, which may be performed by vergence depth estimation module **728** of FIG. 7A, may be based in part upon the interpupillary distance (IPD) of the current user. In particular and in some embodiments, determining the vergence depth may involve projecting the optical and/or visual axes of the user's left and right eyes (to determine their respective gazes) and determining where those axes intersect in space and thus where the user's fixation point or vergence depth is. Geometrically, the optical and/or visual axes are the legs of a triangle, where the base of the triangle is the user's IPD and the tip of the triangle is the user's fixation point or vergence depth. Thus, it should be understood that user's IPD is useful in determining the vergence depth.

[0232] In various embodiments, the wearable system may be calibrated to a particular main user. The calibration may include various processes and may include determining how the user's eyes move as the user focuses on objects at different locations and depths. The calibration may also include identifying the IPD of the user (e.g., the distance between the user's pupils when the user is focused on optical infinity). The calibration may also include determining the user's pupil distance when the user is focused at objects closer than optical infinity, such as an object in the near field (e.g., less than 2.0 meters) and objects in an intermediate field (e.g., between about 2.0 and 3.0 meters). Based on such calibration data, the wearable system may be able to determine the depth the user is looking at by monitoring the pupil distance of the user. In other words, when the user's pupil distance is at its maximum (e.g., is equal to or close to the user's IPD), the wearable system may be able to infer that the user's vergence distance is at or near optical infinity. In contrast, when the user's pupil distance is near its minimum, the wearable system may be able to infer that the user's vergence distance is close to the user, at a distance determined by the calibration.

[0233] In some embodiments, the wearable system may utilize one or more alternative processes for depth plane selection. As one example, the wearable system may implement a content-based switching scheme. It will be appreciated that virtual content may include information about the location in virtual space at which that content should be located. Given this location, the virtual content may effectively specify an associated amount of wavefront divergence. Consequently, rather than determining the fixation point of the user's eyes to switch depth planes (e.g., to switch the amount of wavefront divergence of light for forming the virtual object), the display system may be configured to switch depth planes based upon the desired location in virtual space in which to place virtual content.

[0234] In some embodiments, the wearable system may still make a determination as to whether the user is looking at the virtual content in order to switch to the depth plane specified for that virtual content. For example, the display system may still track a gaze of the user to determine whether they are looking at a virtual object, and once that determination is made, may use the depth information associated with the virtual content to determine whether to switch depth planes. As another example, the wearable system may identify the most likely real or virtual object that the user is looking at based on an assumption that the user would look at a particular real or virtual object. For example, the wearable system may present a video to the user on a 2D virtual screen 1 meter away from the user. While the user

could be looking away from the screen at another object, it may be reasonable to assume that the user would be looking at the video screen. In some embodiments, the wearable system may be configured to make the assumption that the user is looking at real or virtual content that has movement or visible changes, or more movement or changes than other real or virtual content; for example, the wearable system may assign a score to movement or the amount of change in visual appearance by real or virtual content within the user's field of view and make the assumption that the user is looking at the real or virtual content with the highest score (e.g., the most movement or visual change, such as a virtual screen displaying a video).

[0235] Another example of an alternative depth plane selection process is dynamic calibration. Dynamic calibration may be beneficial when a current user has not (or has not yet) performed a dedicated calibration process. As an example, dynamic calibration may be utilized when a guest user is wearing the device. In one example of a dynamic calibration system, the wearable system may collect eye tracking data in order to estimate a current user's IPD and may then estimate the user's vergence depth using the estimated IPD (and their eye gaze directions as discussed in connection with module **728** of FIG. **7A**). IPD estimation may be performed by IPD estimation module **726** of FIG. **7A** and additional details and example embodiments are discussed herein in connection with module **726**. Dynamic calibration may occur as a background process, requiring no specific action from users. In addition, dynamic calibration may continuously obtain samples or images of the user's eyes to further refine IPD estimates. As discussed in further detail below, the wearable system may estimate a user's IPD as the 95th percentile (or other percentile) of all measured IPD values. In other words, the largest 5% of the measured IPD values may be excluded, and then the largest remaining measured IPD value may be taken as the user's IPD. An IPD value calculated in this manner may be referred to herein as IPD₉₅.

[0236] It will be appreciated that the display system may be configured to continually monitor IPD. As such, the number of samples or individual IPD measurements used to determine a value associated with a particular percentile may increase over time and potentially increase the accuracy of the IPD determination. In some embodiments, the IPD value (e.g., IPD **95**) may be continually or periodically updated. For example, the IPD value may be updated after a predefined amount of time has elapsed and/or after a predefined number of individual IPD measurements has been made.

[0237] In some embodiments, a dynamic calibration process may seek to identify the user's IPD (e.g., the maximum pupil distance of the user, such as when the user is looking at optical infinity). In such embodiments, the wearable system may be able to calibrate depth plane selection based on the IPD alone.

[0238] As a particular example, it has been determined that if a user's pupil distance is reduced from their maximum IPD by 0.6 mm, the user is likely focused at a depth of approximately 78 mm. 78 mm may correspond to a switch point between depth planes in some embodiments disclosed herein (e.g., the system may prefer to utilize a first depth plane when the user is focused at less than 78 mm and utilize a second depth plane when the user is focused at greater than 78 mm). In embodiments with a switch point that occurs at

a different depth of focus, the associated pupil distance reduction relative to their maximum IPD will change in relation to the change in the switch point (e.g., from 78 mm to whatever the switch point is in such embodiments).

[0239] In some instances, the wearable system may refine its calculation of the user's vergence depth by considering not just the difference between the user's current pupil distance and their maximum IPD, but also by considering how such relationships change in relation to the user's maximum IPD. In particular, the 0.6 mm number discussed above may be an average number that applies to a population as a whole and that takes into account various biases in the wearable system (e.g., on average, a user having a current pupil distance 0.6 mm less than their maximum IPD may be verging at a distance of 78 mm). However, the actual IPD difference (between maximum and current) associated with a vergence distance of 78 mm (or other switch point distance as discussed in the preceding paragraph) may be larger or greater than 0.6 mm and may be a function of the user's anatomical IPD. As particular examples, a person with an IPD of 54 mm may have a vergence distance of 78 mm when their current IPD is 0.73 mm less than their max IPD (e.g., their IPD when looking at distances of at least 10 meters), a person with an IPD of 64 mm may have a vergence distance of 78 mm when their current IPD is 0.83 mm less than their max IPD, and a person with an IPD of 72 mm may have a vergence distance of 78 mm when their current IPD is 0.93 mm less than their max IPD. These numbers may differ from the 0.6 mm number for a variety of reasons including, but not limited to, the 0.6 mm number does not distinguish between users with different IPDs and the 0.6 mm number may be in reference to an IPD₉₅ value (e.g., may be in reference to an IPD value that is actually slightly lower than the user's anatomical IPD when looking at optical infinity). In some embodiments, the 0.6 mm number may vary depending upon the user's max IPD. For example, a predefined number of the deviations from the 0.6 mm number may be available and may be associated with different ranges of max IPD values.

[0240] Using such relations, the wearable system may be able to determine the user's current vergence depth by comparing their maximum IPD with a current interpupillary distance (which may be reduced according to one or more mathematical functions as their vergence distance decreases). Determining the user's maximum IPD may involve, as an example, collecting a data on the user's IPD over time and identifying the maximum IPD within the collected data. In other embodiments, the wearable system may use heuristics or other processes to determine a user's maximum IPD even when the user does not look to optical infinity. In particular, the wearable system may extrapolate the user's maximum IPD from a plurality of pupil distances associated with closer vergence depth. The wearable system may also encourage the user to look at optical infinity by presenting virtual content at optical infinity and asking the user to focus their attention on the virtual content.

[0241] FIG. **14** is a process flow diagram of an example method **1400** for depth plane selection using an existing calibration, a content-based switching scheme, and/or a dynamic calibration. The method **1400** may be performed by the wearable system described herein. Embodiments of the method **1400** can be used by the wearable system to render content on a depth plane that generally reduces or minimizes

any vergence-accommodation mismatches, which could otherwise lead to user discomfort and fatigue.

[0242] At block **1402**, the wearable system may determine that it is not being worn by a user. The wearable system may determine that it is not being worn using one or more sensors such as eye tracking systems like eye cameras **324**. As an example, the wearable system may determine that it is not being worn by a user based on a determination there are no eyes in eye tracking images captured by eye cameras **324**. In particular, the wearable system may determine, after failing to detect eyes in the eye tracking images for at least a given period of time (e.g., a predetermined period of time, a dynamically determined period of time, etc.), that the wearable system is not being worn by a user.

[0243] Upon detection of the one or more of the user's eyes in eye tracking images such as those captured by cameras **324**, the method **1400** may move to block **1404**. In block **1404**, the wearable system may determine that it is being worn. In some embodiments, some or all of the determinations associated with block **1402** and/or block **1406** may be made based at least in part on data from one or more other sensors of the wearable system, such as IMUs, accelerometers, gyroscopes, proximity sensors, touch sensors, and the like. For example, in these embodiments, the wearable system may monitor data from one or more IMUs, accelerometers, and/or gyroscopes for an indication that the wearable system has been placed on or removed from the user's head, may monitor data from one or more proximity sensors and/or touch sensors to detect the physical presence of the user, or both. For example, the wearable system may compare data received from one or more of these sensors, and the wearable system may have threshold values associated with the respective sensors. In one example, the wearable system may then determine that the device has been removed from the user's head based upon values from one or more proximity sensors meeting or exceeding a threshold value, optionally in conjunction with data from IMUs, accelerometers, and/or gyroscopes indicating sufficient movement (e.g., exceeding a threshold value) to support the conclusion that the device has been removed.

[0244] At block **1406**, the wearable system may attempt to identify the current user by performing an identification process. As one example, the wearable system may estimate the IPD of the current user to determine if the current user's IPD matches a calibrated user's IPD (e.g., if the two IPDs are within some threshold of each other). In some embodiments, the wearable system may determine that the current user is a calibrated user if the calibrated IPD and the current user's IPD are within a threshold of, e.g., 0.5 mm, 1.0 mm, 1.5 mm, or 2.0 mm, of the calibrated user's IPD. In general, larger thresholds may facilitate faster determinations and help to ensure that calibrated users are identified and their calibration parameters used; for example, larger thresholds are biased towards finding that the current user is the calibrated user. In at least some embodiments, the wearable system may be able to determine a current user's IPD with relatively high accuracy (e.g., 95%) within a relatively short time frame (e.g., 5-7 seconds of eye tracking data, which may correspond to between about 150 and 200 frames of eye tracking images).

[0245] If there is a match between the IPD of the current user and the IPD of the calibrated user, the wearable system may assume that the current user is the calibrated user and load an existing calibration at block **1408**. The existing

calibration may be calibration parameters or data generated during a calibration process with the calibrated user wearing the wearable system. In the event that the current user is not actually the calibrated user, but merely has a similar IPD, the calibration data loaded at block **1408** may be an acceptable calibration that provides reasonable performance for the current user, while allowing the current user to use the wearable system without performing a more detailed calibration.

[0246] In some embodiments, the wearable system may identify the current user using measures other than (or in addition to) the user's IPD. As examples, the wearable system may ask the user for a username and/or password, the wearable system may perform iris scanning (e.g., comparing a current image of the user's iris with a reference image to determine whether there is a match, with a match being interpreted to mean that the current user is a calibrated user), voice recognition (e.g., by comparing a current sample of the user's voice with a reference voice file, with a match being interpreted to mean that the current user is a calibrated user), or some combination of these and other authentication or identification techniques. In some embodiments, two or more identification processes may be performed to increase the accuracy of the determination of whether or not the current user is a calibrated user. For example, the current user may be assumed to be a calibrated user, but may be determined to not be a calibrated user if all of the performed identification processes failed to identify the current user as a calibrated user. As another example, the results from the various performed identification processes may be aggregated into a combined score and the current user may be assumed to be a calibrated user unless the combined score exceeds a predetermined threshold value.

[0247] In some embodiments, if the display system includes multiple calibration files for multiple users, then additional criteria may be needed to select the appropriate calibration file. For example, the user may be prompted to select the appropriate calibration file, and/or multiple ones of the identification schemes disclosed herein may be utilized.

[0248] In some embodiments, multiple identification schemes may be utilized to increase the accuracy of user identification. For example, it will be appreciated that IPD is a relatively coarse identification criteria. In some embodiments, IPD may be used as a first criteria to identify whether the current user is likely a calibrated user, and then a more precise or accurate identification scheme may be utilized (e.g., iris scanning). Such a multistep identification scheme advantageously may conserve processing resources since the more accurate identification scheme may be more resource intensive. Consequently, processing resources may be saved by delaying use of the more accurate resource-intensive identification scheme until the IPD determination indicates that a calibrated user is present.

[0249] With continued reference to FIG. **14**, if the current user's IPD does not match the calibrated user's IPD, then the wearable may perform a dynamic calibration in block **1410**. A dynamic calibration may, as discussed herein, involve monitoring eye tracking data to estimate how a user's interpupil distance changes as a function of their vergence distance. As one example, the dynamic calibration may involve estimating a user's maximum IPD and then using the maximum IPD together with a current interpupil distance to estimate a current vergence distance.

[0250] In block 1412, the wearable system may implement content-based switching. With content-based switching, depth plane selection is based on the depth of whatever virtual content the user is determined to be looking at (e.g., the most important or interesting content being displayed, which may be identified by content creators; based on a gaze of the user's eyes, etc.). In at least some embodiments, block 1412 may be performed whenever selected by content creators or other designers of the wearable system and regardless of whether or not the current user is a calibrated user. In various embodiments, content-based switching in block 1412 may be performed when there is no calibrated user. In such embodiments, block 1406 may be skipped, if desired. In addition, in some embodiments, the content-based switching of block 1412 may be performed whether or not block 1408 and/or 1410 (and blocks related to these blocks) are performed or available to the display system; for example, in some embodiments, the display system may perform only block 1412 for determining depth plane switching.

[0251] As described herein, each virtual object may be associated with location information, such as three-dimensional location information. The wearable system may present each virtual object to the user based on the location information. For example, the location information for a particular virtual object may indicate X, Y, and Z coordinates at which the object is to be presented (e.g., a center or centroid of the object may be presented at the coordinates). Thus, the wearable system may obtain information indicating a depth plane at which each virtual object is to be presented.

[0252] As will be described in more detail below with respect to FIGS. 16A-18, the wearable system may assign a respective volume of space (also referred to herein as a "zone" or marker) as surrounding each object. These volumes of space may preferably not overlap. The wearable system may identify a gaze of the user, and identify a zone which includes the user's gaze. For example, the gaze may indicate a three-dimensional location at which the user is fixating (e.g., an approximate three-dimensional location). The wearable system may then cause presentation of virtual content at a depth plane associated with the virtual object included in the identified zone. Thus, in some embodiments, once the display system makes a determination as to which object the user is looking at, switching between depth planes may occur based on the location or depth plane associated with virtual object rather than the fixation point of the user's eyes.

[0253] In block 1414, the wearable system may perform depth plane switching. The depth plane switching of block 1414 may be performed with the configuration parameters loaded or generated in blocks 1408, 1410, or 1412. In particular, if the current user is a calibrated user, then block 1414 may perform depth plane switching according to the configuration parameters generated during calibration with that user. If the current user is not identified as a calibration user and dynamic calibration was performed in block 1410, then block 1412 may involve depth plane switching according to calibration parameters generated in block 1410 as part of dynamic calibration. In at least some embodiments, the calibration parameters generated at block 1410 may be a user's IPD and the depth plane switching in block 1414 may be based on the user's IPD. If the wearable system is implementing content-based switching in block 1412 (which

may occur when there is no calibrated user and/or when a content creator or user prefers to utilize content-based switching), the depth plane switching may be performed according to the depth of the content the user is assumed to be looking at.

[0254] With continued reference to FIG. 14, it will be appreciated that the display system may be configured to continually confirm the identity of the user. For example, after block 1414, the display system may be configured to return to block 1406 to identify the user. In some instances, the display system may lose track of a calibrated user that was previously detected at block 1406 to be wearing the display device. This may occur, for example, because the calibrated user has taken off the display device, or may be due to latency issues, or sensing errors that erroneously indicate that the display device is no longer being worn by the user, even where the calibrated user is still wearing the display device. In some embodiments, even when the display device detects that the calibrated user is no longer wearing the display device, the display system may continue to use the calibrated user's calibration profile for a predetermined amount of time or for a predetermined number of frames before switching to the content-based depth plane switching scheme or the dynamic calibration scheme of blocks 1412 and 1410, respectively. As discussed herein, the display system may be configured to continuously perform block 1406 and may continue to not detect the calibrated user. In response to determining that the calibrated user has gone undetected for the predetermined amount of time or the predetermined number of frames, then the system may switch to the content-based depth plane switching scheme or the dynamic calibration scheme of blocks 1412 and 1410, respectively. Advantageously, because the calibrated user is typically the most likely user of the display device, by continuing to use the calibrated user's calibration profile, the calibrated user may not experience a significant degradation in user experience in the event that the system erroneously fails to detect the calibrated user.

[0255] Another example of a method for depth plane selection is shown in FIG. 15. FIG. 15 is a process flow diagram of an example method 1500 for depth plane selection based on a user's interpupillary distance. The method 1500 may be performed by the wearable system described herein. Embodiments of the method 1500 can be used by the wearable system to render content on a depth plane that generally reduces or minimizes any vergence-accommodation mismatches, which could otherwise lead to user discomfort and fatigue.

[0256] At block 1502, the wearable system may determine that it is not being worn by a user. As an example, the wearable system may determine it is not being worn after failing to detect a user's eyes via eye tracking systems for more than a threshold period of time (such as 5 seconds, 10 seconds, 20 seconds, etc.).

[0257] At block 1504, the wearable system may determine that it is being worn by a user. The wearable system may determine it is being worn using one or more sensors. As an example, the wearable system may include a proximity sensor or a touch sensor that is triggered when the wearable system is placed on a user's head and may determine the system is being worn based on signals from such sensors. As another example, the wearable system may determine it is worn after identifying the presence of a user's eyes in eye tracking images. In some instances, the wearable system

may be able to distinguish between 1) failure to detect a user's eyes because the user has shut their eyes and 2) failure to detect a user's eyes because the user has removed the wearable system. Thus, when the wearable system detects a user's closed eyes, the wearable system may decide the device is being worn. In some embodiments, some or all of the determinations associated with block **1502** and/or block **1504** may be similar to or substantially the same as those associated with block **1402** and/or block **1404** as described above with reference to FIG. 14, respectively.

[0258] At block **1506**, the wearable system may check to see if it has been previously calibrated to any users. Block **1506** may involve, in various embodiments, obtaining the IPD of the calibrated user(s), as part of identifying whether or not the current user is the calibrated user (or one of the calibrated users, if there are multiple calibrated users). In some embodiments, the wearable system may implement content-based switching, as discussed herein, and may activate content-based switching at block **1516** if the wearable system has not been previously calibrated to any users (or if any such calibrations have been deleted or removed from the wearable system). The wearable system may also implement content-based switching for specific content and/or upon request by users or content creators. In other words, content creators or users may be able to specify or request the use of content-based switching, even when the wearable system has been previously calibrated and even when the wearable system has been calibrated for the current user.

[0259] At block **1508**, the wearable system may estimate a current user's IPD and, at block **1510**, the wearable system may accumulate eye tracking data on the current user's IPD. The IPD data accumulated in block **1510** may be measurements of the user's interpupillary distance over time, measurements of the user's left and right centers of rotation of time, an indication of the maximum interpupillary distance measured over time, or any other relevant data. In at least some embodiments, the wearable system may be able to determine a current user's IPD in block **1508** with relatively high accuracy (e.g., 95%) within a relatively short time frame (e.g., 5-7 seconds of eye tracking data, which may correspond to between about 150 and 200 frames of eye tracking images).

[0260] In some embodiments, the wearable system may estimate a user's IPD as a particular percentile (e.g., the 95th percentile) of all IPD values collected (e.g., during dynamic calibration at block **1518** and/or during the accumulation of IPD data at block **1510**). For example, for the 95th percentile, the largest 5% of the measured IPD values may be excluded, and then the largest remaining measured IPD value may be taken as the user's IPD. The IPD value calculated in this manner may be referred to herein as IPD₉₅. One benefit of calculating a user's IPD value in this manner is that outlying values, which may be greater than the user's anatomical IPD, can be excluded. Without being limited by theory, it is believed that with there is a sufficient number of measured IPD values near or at the user's anatomical IPD that the IPD₉₅ value to accurately reflects the user's anatomical IPD. As an example, if roughly 10% of the IPD measures are close to the user's anatomical IPD, then the IPD₉₅ value should still reflect the user's anatomical IPD value even though the largest 5% values were excluded. If desired, other IPD calculations, which may involve excluding a different percentage of the largest measured IPD values may be used. As examples, an IPD₁₀₀

value may be used in which no IPD values are excluded or an IPD₉₈ value may be used in which only the largest 2% of values are excluded (which may be preferable in a system with eye tracking systems that produce relatively few outlying IPD measurements). As additional examples, an IPD₉₀ value may be used, an IPD₈₅ value may be used, or other IPD value with a desired percentage of the measured values excluded.

[0261] At block **1512**, the wearable system may determine if the estimated IPD of the current user is within a threshold value of the IPD of the calibrated user (or of one of the IPDs of one of the calibrated users, if there are multiple such users). The threshold value may be, as examples, 0.5 mm, 1.0 mm, 1.5 mm, 2.0 mm, or 2.5 mm. As one particular example, the threshold value may be 1.0 mm, which may be sufficiently large that a calibrated user is recognized as a calibrated user quickly and accurately, while non-calibrated users can generally be identified as non-calibrated users. In general, it may be preferable to falsely identify non-calibrated users as calibrated users, than to risk failing to identify calibrated users as calibrated users. In other words, false negatives in user identification for calibration may be worse than false positives, as the desire for optimal performance for calibrated users may outweigh minor sacrifices for non-calibrated or guest users, and as the calibrated users may be more likely or more frequent users of a given display system.

[0262] If the current user's IPD is within the threshold of the calibrated user's IPD, the wearable system may assign the current user as the calibrated user and may load the associated calibration parameters in block **1514**. As an example, the calibration parameters loaded in block **1514** may include the calibrated user's IPD, a visual-optical axis offset parameter, and other available calibration parameters.

[0263] If the current user's IPD is not within the threshold of the calibrated user's IPD, the wearable system may assume that the current user is a guest user and may perform dynamic calibration in block **1518** (or content-based switching in block **1516**). At block **1518**, the wearable system may activate dynamic calibration (e.g., dynamically generate calibration parameters), which may include an on-the-fly calibration based on the estimated IPD of the current user, as discussed herein.

[0264] At block **1516**, the wearable system may implement content-based switching under circumstances as discussed herein, such as when no prior calibration data is available. At this block, as discussed herein, depth plane selection may be made according to the depth associated with virtual content being displayed as opposed to tracking the vergence depth of the user. Content-based depth plane switching is further discussed regarding FIGS. 16-17B.

[0265] In block **1520**, the wearable system may perform depth plane switching. The depth plane switching of block **1520** may be performed with the configuration parameters loaded or generated in blocks **1514**, **1516**, or **1518**. In particular, if the current user is a calibrated user, then block **1520** may perform depth plane switching according to the configuration parameters generated during calibration with that user. If the current user is not identified as a calibration user and dynamic calibration was performed in block **1518**, then block **1520** may involve depth plane switching according to calibration parameters generated in block **1520** as part of dynamic calibration. In at least some instances, the calibration parameters generated at block **1518** may be a

user's IPD and the depth plane switching in block 1520 may be based on the user's IPD. If the wearable system is implementing content-based switching in block 1516 (which may occur when there is no calibrated user and/or when a content creator or user selects utilization of content-based switching), the depth plane switching of block 1520 may be performed according to the depth of the content the user is determined to be looking at.

Content-Based Switching

[0266] As described herein, the wearable system may present virtual content via a particular depth plane. As the virtual content is updated, for example as virtual objects move and/or are replaced with different virtual objects, the wearable system may select different depth planes at which to present the virtual content. As additional nonlimiting examples, as a user of the wearable system focuses on different virtual objects, or at different locations with the user's field of view, the wearable system may select different depth planes at which to present the virtual content.

[0267] An example scheme to select a depth plane, referred to herein as content-based switching, is further discussed below with reference to FIGS. 16A-17B. In content-based switching, the wearable system may select a depth plane at which to present virtual content based on a determination of which virtual object the user is likely looking. In some embodiments, this determination may be made depending on whether the user is fixating within a particular zone. For example, the wearable system may associate a zone with each virtual object to be presented as virtual content to the user. A zone may be, for example, a volume of space surrounding a virtual object. The volume of space may be a sphere, cube, hyperrectangle, pyramid, or any arbitrary three-dimensional polygon (e.g., a polyhedron). As will be described, the zones are preferably non-overlapping and, thus, the space encompassed by each zone may only be associated with a single virtual object.

[0268] In some embodiments, the wearable system may monitor a gaze of the user, for example, to identify locations at which the user is viewing (e.g., within the user's field of view). For an example gaze, the wearable system may identify a zone which includes a location being viewed by the user. In this example, having identified the zone and having only one virtual object in that zone, the wearable system may select a depth plane corresponding to the virtual object included in the identified zone. The wearable system may then present the virtual content at the selected depth plane, which is the depth plane associated with the virtual object.

[0269] Advantageously, the wearable system may utilize higher latency eye tracking, or otherwise less accurate eye tracking schemes than may be required if depth plane switching were dependent upon accurately determining the depth of the fixation point. In some embodiments, for scenarios with multiple virtual objects displayed at disparate depths, statistical probability or spatial correlation may be utilized to distinguish what content a user is likely to be observing or directing attention to. Thus, the wearable system may determine a fixation area or fixation volume (e.g., rather than a precise fixation point), and select a depth plane based on the fixation area or volume. In some embodiments, weighted factors such as a last application utilized, or degree to which the user's gaze changed, can adjust the statistical probability and, thus, the size/shape of the zone. In

some embodiments, the size/shape of the zone is correlated with the statistical probability that a user is fixating within a particular volume; for example, where there is a large amount of uncertainty (e.g., where the uncertainty exceeds a threshold due to noise, low tolerances in the eye-tracking system, the tracking particulars of an application that was most-recently used, the speed of changes in the user's gaze, etc.) the size of the zone may increase, and where there is less uncertainty, the size of the zone may decrease.

[0270] As will be described in FIG. 16A, in some embodiments, the zones may encompass a respective angular distance (e.g., extending from a user's eyes to an infinite distance from the user). In this example, the wearable system may therefore require accuracy only sufficient to place the user's gaze within particular portions of an X and Y plane.

[0271] In some embodiments, the display system may transition from performing content-based depth plane switching to performing depth plane switching based on a dynamic calibration. Such a transition may occur, for example, when data obtained for performing the content-based depth plane switching scheme is determined to be unreliable, or where content is provided at a range of different depths spanning multiple depth planes. As noted herein, the amount of calculated uncertainty in the determination of the zone in which a user fixating may vary based on various factors such noise, tolerances of the eye-tracking system, the tracking particulars of an application that was most-recently used, the speed of changes in the user's gaze, etc. In some embodiments, the display system may be configured to transition to depth plane switching based on dynamic calibration when the uncertainty (e.g., the uncertainty associated with determining a position of a user's fixation point) exceeds a threshold value. In some other embodiments, particular virtual content may span across multiple depth planes. For content spanning across a threshold number of depth planes (e.g., three depth planes), the display system may be configured to transition to depth plane switching based on dynamic calibration.

[0272] In some embodiments, the display system may be configured to transition performing content-based depth plane switching to performing depth plane switching based on dynamic calibration in response to determining that the uncertainty associated with the depth of virtual content exceeds a threshold value. For example, the display system may determine that the depth or location information for virtual content that is being made available by a particular application that is running on the display system is relatively unreliable (e.g., the particular application indicates that the depth or location of mixed reality virtual content relative to the user is static even when the user's position changes or the user moves beyond a predetermined level of position change or movement, respectively) and, in turn, may transition to depth plane switching based on dynamic calibration. Such a transition may facilitate a more comfortable and/or realistic viewing experience.

[0273] Although described primarily within the context of depth plane switching, it is to be understood that one or more of the techniques described herein with reference to FIG. 14 and/or FIG. 15 may be leveraged in any of a variety of different display systems that are capable of outputting light to an eye of a user with different amounts of wavefront divergence. For example, in some embodiments, one or more of the techniques described herein with reference to

FIG. 14 and/or FIG. 15 may be leveraged in a head-mounted display system that includes one or more variable focus elements (VFEs). For example, one or more of the lenses 458, 456, 454, 452 (FIG. 4) may be VFEs, as discussed herein. In these embodiments, the head-mounted display system may control operation of its one or more VFEs in real-time based at least in part on whether a wearer of the head-mounted display system is determined to be a calibrated user or a guest user. That is, the head-mounted display system may control or adjust an amount of wavefront divergence with which light is output to the wearer (the focal distance of the light) based at least in part on whether a wearer of the head-mounted display system is determined to be a calibrated user or a guest user. Examples of architectures and control schemes for variable focus eyepieces are disclosed in U.S. Publication No. 2016/0110920, published Apr. 21, 2016 and in U.S. Publication No. 2017/0293145, published Oct. 12, 2017, each of which is incorporated by reference herein in its entirety. Other configurations are possible.

[0274] FIG. 16A illustrates a representation of a user's field of view 1602. The user's field of view 1602 (a three-dimensional frustum in the illustrated embodiment) may include a plurality of depth planes 1604A-D extending in a z-direction from the user's eyes 1606. The field of view 1602 may be separated into different zones 1608A-D. Each zone may therefore encompass a particular volume of space included in the field of view 1602. In this example, each zone encompasses a volume of space that preferably includes all of the different depth planes 1604A-D. For example, zone 1608A may extend along the z-direction from the user's eyes 1606 to, for example, an infinite distance from the eyes 1606. Along an orthogonal direction, such as the x-direction, zone 1608A may extend from an extremity of the field of view 1602 until the border between zones 1608A and 1608B.

[0275] In the illustrated embodiment, zone 1608B includes tree virtual object 1610. Tree virtual object 1610 is illustrated as being presented at depth plane C 1604C. For example, the tree virtual object 1610 may be associated with location information (e.g., stored, or otherwise accessible to, the wearable system). As described above, this location information may be utilized to identify a three-dimensional location at which the tree virtual object 1610 is to be presented. As illustrated, the display system may also display a book as virtual object 1612. Zone 1608C includes the book virtual object 1612. Thus, the wearable system is presenting virtual content comprising the tree virtual object 1610 and the book virtual object 1612.

[0276] The user's eyes 1606 may move about the field of view 1602, for example fixating on, or viewing, different locations with the field of view 1602. If the user's eyes 1606 fixate on the tree virtual object 1610, or on the depth plane C, the wearable system may determine to select depth plane C 1604C to present the virtual content. Similarly, if the user's eyes 1606 fixate on the book virtual object 1612, more on the depth plane B, the wearable system may determine to select depth plane B 1604B to present the virtual content. As described in, at least, FIG. 14, the wearable system may utilize different schemes to cause selection of depth planes.

[0277] With respect to content-based switching, the wearable system may identify a zone which includes a location being fixated upon by the user's eyes 1606. The wearable system may then select a depth plane corresponding to the

virtual object included in the identified zone. For example, FIG. 16A illustrates an example fixation point 1614. The wearable system may determine, based on the user's eyes 1606, that the user is viewing fixation point 1614. The wearable system may then identify that fixation point 1614 is included in zone 1608C. Since this zone 1608C includes the book virtual object 1612, the wearable system may cause presentation of the virtual content at depth plane C 1604C. Without being limited by theory, this is believed to provide a comfortable viewing experience, since the only virtual object that may be fixate upon in the zone 1608C is the book virtual object 1612 and, as such, it would be appropriate to switch to the depth plane of that book virtual object 1612.

[0278] Consequently, if the user adjusts fixation to fixation point 1616 or 1618, the wearable system may maintain presentation at depth plane C 1604B. However, if the user adjusts fixation to within zone 1608B, the wearable system may select depth plane C 1604C at which the present the virtual content. While the example of FIG. 16A includes four zones, it should be understood that fewer zones, or a greater number of zones, may be utilized. Optionally, the number of zones may be the same as the number of virtual objects, with a unique zone for each virtual object, and, in some embodiments, the number of zones may vary dynamically as the number virtual objects changes (preferably so long as the zones can extend from the front to the back of the display frustum without more than one object occupying the same zone e.g., without two or more objects being roughly in the same line of sight relative of the user). For example, two zones may be utilized in the example of FIG. 16A, which has two virtual objects that do not have overlapping lines of sight to the user. Optionally, the wearable system may dynamically adjust a volume of space encompassed by each zone. For example, as accuracy of gaze detection increases or decreases the volumes of space may be increased, decreased, or the zones adjusted in number.

[0279] FIG. 16B illustrates an example, shown in perspective view, of the volume of space encompassed by each zone of FIG. 16A. As illustrated, each zone 1608A, 1608B, 1608C, and 1608D extends from the front of the display frustum 1602 to the back in the z-direction. As illustrated, on the x-y axis, the zones extend vertically to segment or divide the field of view along the x-axis. In some other embodiments, on the x-y axis, the zones may extend horizontally to segment the field of view along the y-axis. In such embodiments, the illustrated frustum may be effectively rotated 90° on the x-y plane. In yet other embodiments, on the x-y axis, the zones may segment or divide the field of view on both the x and the y axes, thereby forming a grid on the x-y axis. In all these embodiments, each zone still preferably extends from the front to the back of the display frustum 1602 (e.g., from the nearest plane that display system may display content to the furthest plane).

[0280] FIG. 17A illustrates another representation of a user's field of view 1602 for content-based switching. In the example of FIG. 17A, as in FIG. 16A, the user's field of view 1602 includes depth planes A-D 1604A-D. However, the volume of space associated with each virtual object is different from that of FIGS. 16A-B. For example, the zone 1702 in which book virtual object 1612 is located is illustrated as encompassing an enclosed shape, such as a spherical volume of space, a cylindrical volume of space, a cube, a polyhedron, etc. Preferably, the zone 1702 extends, on the z-axis, less than the entire depth of the display frustum 1602.

Advantageously, such an arrangement of zones allows differentiation between objects that may be in similar lines of sight to the user.

[0281] Optionally, the wearable system may adjust the zones of FIG. 17A based on the eye tracking accuracy exceeding one or more thresholds. For example, the sizes of the zones may decrease as confidence in tracking accuracy increases. As another example, it will be appreciated that each virtual object has an associated zone. The wearable system may adjust the zones of FIG. 17A to avoid overlapping zones. For example, if a new virtual object is to be displayed within a zone corresponding to a different virtual object, then the zones for one or both virtual objects may be decreased in size to avoid overlap. For example, if the book virtual object 1612 in FIG. 17A is moved in front of the tree virtual object 1610 (e.g., in zone 1608A), the wearable system may adjust the zones in FIG. 17A to adhere closer to the virtual objects (e.g., as illustrated in FIG. 17A) to prevent overlap.

[0282] With continued reference to FIG. 17A, the tree virtual object 1610 is illustrated as being included in a first zone 1706 and a second zone 1708. Optionally, the first zone 1706 may represent a relatively small volume of space surrounding the tree virtual object 1610. The second zone 1708 may represent a current volume of space surrounding the tree virtual object 1610. For example, as virtual objects move around within the field of view 1602, the volume of space encompassed by a zone may be adjusted (e.g., in real-time). In this example, the adjustment may ensure that each zone does not overlap with any other zone. Thus, if the book virtual object 1612 moves closer to the tree virtual object 1610, the wearable system may reduce the volume of space of the zone surrounding the tree virtual object 1610, the book virtual object 1612, or both. For example, the second zone 1708 may be reduced to be closer in volume to the first zone 1706.

[0283] As noted herein, in content-based depth plane switching, the depth plane associated with a virtual object governs, rather than the fixation point. As example of this is illustrated with respect to fixation point 1704. As illustrated, the example zone 1702 of the book virtual object 1612 may encompass a volume of space which includes a portion defined by depth plane B 1604B and depth plane 1604C. In some cases, the wearable system determines that the user is fixating at point 1704 in depth plane C, and also identifies this fixation point 1704 as being included in zone 1702. Because the depth plane associated with the book virtual object 1612 governs the depth plane switching, the wearable system switches to the depth plane B of the book virtual object 1612 rather than the depth plane C of the fixation point 1704. FIG. 17B illustrates an example of a perspective view of the representation of FIG. 17A.

[0284] FIG. 18 illustrates a flowchart of an example process to select a depth plane based on content-based switching. For convenience, the process 1800 will be described as being performed by a wearable system of one or more processors (e.g., the wearable system, such as wearable system 200, described above).

[0285] At block 1802, the wearable system presents virtual content at a particular depth plane. As described above, depth planes may be utilized to provide accommodation cues to a user of the wearable system. For example, each depth plane may be associated with an amount of wavefront

divergence of light presented by the wearable system to the user. The virtual content may comprise one or a plurality of virtual objects.

[0286] At block 1804, the wearable system determines a fixation point. The wearable system may utilize sensors, such as cameras, to estimate a three-dimensional location at which the user is fixating. These cameras may update at a particular rate, such as 30 Hz, 60 Hz, and so on. The wearable system may determine vectors extending from the user's eyes (e.g., from the center or pupil of the eye), and estimate a three-dimensional location at which the vectors intersect. In some embodiments, the fixation point may be estimated based on IPD, with a predefined change in IPD from a maximum IPD being assumed to correlate with fixation at a particular depth plane. In addition, optionally in addition to IPD, the gaze of the user may be determined to further locate the approximate position of the fixation point. This estimated location may have certain error associated with it, and the wearable system may therefore determine a volume of space in which the fixation point likely lies relative to virtual content.

[0287] At block 1806, the wearable system identifies a zone that includes the fixation point. As discussed regarding FIGS. 16A-17B, the wearable system may associate a zone with each virtual object. For example, a zone may include a particular (e.g., single) virtual object. Preferably, the zones for different virtual objects do not overlap.

[0288] At block 1808, the wearable system selects a depth plane associated with virtual content included in the identified zone. The wearable system may identify the virtual content associated with the zone identified in block 1806. The wearable system may then obtain information indicating a depth plane at which the virtual content is to be presented. For example, the virtual content may be associated with location information, indicating a three-dimensional location. The wearable system may then identify a depth plane associated with the three-dimensional location. This depth plane may be selected to present the virtual content.

[0289] FIG. 19 illustrates a flowchart of an example process to adjust zones based on content-based switching. For convenience, the process 1800 will be described as being performed by a wearable system of one or more processors (e.g., the wearable system, such as wearable system 200, described above).

[0290] At block 1902, the wearable system presents virtual objects. The wearable system may, as described herein, have zones associated with presented virtual objects. For example, a zone may encompass a volume of space and include a particular virtual object (e.g., a single virtual object). The zones may be any shape or polyhedron, and may extend infinitely in one or more directions in some embodiments. Examples of zones are illustrated in FIGS. 16A-17B.

[0291] At block 1904, the wearable system adjusts a number of virtual objects or adjusts locations of the virtual objects. The wearable system may present additional virtual objects, for example at block 1902 5 virtual objects may be presented. At block 1904, 8 virtual objects may be presented. Optionally, the wearable system may update locations of the virtual objects. For example, a virtual object presented in block 1902 may be a bee. The virtual object may thus travel about the user's field of view.

[0292] At block 1906, the wearable system updates the zones associated with one or more virtual objects. With

respect to the example of additional virtual objects being presented, the virtual objects may become closer together. Since each zone may only be allowed to include a single virtual object, in some embodiments, the increase in the number of virtual objects may require an adjustment to the zones. For example, FIG. 16A illustrates two virtual objects. If additional virtual objects are included, they may be included in a zone in which either of the two virtual objects are also included. Thus, the wearable system may adjust the zones—for example, adjust a volume of space assigned to each zone (e.g., reduce the volume of space). In this way, each zone may include a single virtual object. With respect to the example of virtual objects moving, a moving virtual object may get closer to another virtual object. Thus, the moving virtual object may extend into a zone associated with the other virtual object. Similar to the above, the wearable system may adjust the zones to ensure that each virtual object is included in its own zone.

[0293] As an example of updating, the wearable system may associate zones with the virtual objects identified in block 1902. For example, the zones may be similar to the zones 1608A-1608D illustrated in FIG. 16A. The wearable system may adjust the zones to encompass smaller volumes of space. For example, the wearable system may update the zones to be similar to the zones 1702, 1706 illustrated in FIG. 17A.

[0294] As another example of updating, the zones may be similar to the zones of FIG. 17A. With respect to FIG. 17A, if the book virtual object 1612 moves closer to the tree virtual object 1610, a zone 1708 surrounding the tree virtual object 1610 may include the book virtual object 1612. Thus, the wearable system may update the zone 1708 to reduce the volume of space it encompasses. For example, the zone 1708 may be adjusted to be zone 1706. As another example, zone 1708 may be adjusted to be closer to zone 1706. As described in FIG. 17A, zone 1706 may optionally reflect a minimum zone around the tree virtual object 1610.

Computer Vision to Detect Objects in Ambient Environment

[0295] As discussed above, the display system may be configured to detect objects in or properties of the environment surrounding the user. The detection may be accomplished using a variety of techniques, including various environmental sensors (e.g., cameras, audio sensors, temperature sensors, etc.), as discussed herein.

[0296] In some embodiments, objects present in the environment may be detected using computer vision techniques. For example, as disclosed herein, the display system's forward-facing camera may be configured to image the ambient environment and the display system may be configured to perform image analysis on the images to determine the presence of objects in the ambient environment. The display system may analyze the images acquired by the outward-facing imaging system to perform scene reconstruction, event detection, video tracking, object recognition, object pose estimation, learning, indexing, motion estimation, or image restoration, etc. As other examples, the display system may be configured to perform face and/or eye recognition to determine the presence and location of faces and/or human eyes in the user's field of view. One or more computer vision algorithms may be used to perform these tasks. Non-limiting examples of computer vision algorithms include: Scale-invariant feature transform (SIFT), speeded up robust features (SURF), oriented FAST and

rotated BRIEF (ORB), binary robust invariant scalable keypoints (BRISK), fast retina keypoint (FREAK), Viola-Jones algorithm, Eigenfaces approach, Lucas-Kanade algorithm, Horn-Schunck algorithm, Mean-shift algorithm, visual simultaneous location and mapping (vSLAM) techniques, a sequential Bayesian estimator (e.g., Kalman filter, extended Kalman filter, etc.), bundle adjustment, Adaptive thresholding (and other thresholding techniques), Iterative Closest Point (ICP), Semi Global Matching (SGM), Semi Global Block Matching (SGBM), Feature Point Histograms, various machine learning algorithms (such as e.g., support vector machine, k-nearest neighbors algorithm, Naive Bayes, neural network (including convolutional or deep neural networks), or other supervised/unsupervised models, etc.), and so forth.

[0297] One or more of these computer vision techniques may also be used together with data acquired from other environmental sensors (such as, e.g., microphone) to detect and determine various properties of the objects detected by the sensors.

[0298] As discussed herein, the objects in the ambient environment may be detected based on one or more criteria. When the display system detects the presence or absence of the criteria in the ambient environment using a computer vision algorithm or using data received from one or more sensor assemblies (which may or may not be part of the display system), the display system may then signal the presence of the object.

Machine Learning

[0299] A variety of machine learning algorithms may be used to learn to identify the presence of objects in the ambient environment. Once trained, the machine learning algorithms may be stored by the display system. Some examples of machine learning algorithms may include supervised or non-supervised machine learning algorithms, including regression algorithms (such as, for example, Ordinary Least Squares Regression), instance-based algorithms (such as, for example, Learning Vector Quantization), decision tree algorithms (such as, for example, classification and regression trees), Bayesian algorithms (such as, for example, Naive Bayes), clustering algorithms (such as, for example, k-means clustering), association rule learning algorithms (such as, for example, a-priori algorithms), artificial neural network algorithms (such as, for example, Perceptron), deep learning algorithms (such as, for example, Deep Boltzmann Machine, or deep neural network), dimensionality reduction algorithms (such as, for example, Principal Component Analysis), ensemble algorithms (such as, for example, Stacked Generalization), and/or other machine learning algorithms. In some embodiments, individual models may be customized for individual data sets. For example, the wearable device may generate or store a base model. The base model may be used as a starting point to generate additional models specific to a data type (e.g., a particular user), a data set (e.g., a set of additional images obtained), conditional situations, or other variations. In some embodiments, the display system may be configured to utilize a plurality of techniques to generate models for analysis of the aggregated data. Other techniques may include using predefined thresholds or data values.

[0300] The criteria for detecting an object may include one or more threshold conditions. If the analysis of the data acquired by the environmental sensor indicates that a thresh-

old condition is passed, the display system may provide a signal indicating the detection the presence of the object in the ambient environment. The threshold condition may involve a quantitative and/or qualitative measure. For example, the threshold condition may include a score or a percentage associated with the likelihood of the reflection and/or object being present in the environment. The display system may compare the score calculated from the environmental sensor's data with the threshold score. If the score is higher than the threshold level, the display system may detect the presence of the reflection and/or object. In some other embodiments, the display system may signal the presence of the object in the environment if the score is lower than the threshold. In some embodiments, the threshold condition may be determined based on the user's emotional state and/or the user's interactions with the ambient environment.

[0301] In some embodiments, the threshold conditions, the machine learning algorithms, or the computer vision algorithms may be specialized for a specific context. For example, in a diagnostic context, the computer vision algorithm may be specialized to detect certain responses to the stimulus. As another example, the display system may execute facial recognition algorithms and/or event tracing algorithms to sense the user's reaction to a stimulus, as discussed herein.

[0302] It will be appreciated that each of the processes, methods, and algorithms described herein and/or depicted in the figures may be embodied in, and fully or partially automated by, code modules executed by one or more physical computing systems, hardware computer processors, application-specific circuitry, and/or electronic hardware configured to execute specific and particular computer instructions. For example, computing systems may include general purpose computers (e.g., servers) programmed with specific computer instructions or special purpose computers, special purpose circuitry, and so forth. A code module may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language. In some embodiments, particular operations and methods may be performed by circuitry that is specific to a given function.

[0303] Further, certain embodiments of the functionality of the present disclosure are sufficiently mathematically, computationally, or technically complex that application-specific hardware or one or more physical computing devices (utilizing appropriate specialized executable instructions) may be necessary to perform the functionality, for example, due to the volume or complexity of the calculations involved or to provide results substantially in real-time. For example, a video may include many frames, with each frame having millions of pixels, and specifically programmed computer hardware is necessary to process the video data to provide a desired image processing task or application in a commercially reasonable amount of time.

[0304] Code modules or any type of data may be stored on any type of non-transitory computer-readable medium, such as physical computer storage including hard drives, solid state memory, random access memory (RAM), read only memory (ROM), optical disc, volatile or non-volatile storage, combinations of the same and/or the like. In some embodiments, the non-transitory computer-readable medium may be part of one or more of the local processing and data module (140), the remote processing module (150),

and remote data repository (160). The methods and modules (or data) may also be transmitted as generated data signals (e.g., as part of a carrier wave or other analog or digital propagated signal) on a variety of computer-readable transmission mediums, including wireless-based and wired/cable-based mediums, and may take a variety of forms (e.g., as part of a single or multiplexed analog signal, or as multiple discrete digital packets or frames). The results of the disclosed processes or process steps may be stored, persistently or otherwise, in any type of non-transitory, tangible computer storage or may be communicated via a computer-readable transmission medium.

[0305] Any processes, blocks, states, steps, or functionalities in flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing code modules, segments, or portions of code which include one or more executable instructions for implementing specific functions (e.g., logical or arithmetical) or steps in the process. The various processes, blocks, states, steps, or functionalities may be combined, rearranged, added to, deleted from, modified, or otherwise changed from the illustrative examples provided herein. In some embodiments, additional or different computing systems or code modules may perform some or all of the functionalities described herein. The methods and processes described herein are also not limited to any particular sequence, and the blocks, steps, or states relating thereto may be performed in other sequences that are appropriate, for example, in serial, in parallel, or in some other manner. Tasks or events may be added to or removed from the disclosed example embodiments. Moreover, the separation of various system components in the embodiments described herein is for illustrative purposes and should not be understood as requiring such separation in all embodiments. It should be understood that the described program components, methods, and systems may generally be integrated together in a single computer product or packaged into multiple computer products.

Other Considerations

[0306] Each of the processes, methods, and algorithms described herein and/or depicted in the attached figures may be embodied in, and fully or partially automated by, code modules executed by one or more physical computing systems, hardware computer processors, application-specific circuitry, and/or electronic hardware configured to execute specific and particular computer instructions. For example, computing systems can include general purpose computers (e.g., servers) programmed with specific computer instructions or special purpose computers, special purpose circuitry, and so forth. A code module may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language. In some implementations, particular operations and methods may be performed by circuitry that is specific to a given function.

[0307] Further, certain implementations of the functionality of the present disclosure are sufficiently mathematically, computationally, or technically complex that application-specific hardware or one or more physical computing devices (utilizing appropriate specialized executable instructions) may be necessary to perform the functionality, for example, due to the volume or complexity of the calculations involved or to provide results substantially in real-

time. For example, animations or video may include many frames, with each frame having millions of pixels, and specifically programmed computer hardware is necessary to process the video data to provide a desired image processing task or application in a commercially reasonable amount of time.

[0308] Code modules or any type of data may be stored on any type of non-transitory computer-readable medium, such as physical computer storage including hard drives, solid state memory, random access memory (RAM), read only memory (ROM), optical disc, volatile or non-volatile storage, combinations of the same and/or the like. The methods and modules (or data) may also be transmitted as generated data signals (e.g., as part of a carrier wave or other analog or digital propagated signal) on a variety of computer-readable transmission mediums, including wireless-based and wired/cable-based mediums, and may take a variety of forms (e.g., as part of a single or multiplexed analog signal, or as multiple discrete digital packets or frames). The results of the disclosed processes or process steps may be stored, persistently or otherwise, in any type of non-transitory, tangible computer storage or may be communicated via a computer-readable transmission medium.

[0309] Any processes, blocks, states, steps, or functionalities in flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing code modules, segments, or portions of code which include one or more executable instructions for implementing specific functions (e.g., logical or arithmetic) or steps in the process. The various processes, blocks, states, steps, or functionalities can be combined, rearranged, added to, deleted from, modified, or otherwise changed from the illustrative examples provided herein. In some embodiments, additional or different computing systems or code modules may perform some or all of the functionalities described herein. The methods and processes described herein are also not limited to any particular sequence, and the blocks, steps, or states relating thereto can be performed in other sequences that are appropriate, for example, in serial, in parallel, or in some other manner. Tasks or events may be added to or removed from the disclosed example embodiments. Moreover, the separation of various system components in the implementations described herein is for illustrative purposes and should not be understood as requiring such separation in all implementations. It should be understood that the described program components, methods, and systems can generally be integrated together in a single computer product or packaged into multiple computer products. Many implementation variations are possible.

[0310] The processes, methods, and systems may be implemented in a network (or distributed) computing environment. Network environments include enterprise-wide computer networks, intranets, local area networks (LAN), wide area networks (WAN), personal area networks (PAN), cloud computing networks, crowd-sourced computing networks, the Internet, and the World Wide Web. The network may be a wired or a wireless network or any other type of communication network.

[0311] The systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combi-

nations and subcombinations are intended to fall within the scope of this disclosure. Various modifications to the implementations described in this disclosure may be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the claims are not intended to be limited to the implementations shown herein, but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features disclosed herein.

[0312] Certain features that are described in this specification in the context of separate implementations also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination. No single feature or group of features is necessary or indispensable to each and every embodiment.

[0313] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise.

[0314] As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: A, B, or C” is intended to cover: A, B, C, A and B, A and C, B and C, and A, B, and C. Conjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be at least one of X, Y or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y and at least one of Z to each be present.

[0315] Similarly, while operations may be depicted in the drawings in a particular order, it is to be recognized that such operations need not be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the

drawings may schematically depict one more example processes in the form of a flowchart. However, other operations that are not depicted can be incorporated in the example methods and processes that are schematically illustrated. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the illustrated operations. Additionally, the operations may be rearranged or reordered in other implementations. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products. Additionally, other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results.

What is claimed is:

1. An augmented reality display system comprising:
 - a head-mounted display configured to present virtual content by outputting light to a wearer, wherein the head-mounted display is configured to output light to an eye of the wearer with different amounts of wavefront divergence corresponding to different perceived depths away from the wearer; and
 - at least one processor communicatively coupled to the head-mounted display, the at least one processor configured to:
 - determine whether the wearer is a calibrated user or a guest user;
 - when the wearer is determined to be a calibrated user:
 - load pre-existing user depth plane switching calibration information; and
 - set depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information; and
 - when the wearer is determined to be a guest user:
 - identify a virtual object at which the guest user is most likely to be looking, wherein the virtual object has an associated depth plane; and
 - set depth plane switching parameters for the head-mounted display based upon the associated depth plane, wherein the display system is configured to:
 - determine whether the guest user is most likely looking at the virtual object by determining whether the guest user's eyes are fixating within a volume encompassing the virtual object;
 - determine an uncertainty associated with determining a position of a fixation point of the wearer; and
 - vary a size of the volume encompassing the virtual object based upon the uncertainty.
2. The augmented reality display system of claim 1, wherein the display system is configured to determine whether the wearer is a calibrated user or a guest user by determining the wearer's interpupillary distance.
3. The augmented reality display system of claim 1, wherein, upon detecting that the calibrated user is no longer wearing the head-mounted display after determining that the wearer is a calibrated user, the display system is configured to continue to utilize the calibrated user's depth plane

switching calibration information for a predetermined amount of time or for a predetermined number of image frames.

4. The augmented reality display system of claim 1, wherein the head-mounted display comprises a waveguide stack configured to pass light from real world into an eye of the wearer, wherein the waveguide stack comprises a plurality of waveguides comprising one or more waveguides configured to output light to the eye of the wearer with a different amount of wavefront divergence than one or more other waveguides of the plurality of waveguides.

5. A method for determining parameters for depth plane switching in a display system configured to direct image light to eyes of a user to display virtual image content, wherein the display system comprises a head-mounted display, the display system configured to present the virtual image content on a plurality of depth planes, wherein each depth plane is associated with image light having a different amount of wavefront divergence, wherein the display system is configured to switch the virtual image content between different depth planes by changing a wavefront divergence of the image light, the method comprising:

determining whether the user is a calibrated user or a guest user;

when the user is determined to be a calibrated user:

loading pre-existing user depth plane switching calibration information; and

setting depth plane switching parameters for the head-mounted display based upon the pre-existing user depth plane switching calibration information; and

when the user is determined to be a guest user:

determining whether the guest user is looking at a virtual object, wherein the virtual object has an associated depth plane; and

setting depth plane switching parameters for the head-mounted display based upon the associated depth plane, wherein the display system is further configured to:

determine whether the guest user is most likely looking at the virtual object by determining whether the guest user's eyes are fixating within a volume encompassing the virtual object;

determine an uncertainty associated with determining a position of a fixation point of the user; and

vary a size of the volume encompassing the virtual object based upon the uncertainty.

6. The method of claim 5, further comprising, upon detecting that the calibrated user is no longer wearing a display of the display system after determining that the user is a calibrated user, continue utilizing the calibrated user's depth plane switching calibration information for a predetermined amount of time or for a predetermined number of image frames.

7. The method of claim 5, wherein the display system comprises a waveguide stack configured to pass light from real world into an eye of the user, wherein the waveguide stack comprises a plurality of waveguides comprising one or more waveguides configured to output light to an eye of the user with a different amount of wavefront divergence than one or more other waveguides of the plurality of waveguides.

8. A method for determining parameters for depth plane switching in a display system configured to direct image light to eyes of a user to display virtual image content, the

eyes separated by an interpupillary distance and the display system configured to present the virtual image content on a plurality of depth planes, wherein each depth plane is associated with image light having a different amount of wavefront divergence, wherein the display system is configured to switch the virtual image content between different depth planes by changing a wavefront divergence of the image light, the method comprising:

determining whether the user is a calibrated user or a guest user;

when the user is determined to be a calibrated user:

loading pre-existing user depth plane switching calibration information; and

setting depth plane switching parameters for the display system based upon the pre-existing user depth plane switching calibration information; and

when the user is determined to be a guest user:

determining the interpupillary distance of the guest user; and

setting depth plane switching parameters for the display system based upon the determined interpupillary distance, wherein the display system is further configured to:

determine whether the guest user is most likely looking at the virtual image content by determining whether the guest user's eyes are fixating within a volume encompassing the virtual image content;

determine an uncertainty associated with determining a position of a fixation point of the user; and vary a size of the volume encompassing the virtual image content based upon the uncertainty.

9. The method of claim 8, wherein determining the interpupillary distance of the guest user comprises determining an interpupillary distance of eyes of the guest user focused at optical infinity.

10. The method of claim 8, wherein determining whether the user is the calibrated user or is a guest user comprises determining the interpupillary distance of the user with one or more eye tracking cameras configured to image the eyes of the user.

11. The method of claim 8, wherein determining whether the user is the calibrated user comprises determining the interpupillary distance of the user with one or more eye

tracking cameras configured to image the eyes of the user and determining that the interpupillary distance of the user is within a predetermined range.

12. The method of claim 11, wherein determining whether the user is the guest user comprises determining the interpupillary distance of the user with the one or more eye tracking cameras and determining that the interpupillary distance of the user is outside of the predetermined range.

13. The method of claim 8, wherein the pre-existing user depth plane switching calibration information comprises an measured interpupillary distance of the calibrated user, wherein determining whether the user is the calibrated user or is a guest user comprises determining the interpupillary distance of the user, determining that the user is the calibrated user when the user's interpupillary distance is within 1.0 mm of the measured interpupillary distance of the calibrated user, and determining that the user is a guest user when the user's interpupillary distance is not within 1.0 mm of the measured interpupillary distance of the calibrated user.

14. The method of claim 8, wherein determining whether the user is the calibrated user or is a guest user comprises identifying the user with at least one eye tracking camera.

15. The method of claim 8, further comprising:

with an eye tracking system, determining an optical axis for each of the user's eyes; and

determining a vergence depth of the user based at least on an intersection of the determined optical axes for each of the user's eyes and a comparison of a maximum interpupillary distance and a current interpupillary distance of the user.

16. The method of claim 8, further comprising:

determining a vergence distance of the user based at least in part on a comparison of a maximum interpupillary distance and a current interpupillary distance of the user.

17. The method of claim 8, further comprising:

selecting which depth plane to use to present the virtual image content based at least in part on a current vergence depth of the user.

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